

CODED PHOTOGRAPHY (& INTRODUCTION TO PLENOPTIC IMAGING)

Computational Imaging – Spring 2024

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Slide credits

Many of these slides are based on the following content:

- [Computational Photography \(15-463\)](#). Carnegie Mellon University.
- Computational Plenoptic Imaging. Wetzstein et al. SIGGRAPH Course.

So far

We have been building a basis:

- Image formation and digital photography – the conventional methods upon which computational imaging builds
- Mathematical foundations – (reminder of) the set of tools needed for the rest of the course

Upcoming

- Coded photography

And we will then widen our perspective, into a more general vision

- Plenoptic function and plenoptic imaging
- Compressive sensing

Conventional photography



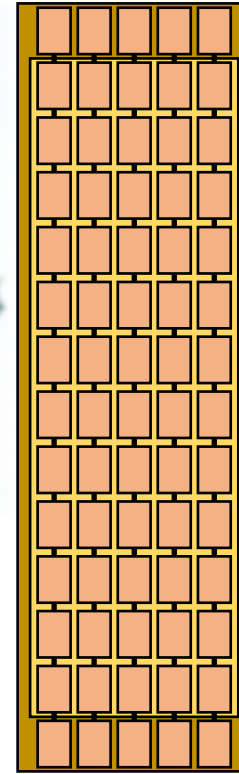
real world



optics



captured image



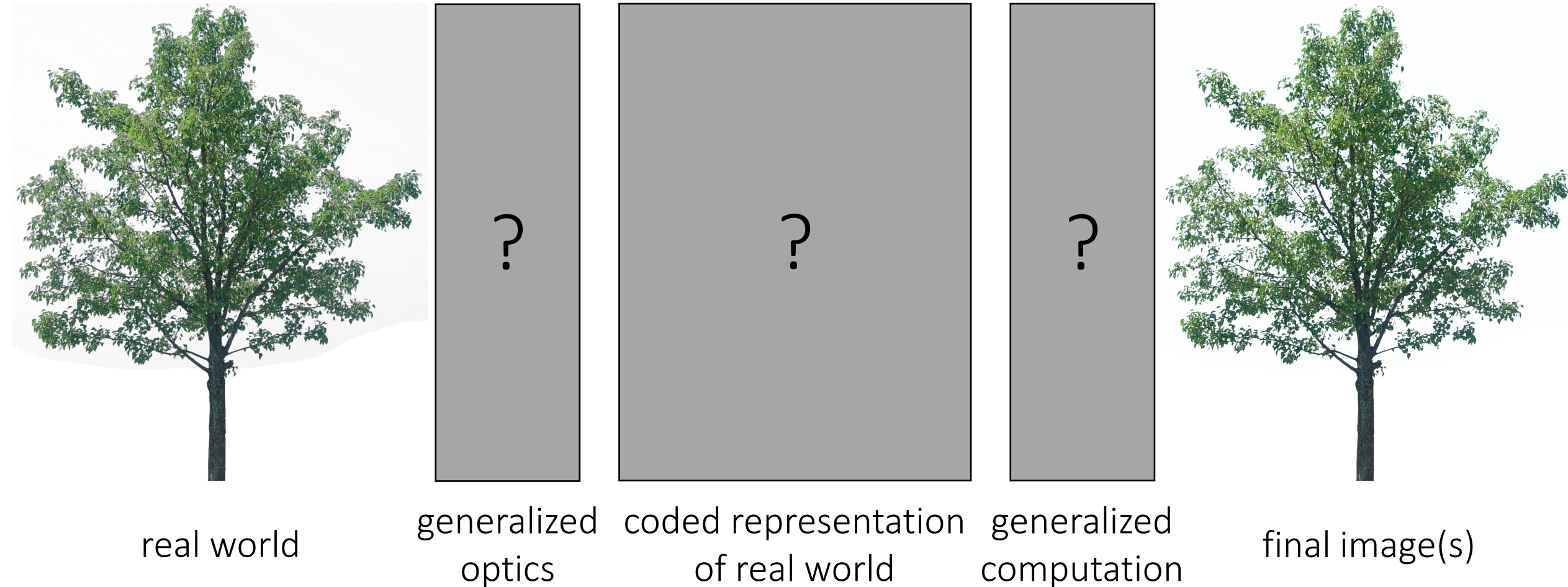
computation



enhanced image

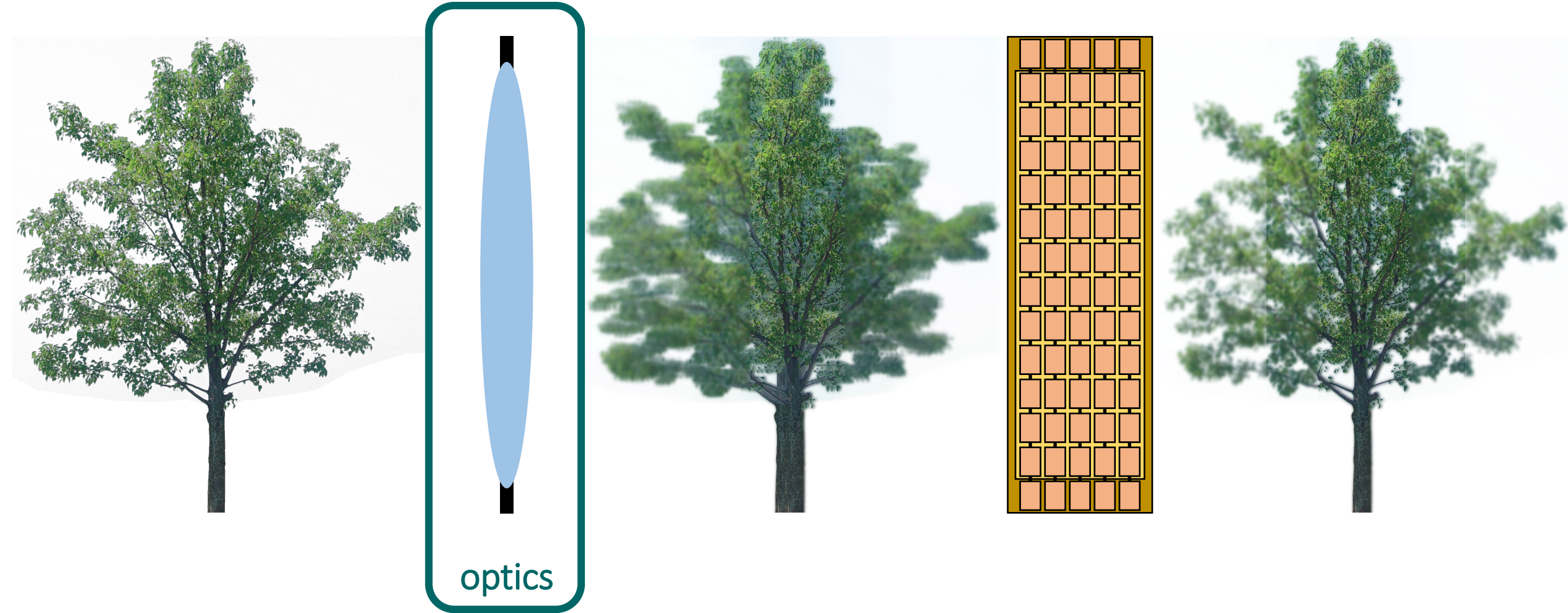
- Optics capture something that is (close to) the final image.
- Computation mostly “enhances” the captured image (e.g., deblur).

Computational photography

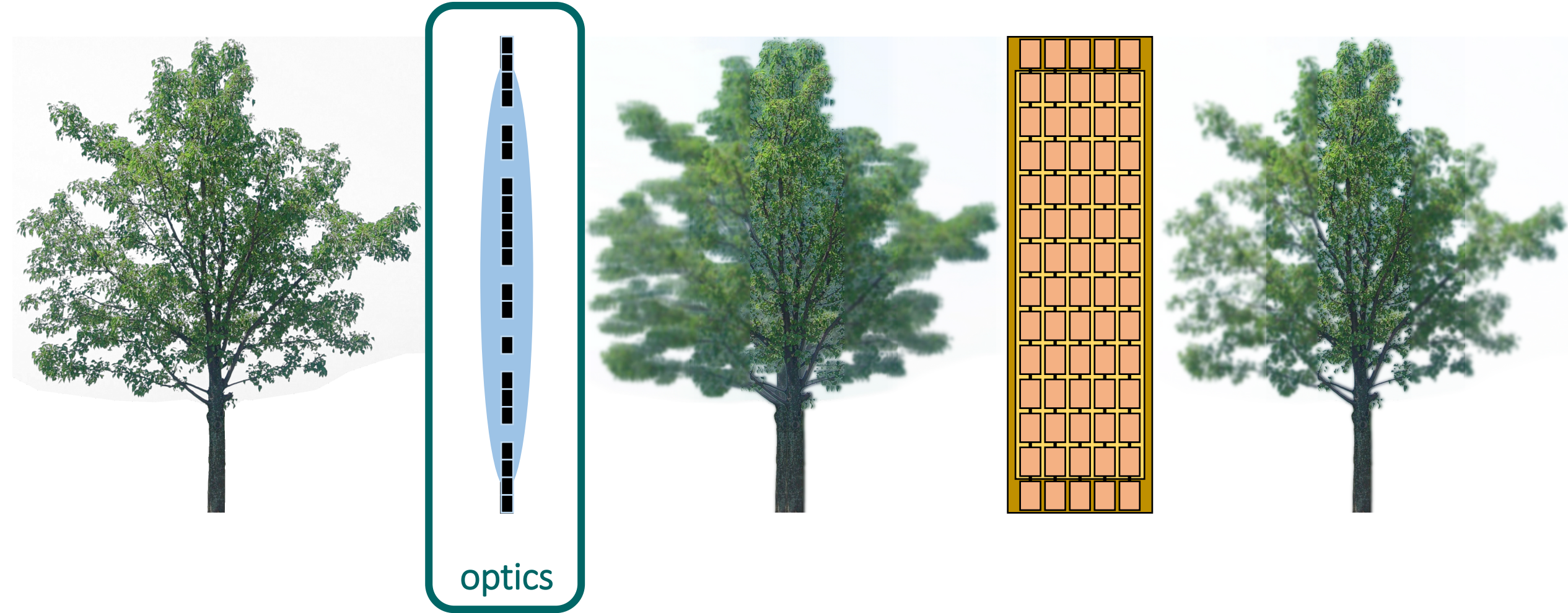


- Generalized optics encode world into intermediate representation.
- Generalized computation decodes representation into one or multiple images.

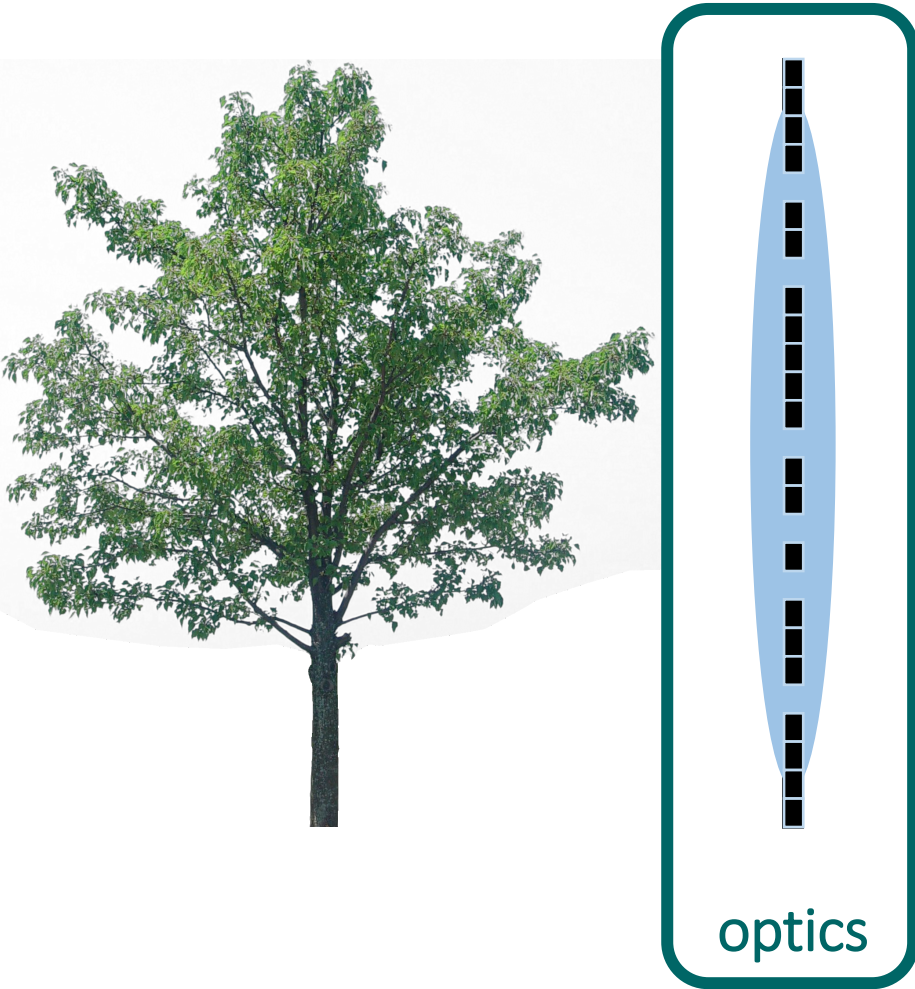
An example of this: Coded Apertures



An example of this: Coded Apertures



An example of this: Coded Apertures



Coded Photography

coded photography

The introduction of codes (or other types of generalized optics) at some point of the imaging pipeline can be used in a wealth of imaging techniques...

- Defocus deblurring
- Depth from defocus

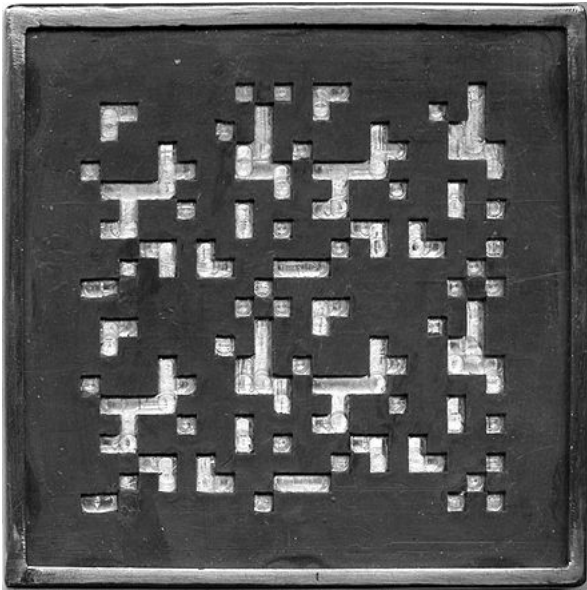
} These two often (not always) go hand-in-hand

- Scene motion & motion deblurring

Coded Apertures for Defocus Deblurring:

Building an Intuition

Coded apertures: An old idea



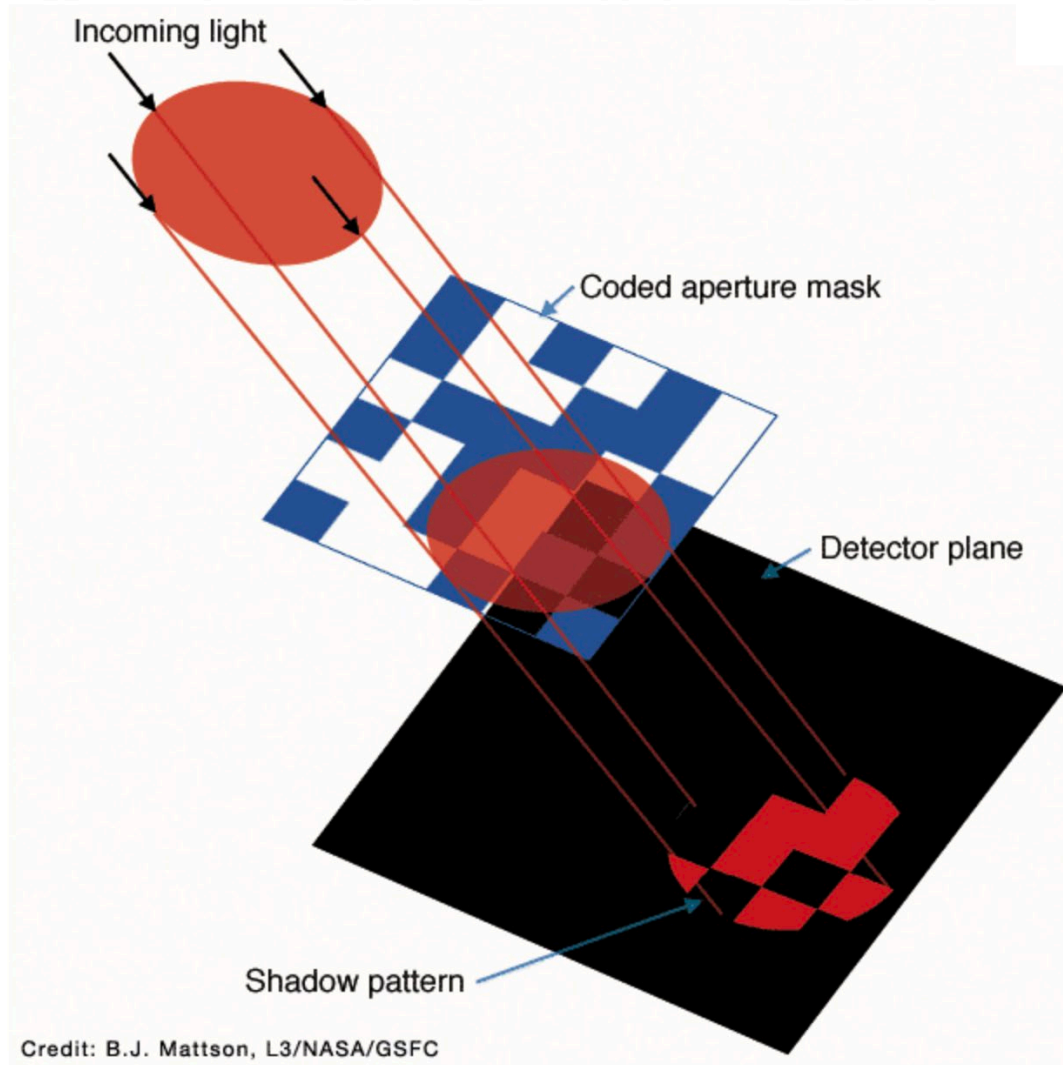
Astronomy!

“Normal” imaging is based on lenses and mirrors

The energy of *X-rays* and *γ -rays* is too high: no refraction nor reflection occurs

(Indices of refraction for energies $> \text{KeV}$ are pretty close to 1)

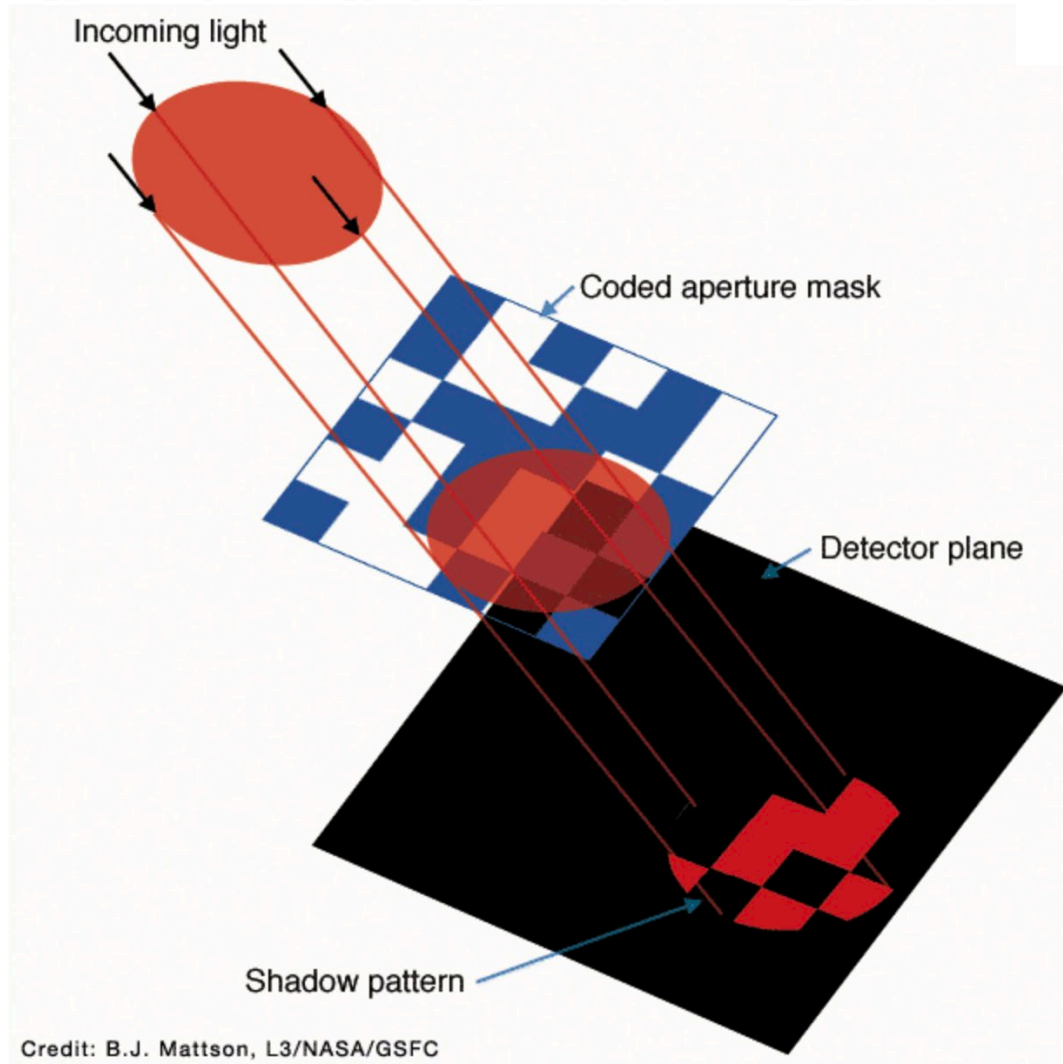
Options to increase refraction include silicon or gold prisms



In a coded-aperture-based telescope, there's a mask mounted above the sensors at a distance of about 1m

It has small lead tiles glued to it in a known pattern. Gamma rays hitting the lead tiles are stopped, but those going through the open areas continue on

It then casts a shadow on the detector, from which the position in the sky of the source may be inferred



Of course the most basic coded aperture would be the **pinhole camera**, but it has a very low throughput (most rays are blocked)

The coded aperture imaging technique is one of the earliest forms of computational photography

*J. G. Ables (1968). "Fourier transform photography: a new method for X-ray astronomy". Publications of the Astronomical Society of Australia. Cambridge University Press. **1** (4): 172–173.*

*R. H. Dicke (1968). "Scatter-hole cameras for x-rays and gamma rays". The Astrophysical Journal. **153**:L101.*

Coded apertures for defocus deblurring

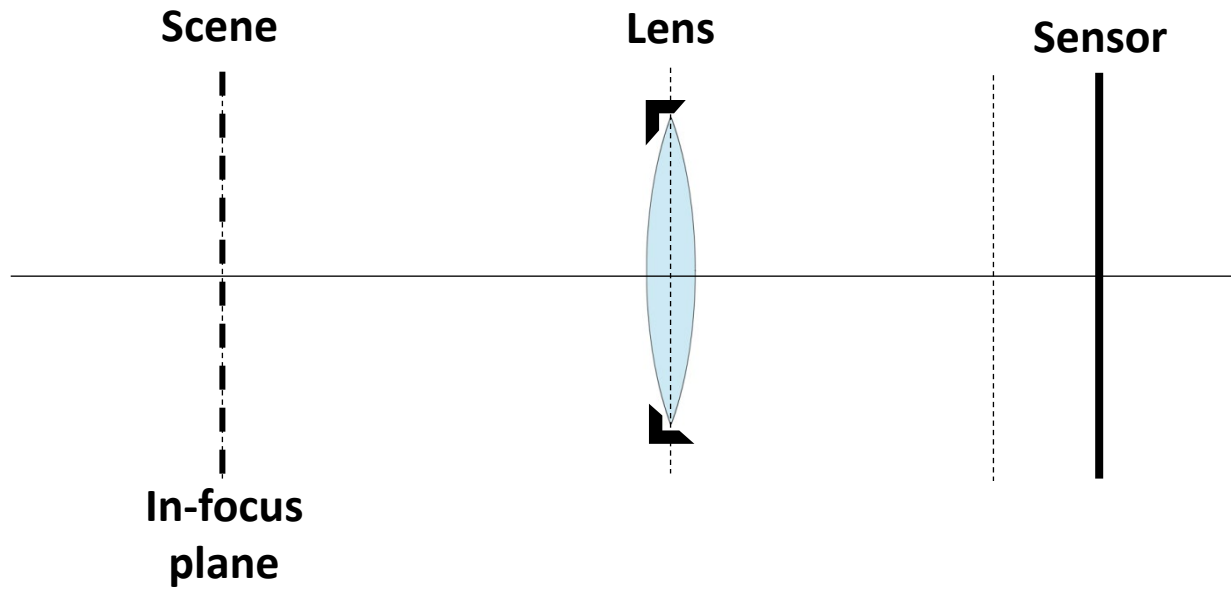


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

In-focus scene

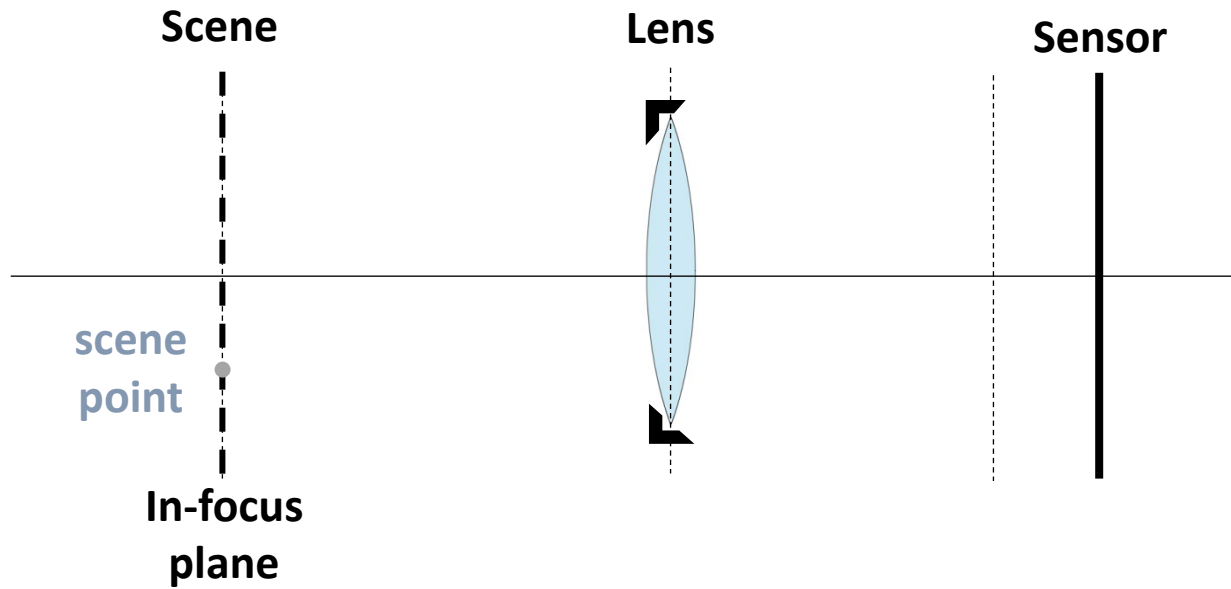


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

In-focus scene

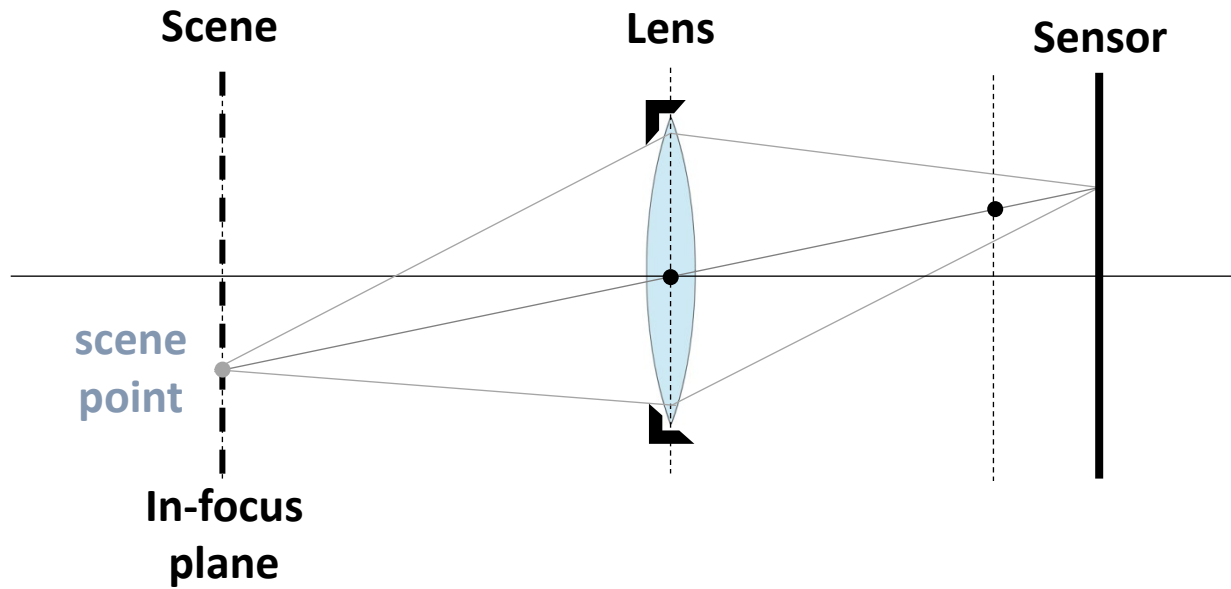


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

In-focus scene

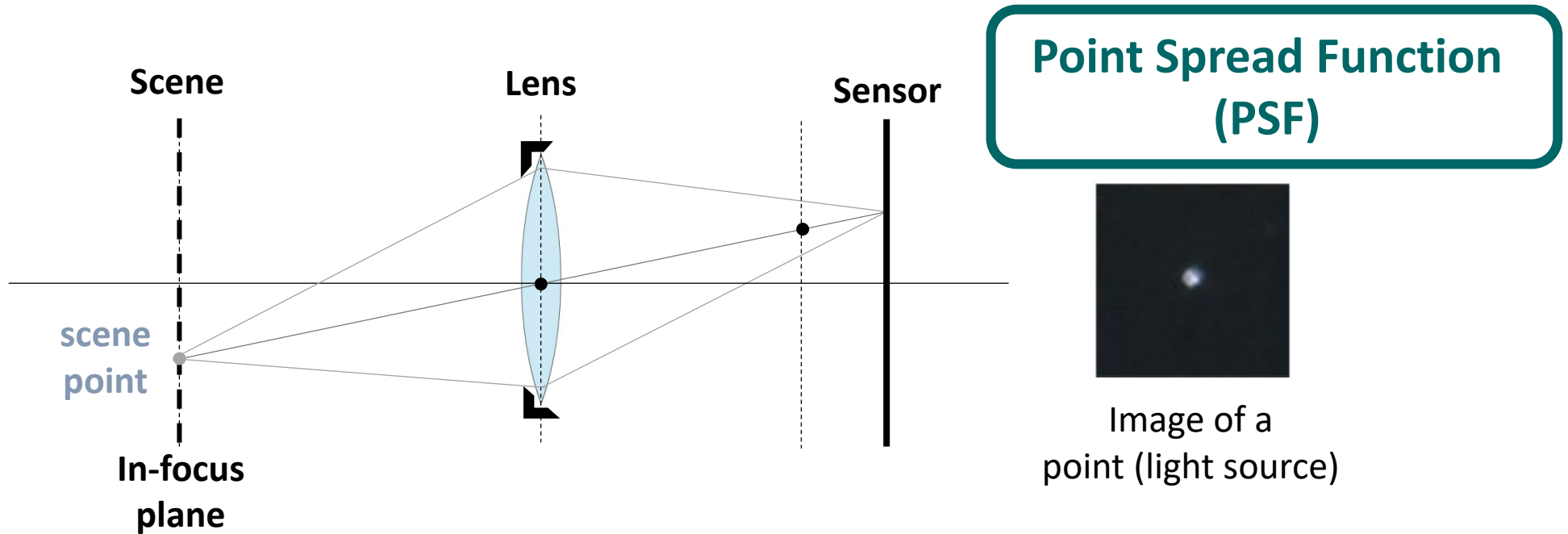


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

In-focus scene

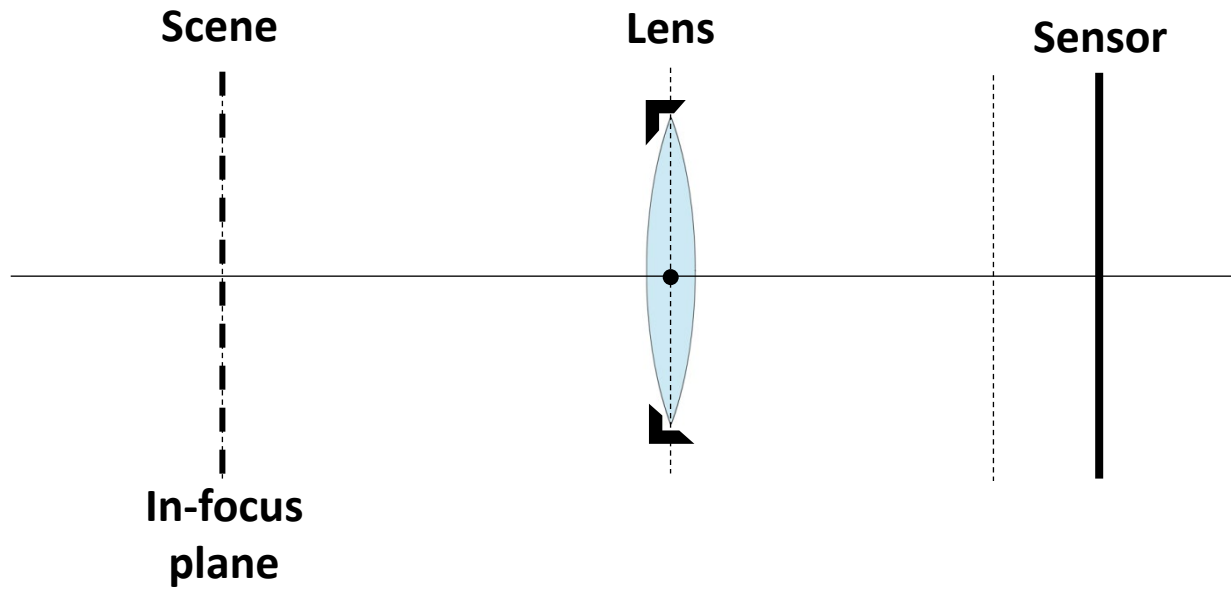


Point Spread Function (PSF): Response of a focused imaging system to a point source. Impulse response of an imaging system.

(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

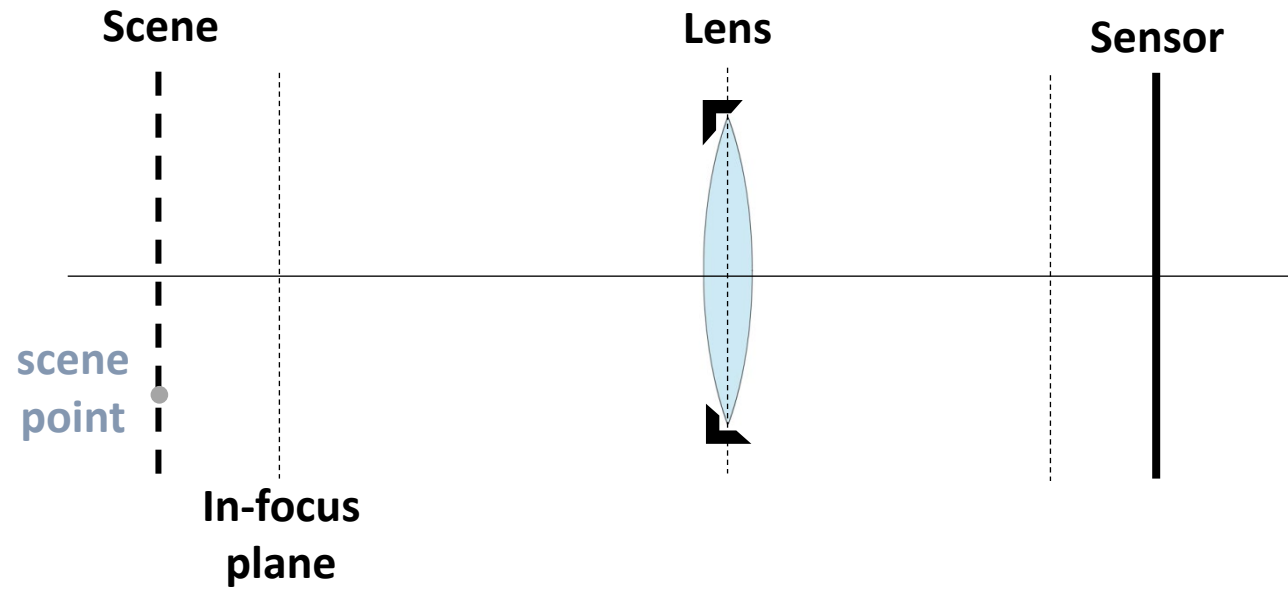


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

Out-of-focus scene

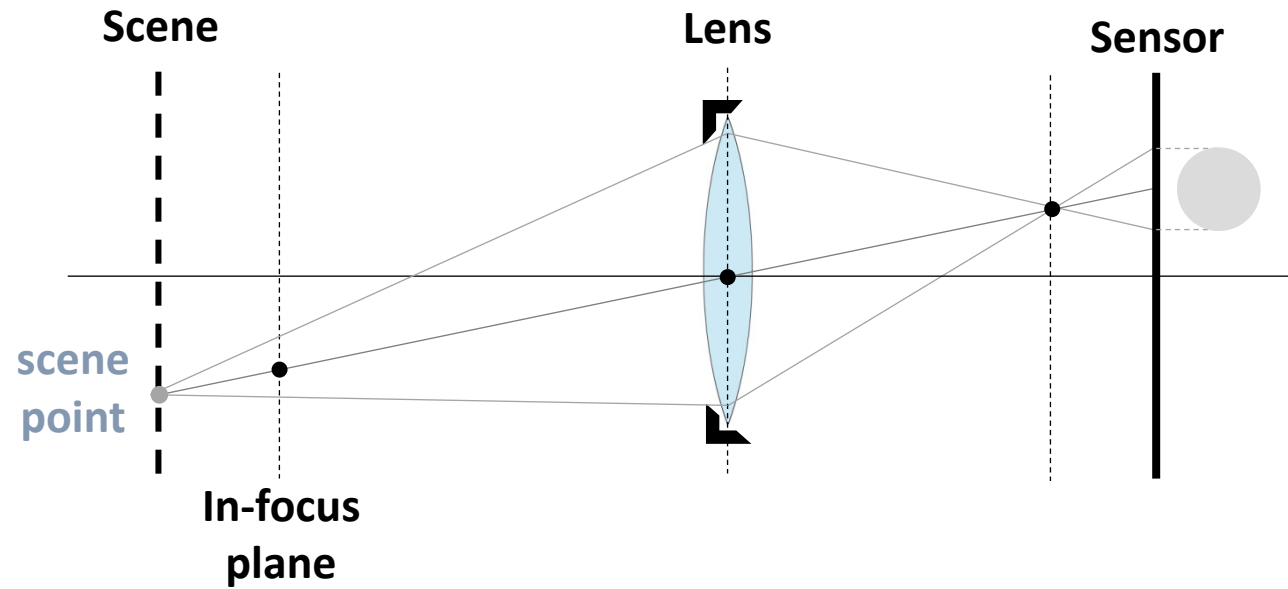


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

Out-of-focus scene

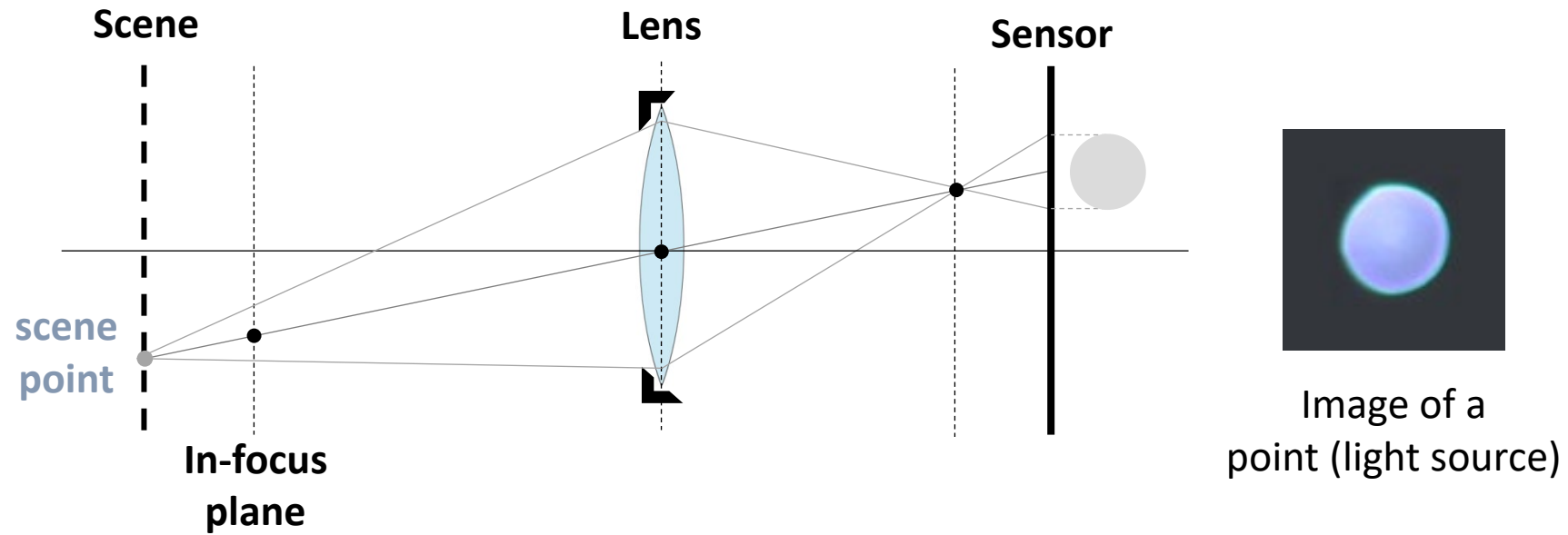


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

Out-of-focus scene

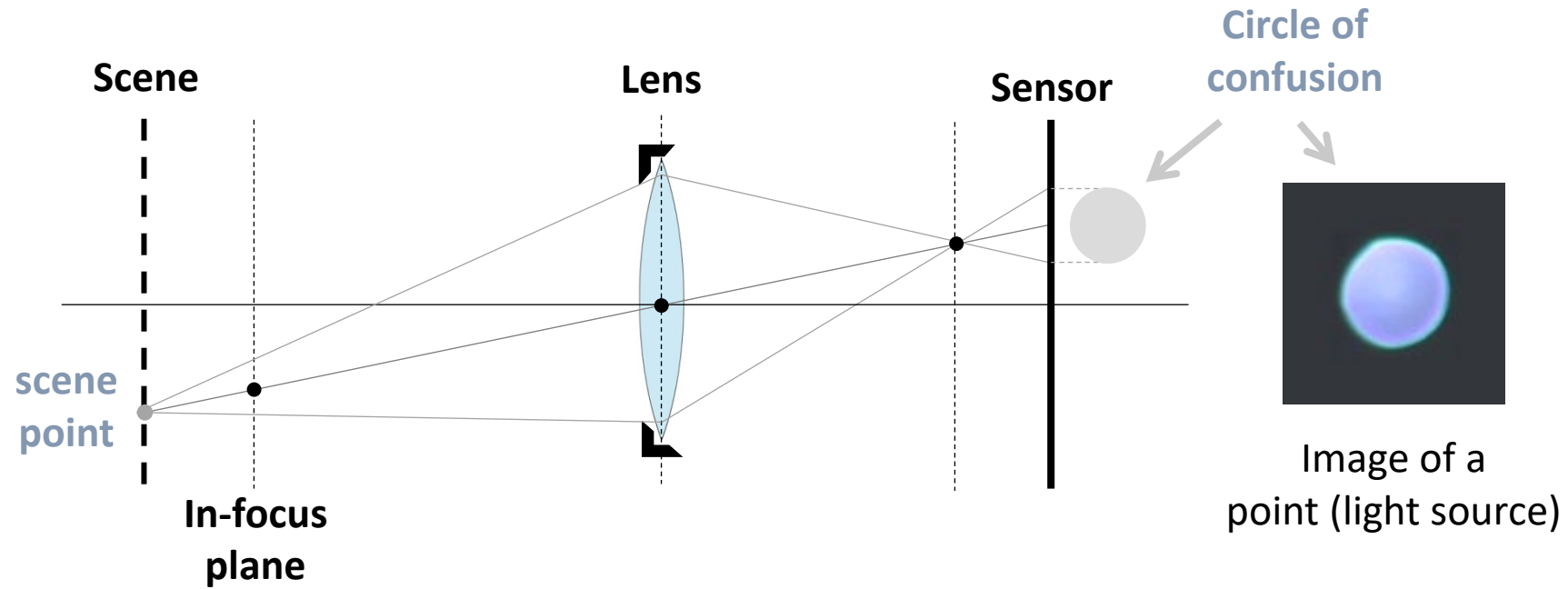


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

Out-of-focus scene

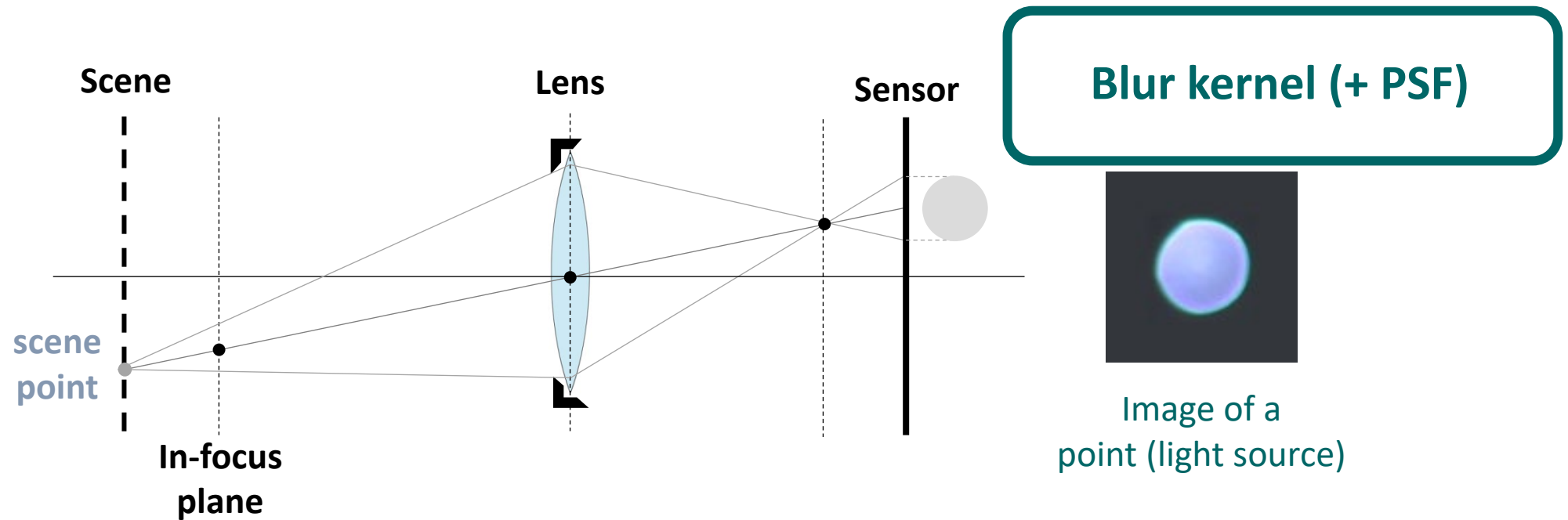


(reminder)

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Defocus blur

Out-of-focus scene



(reminder)

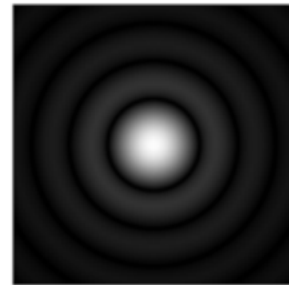
Lens as an optical low-pass filter



image from a perfect lens

I

*



=



image from imperfect lens

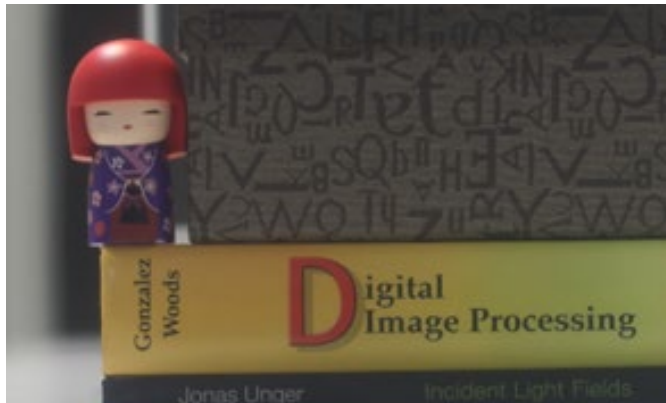
*

k

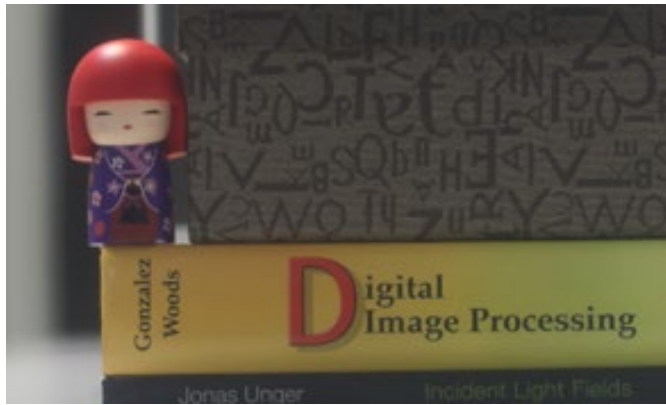
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B

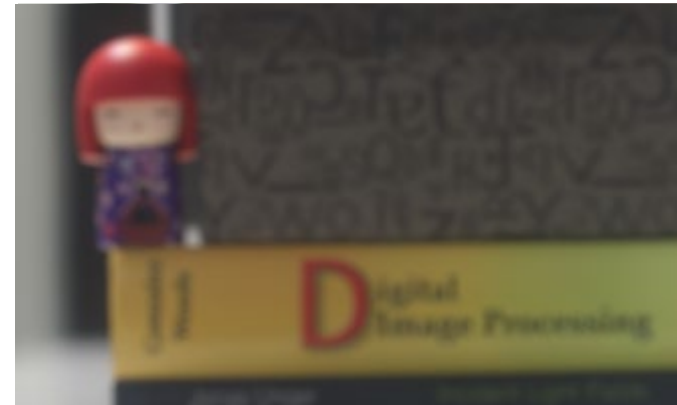
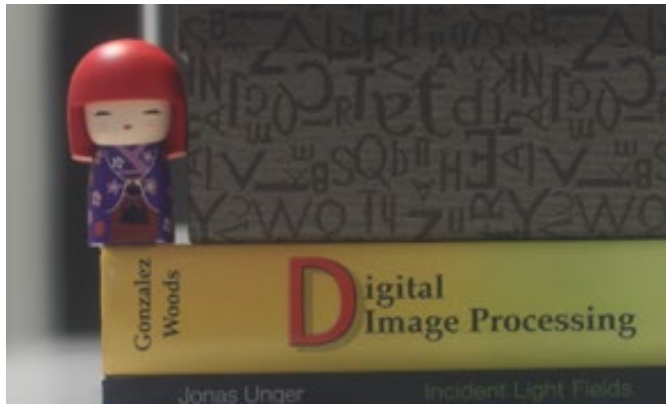
(reminder)



(reminder)

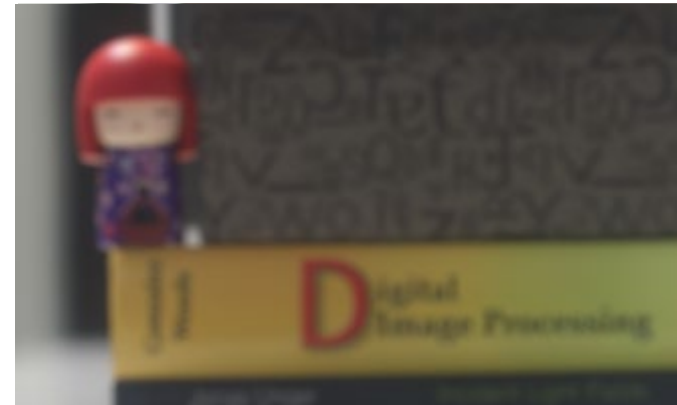
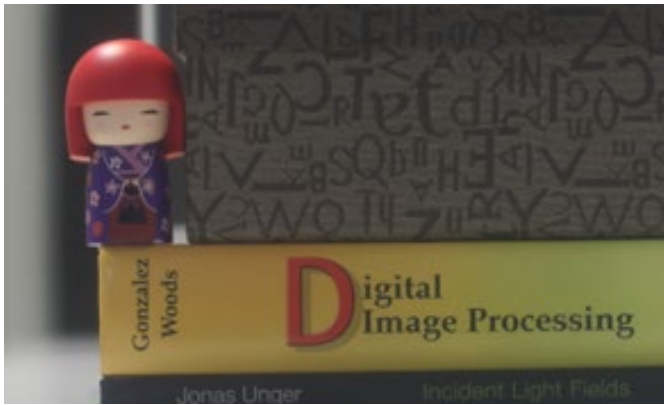


(reminder)



(reminder)

Blur kernel



(reminder)



Image formation is modeled as convolution of the scene by the PSF of the imaging system.

(reminder)

Fourier Transform – *The Convolution Theorem*

- The Fourier transform of the convolution of two functions is the product of their Fourier transforms

$$F[g * h] = F[g]F[h]$$

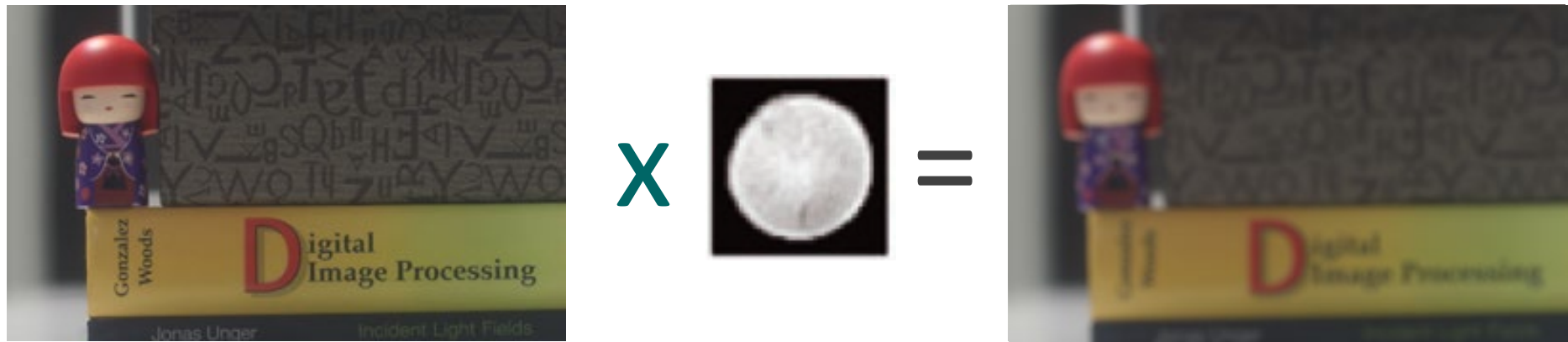
- **Convolution** in spatial domain is equivalent to **multiplication** in frequency domain!

$$g * h = F^{-1}[F[g]F[h]]$$

In the primal domain...



In the Fourier domain...



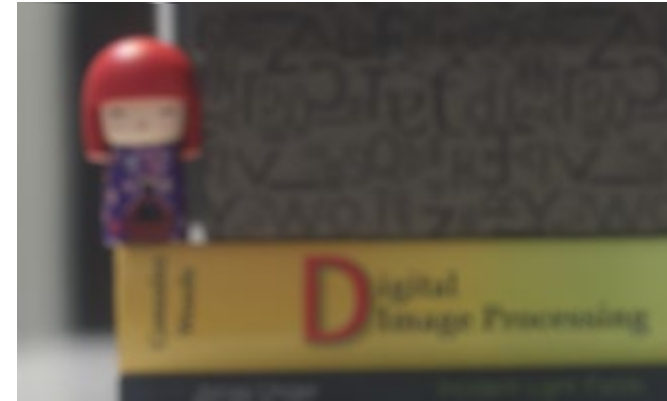
If we want to recover the original (in-focus) image...



(reminder)



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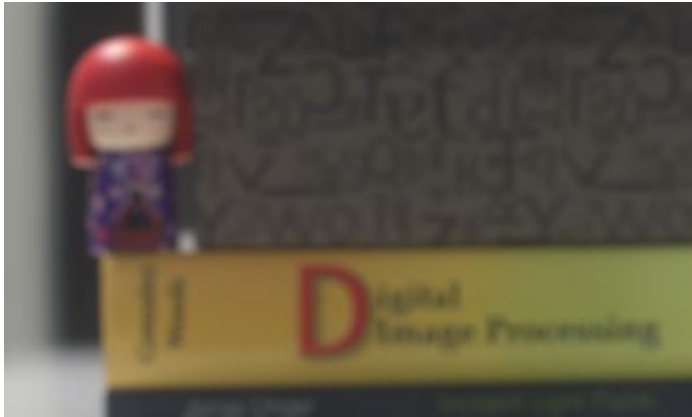


(reminder)



Captured image

(reminder)



Captured image

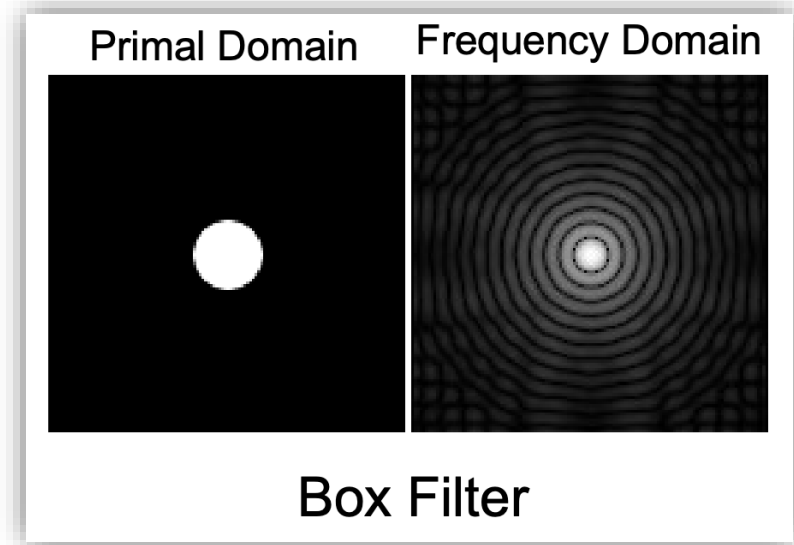


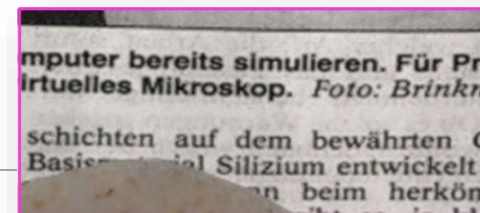
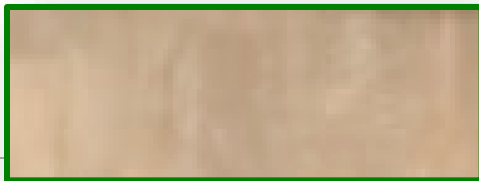
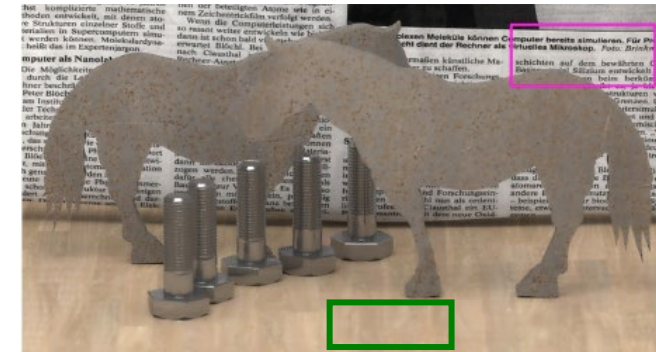
Recovered image

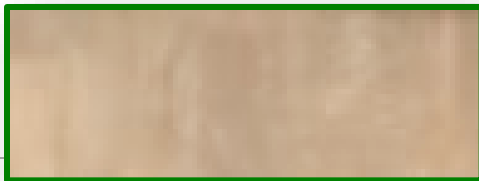




(reminder)

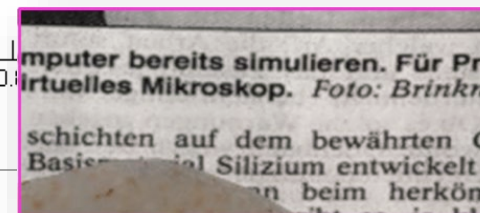


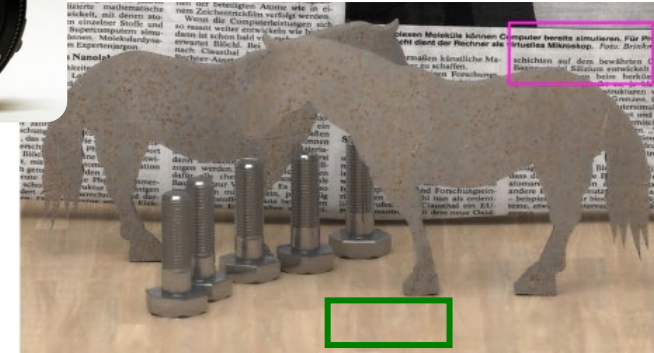




0.4 0.6 0.8

Details



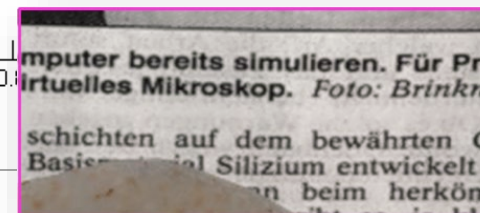


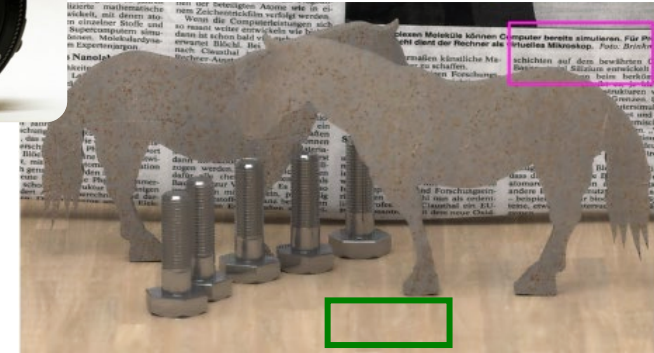
Recoverable energy

0
-2
4
6
8
0
2
4
6
-18

0.4 0.6 0.8

Details



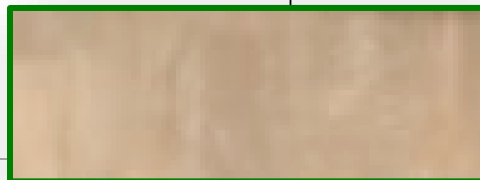
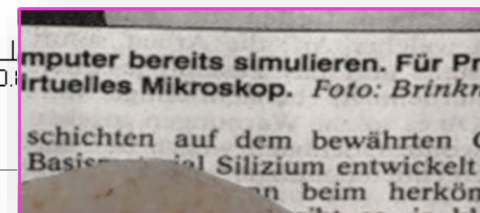


Recoverable energy



0.4 0.6

Details

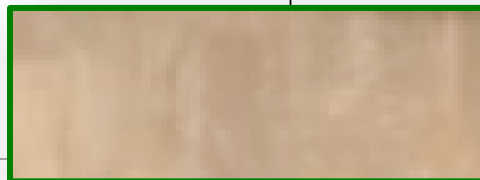
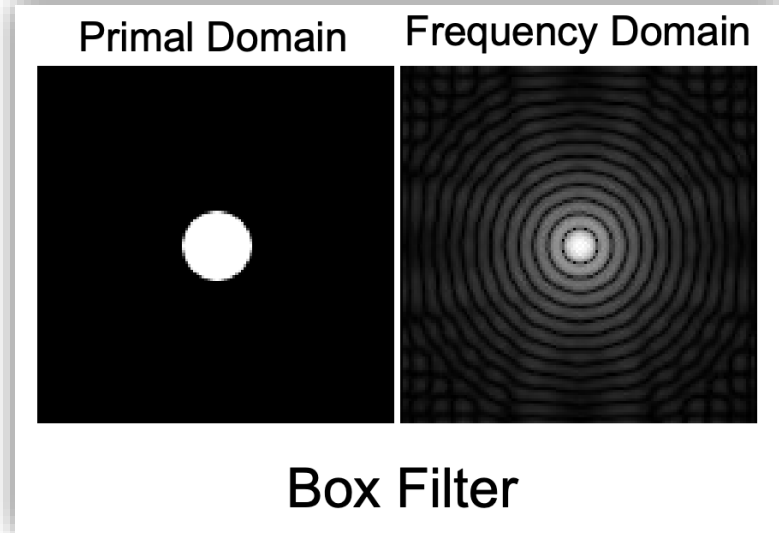




Recoverable energy

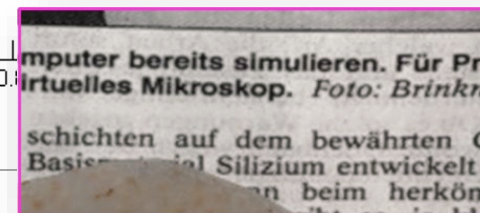


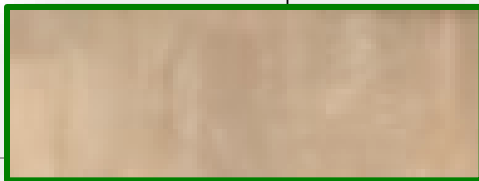
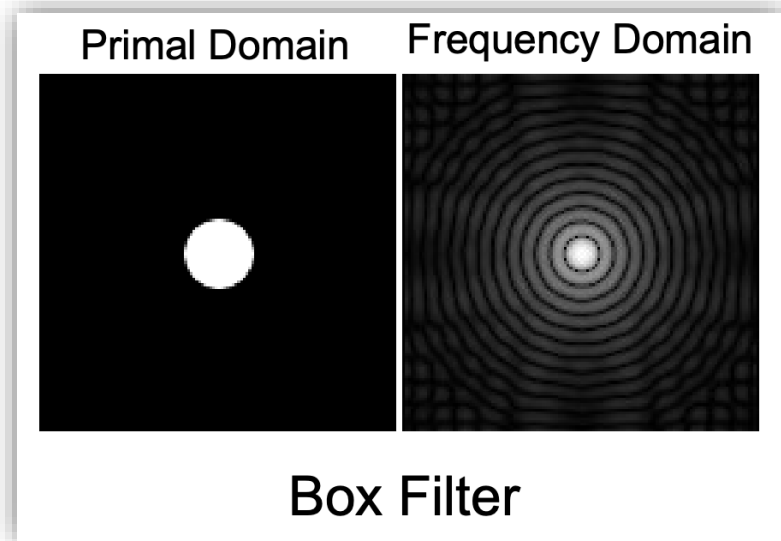
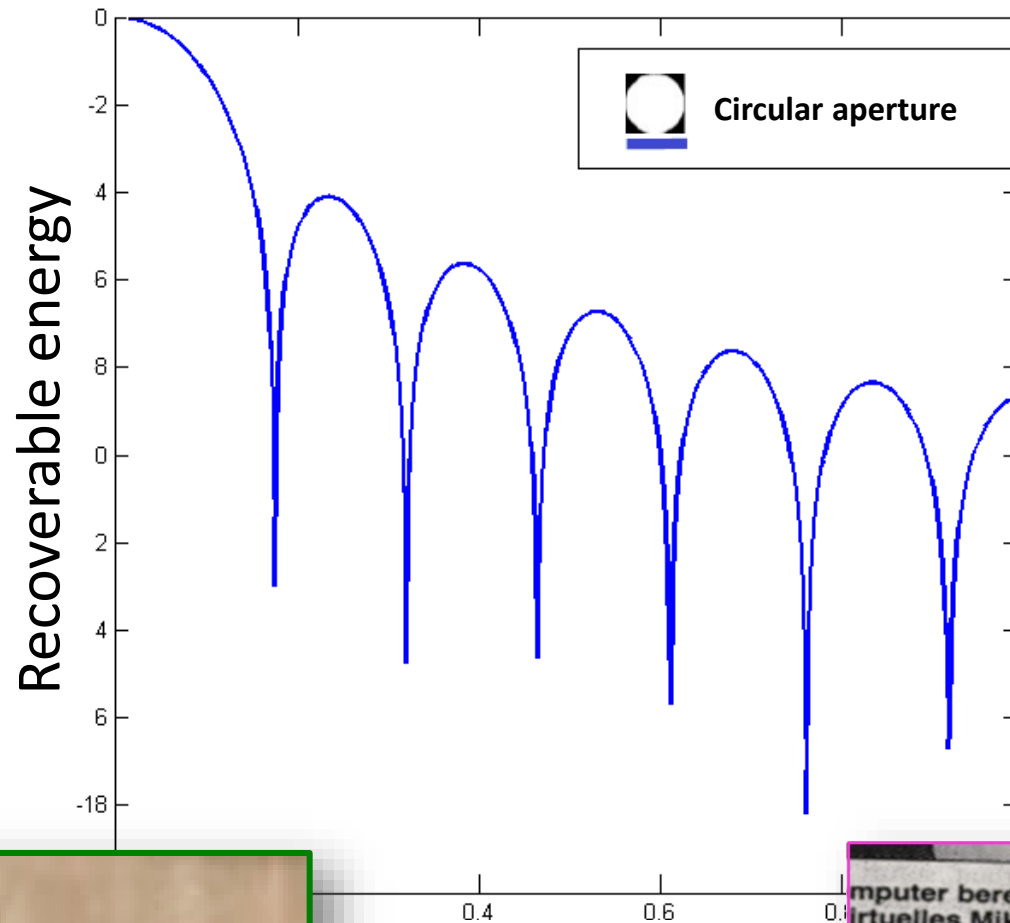
Jean Baptiste Joseph Fourier (1768-1830)



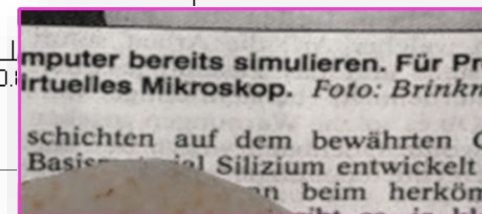
0.4 0.6 0.8

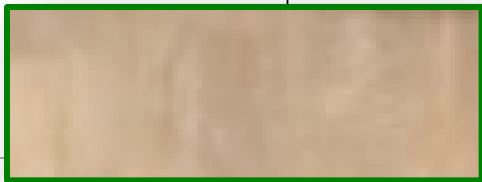
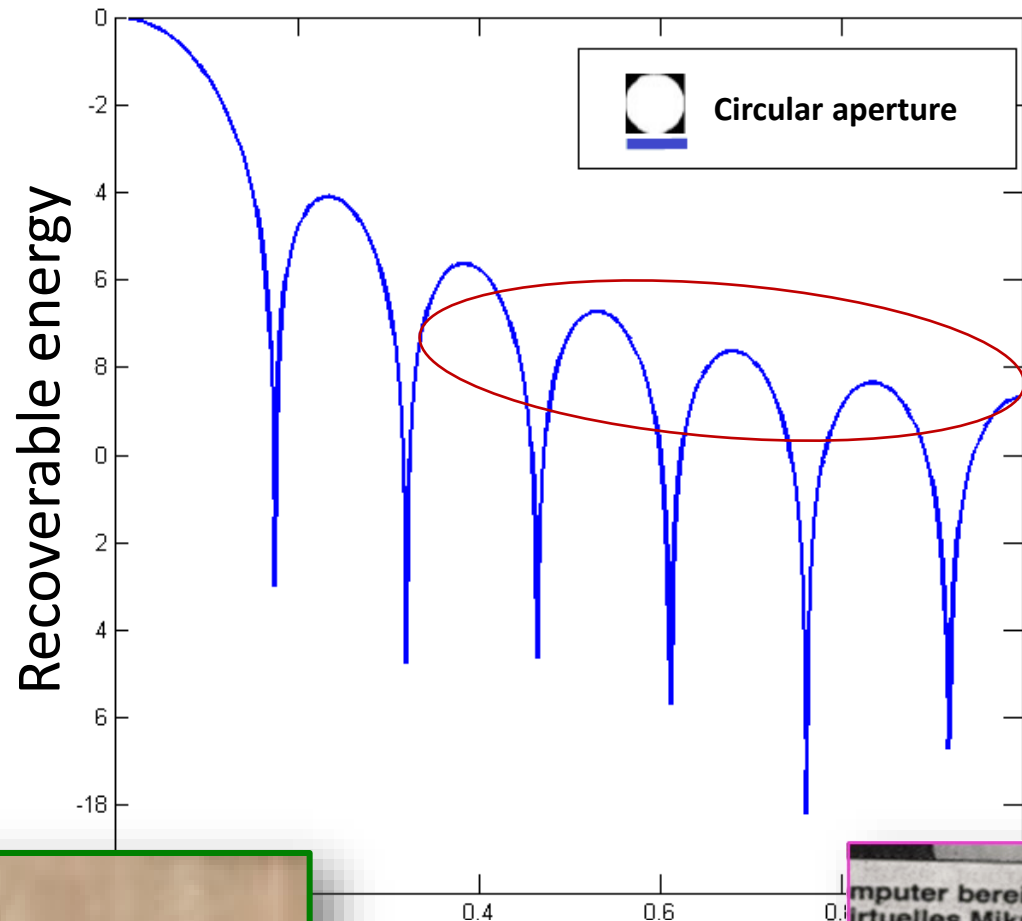
Details



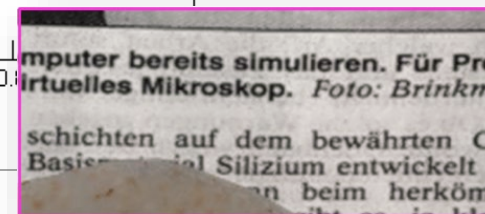
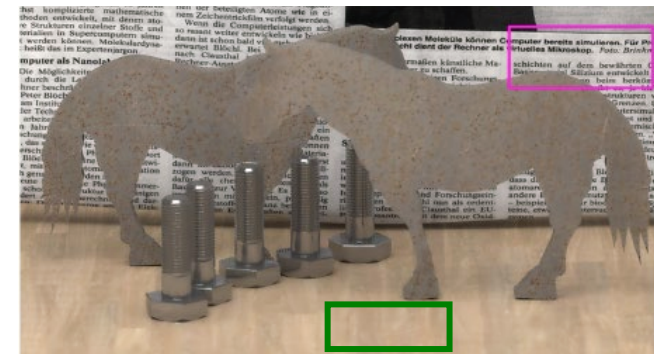


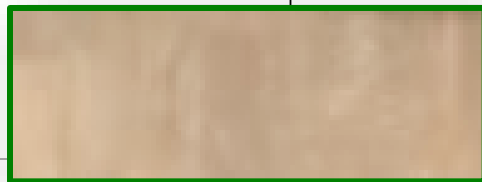
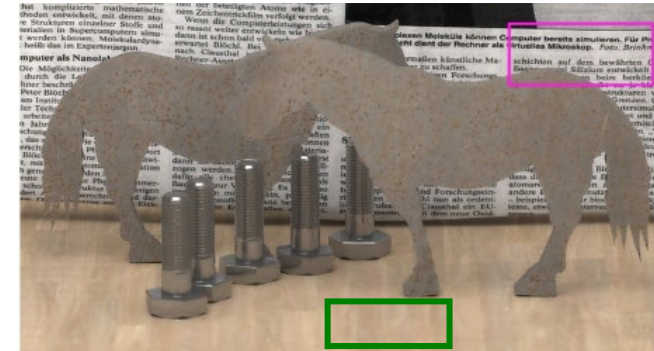
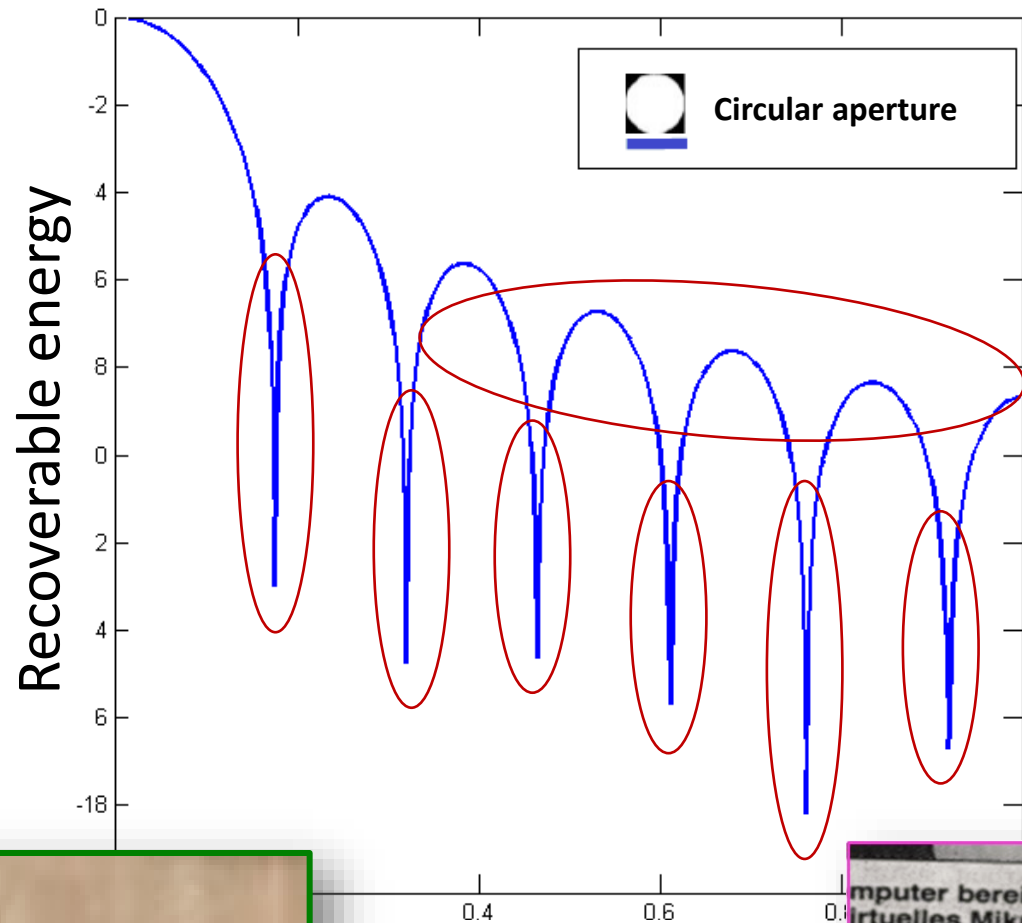
Details



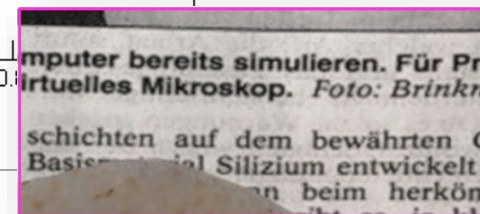


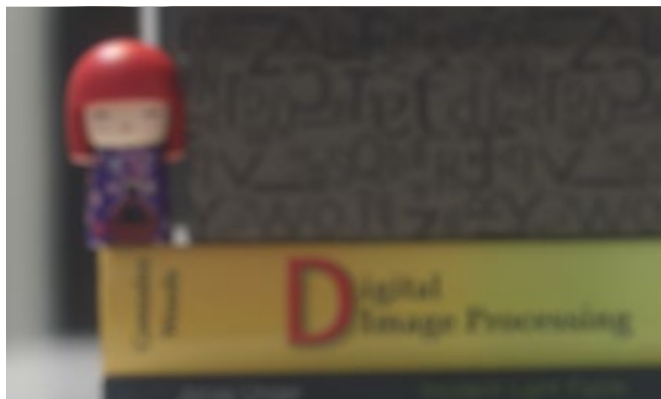
Details





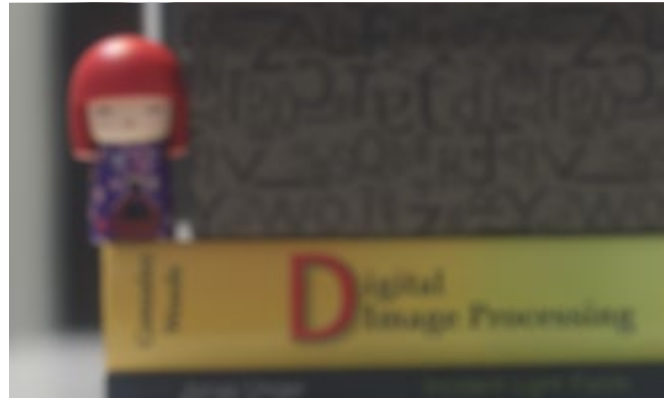
Details





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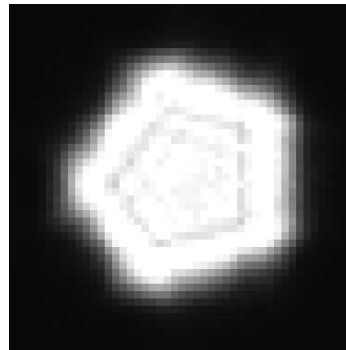
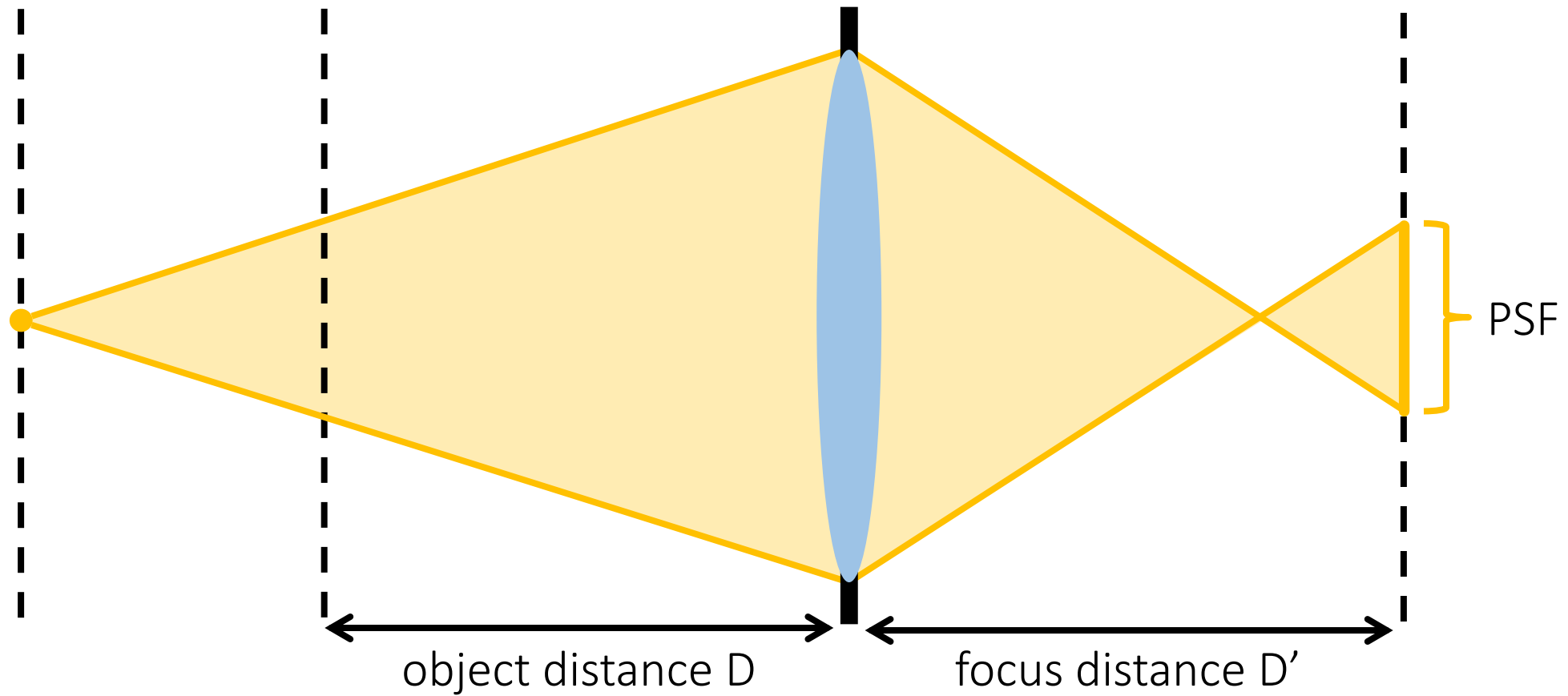
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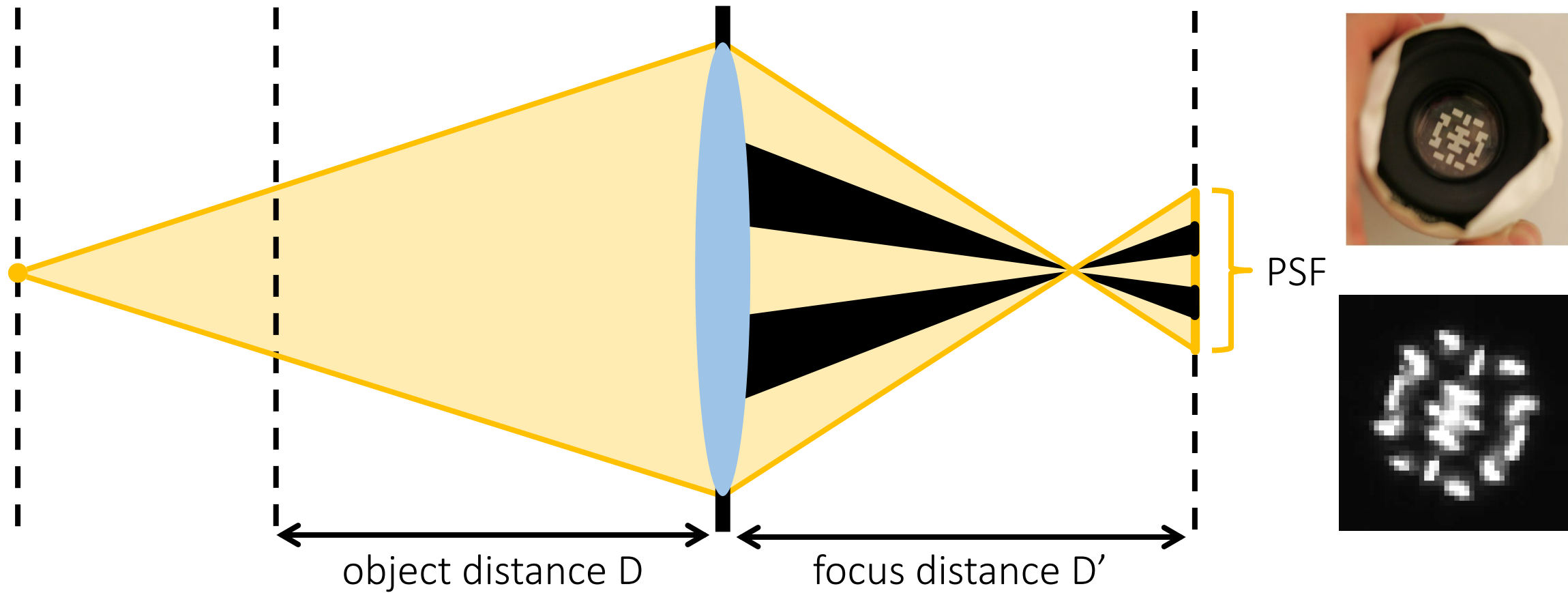


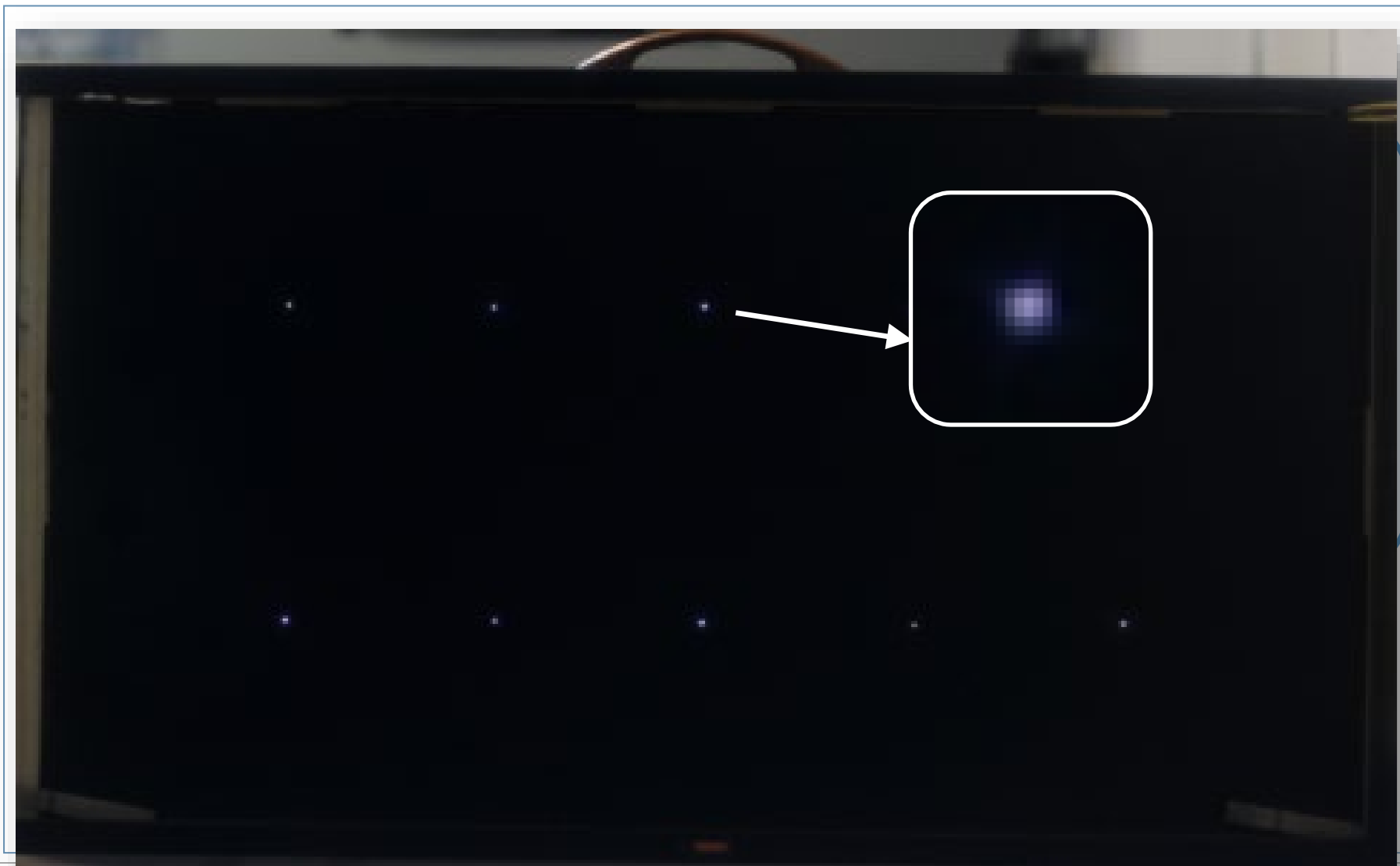


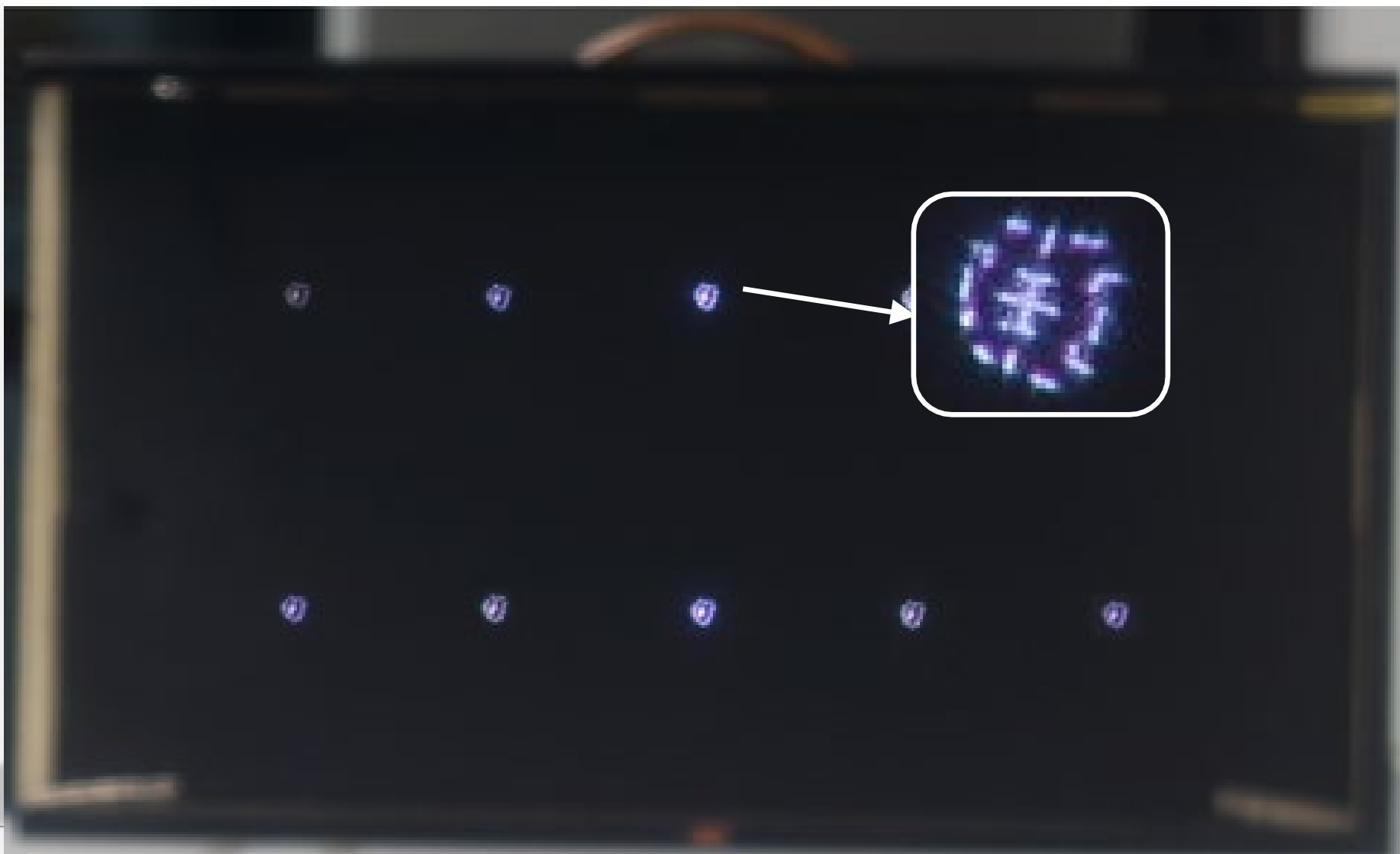
Coded aperture changes shape of kernel

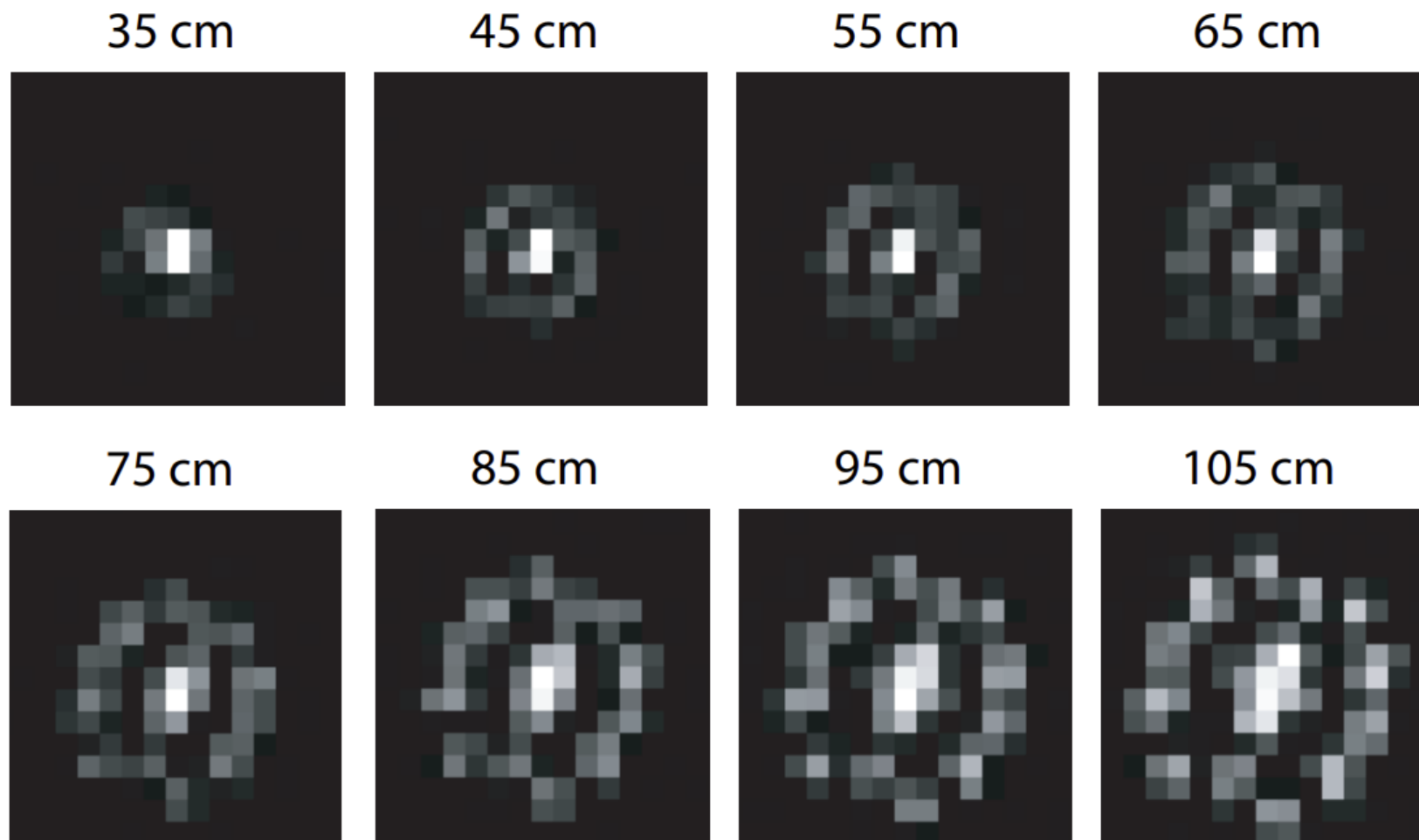


Coded aperture changes shape of kernel







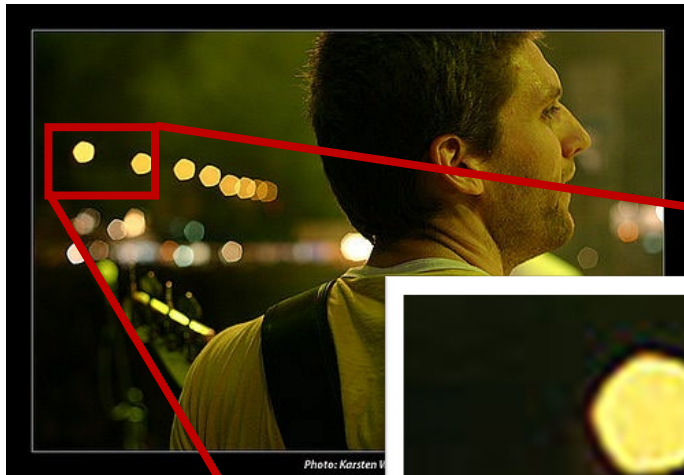


Calibrated kernels at a variety of depths from the focus plane

Image of a point light source



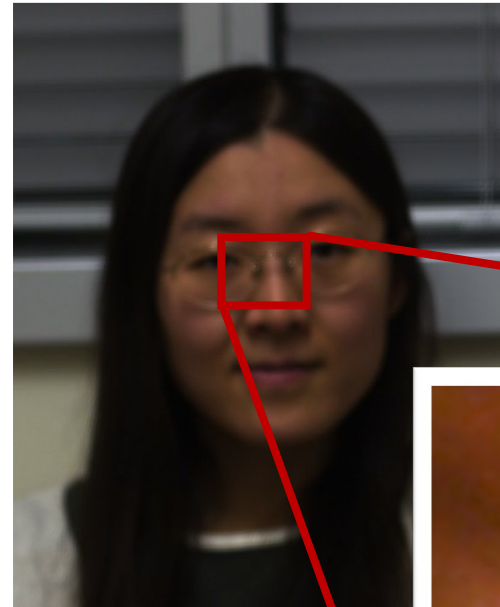
Conventional
Aperture



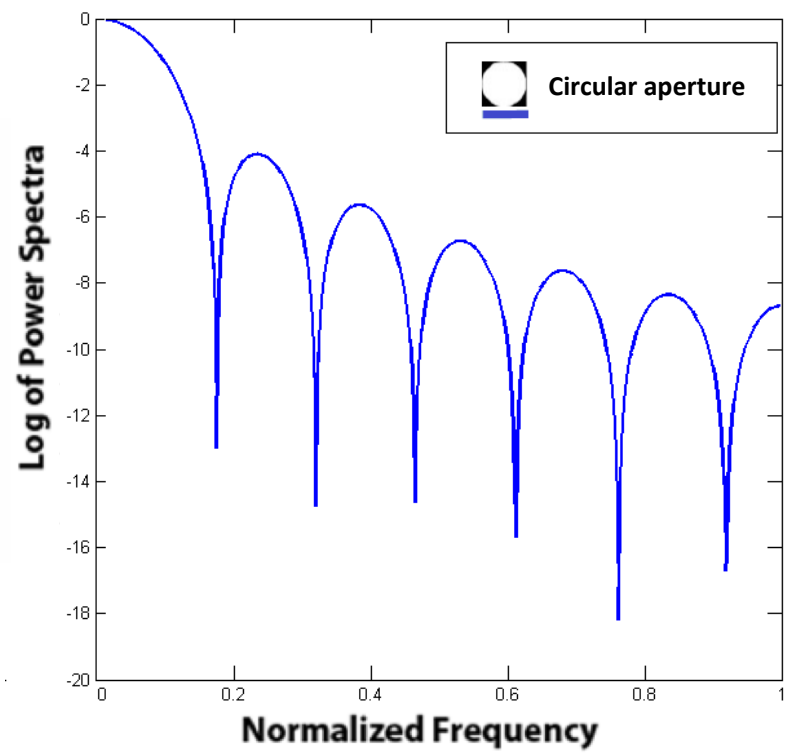
Captured Image

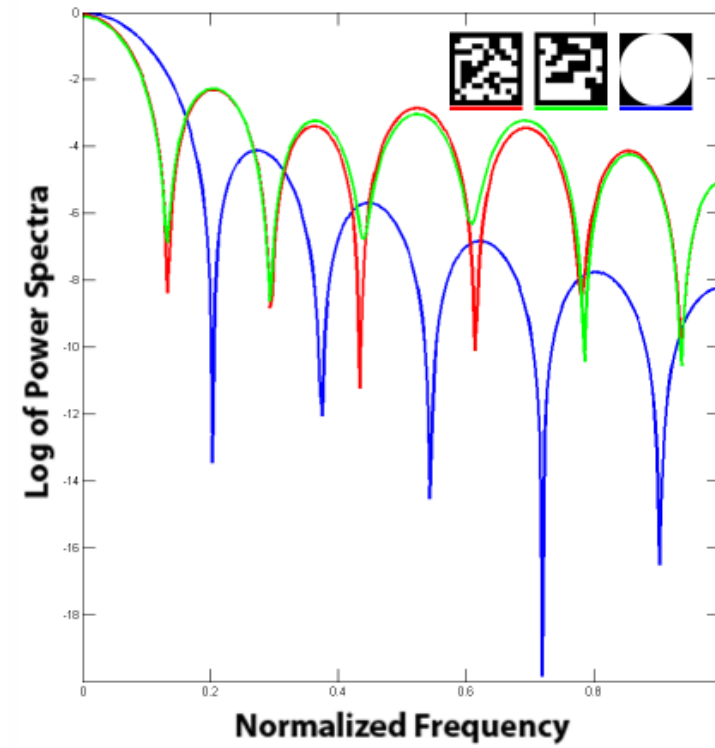
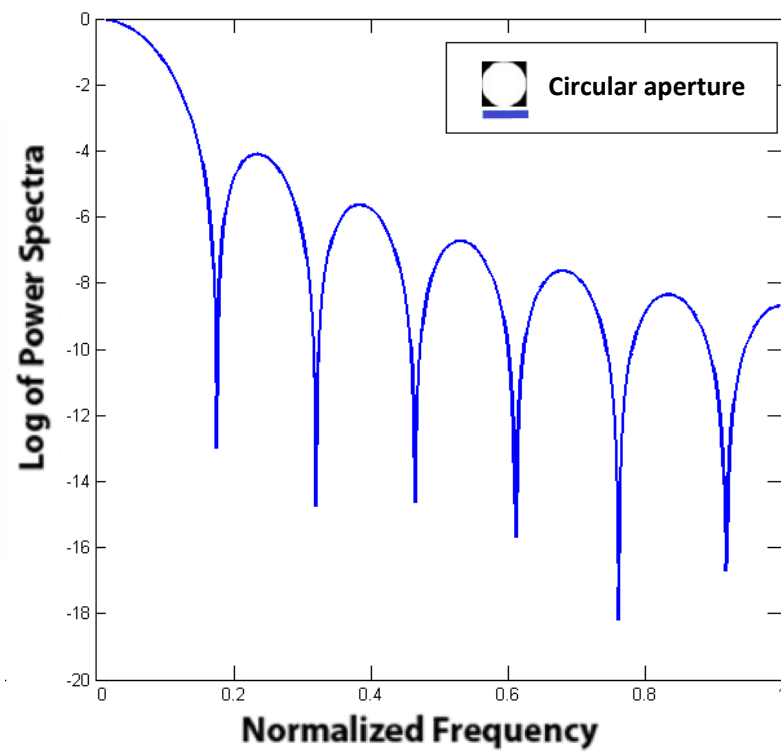


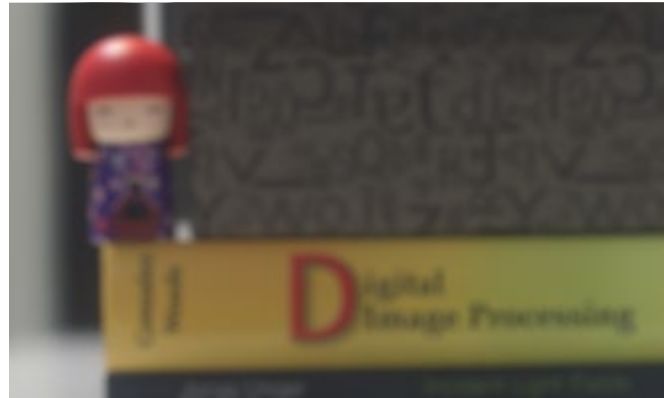
Coded
Aperture



Captured Image

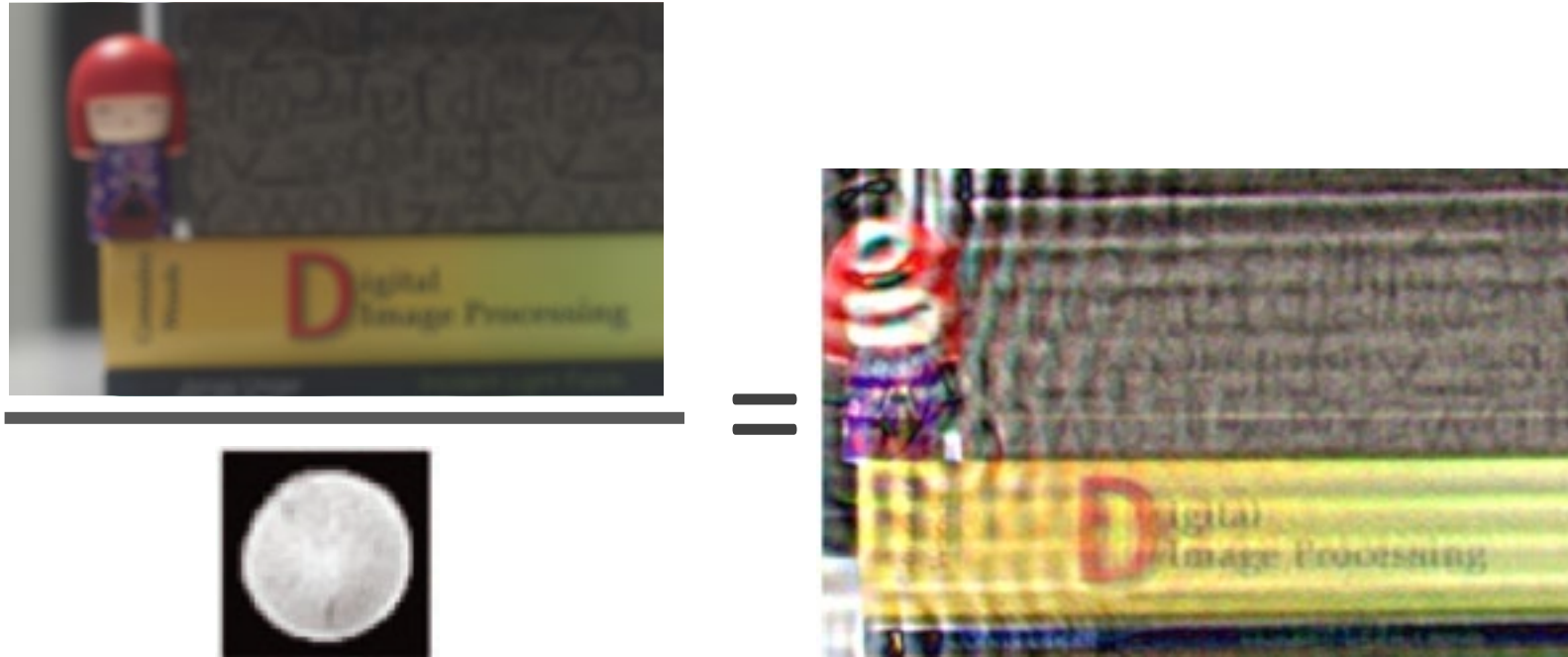




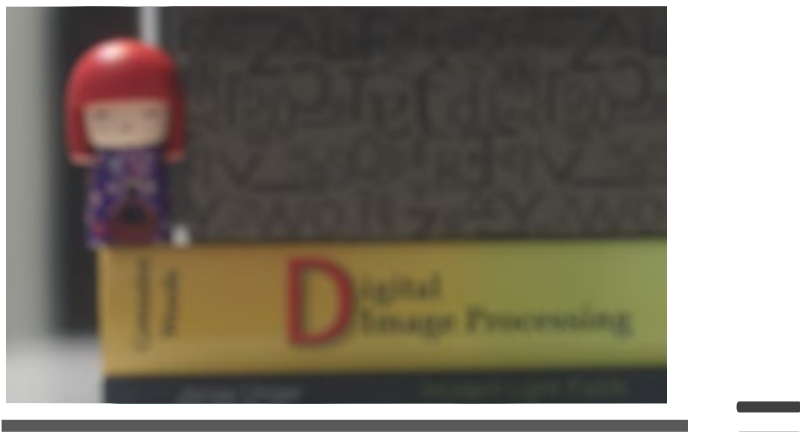


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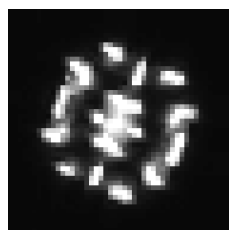




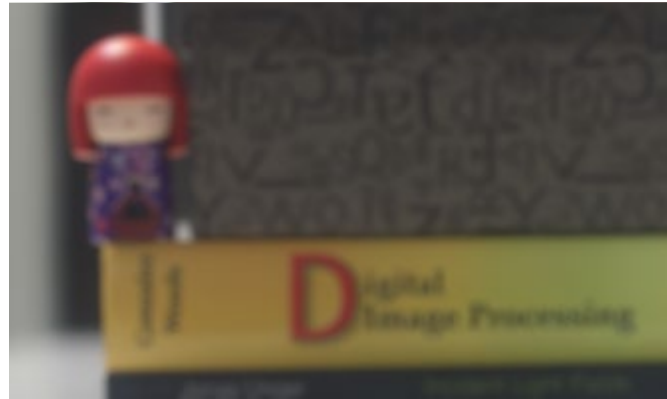
Bad code



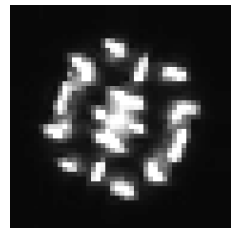
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New code!



=



New code!



Recovered (captured with a circular aperture)



Recovered (captured with a coded aperture)



Recovered (captured with a circular aperture)



Recovered (captured with a coded aperture)

How are these codes obtained?

Usually posed as an optimization problem, defining the appropriate objective function (& constraints).

Levin, A., Fergus, R., Durand, F., & Freeman, W. T. (2007).
Image and depth from a conventional camera with a coded
aperture. *ACM transactions on graphics (TOG)*, 26(3), 70-es.

Coded Photography

coded photography

The introduction of codes (or other types of generalized optics) at some point of the imaging pipeline can be used in a wealth of imaging techniques...

- Defocus deblurring
- Depth from defocus

} These two often (not always) go hand-in-hand

- Scene motion & motion deblurring

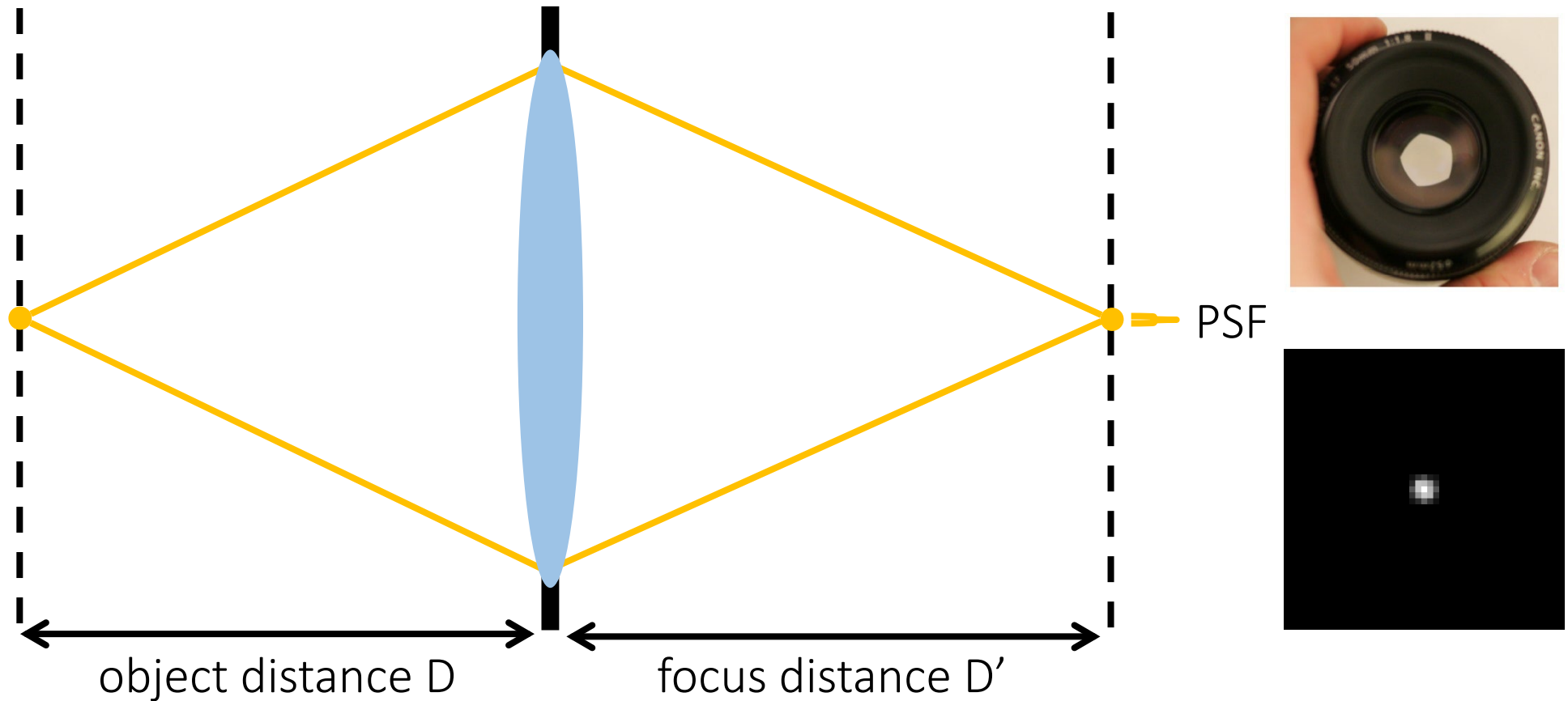
Defocus deblurring & depth from defocus

Why do they go hand-in-hand?

Defocus blur in a conventional camera is depth-dependent.

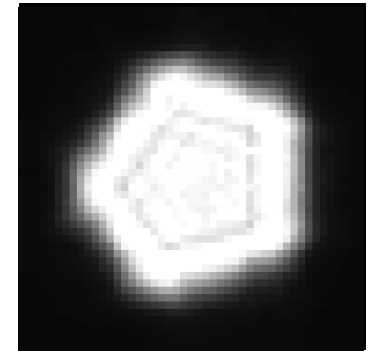
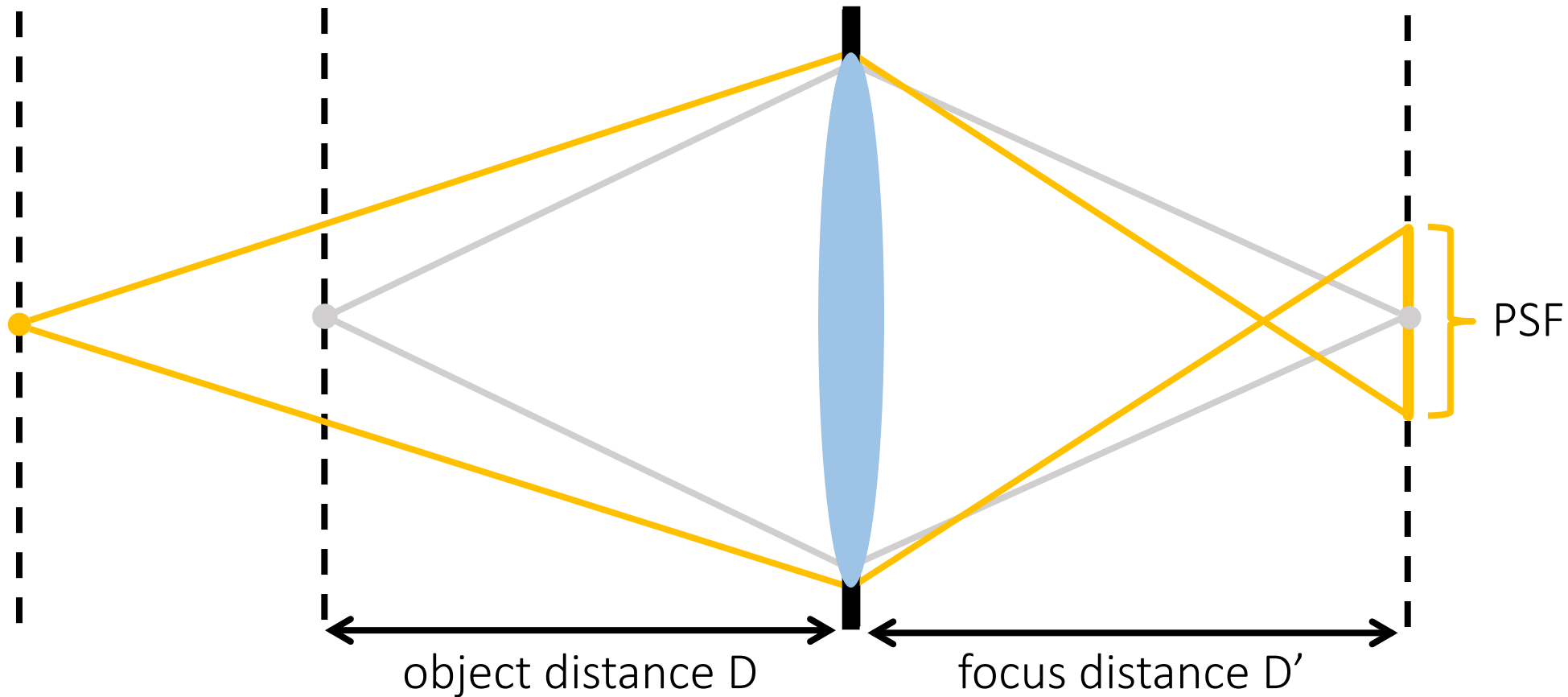
Why do they go hand-in-hand?

The aperture determines the shape of the blur kernel.



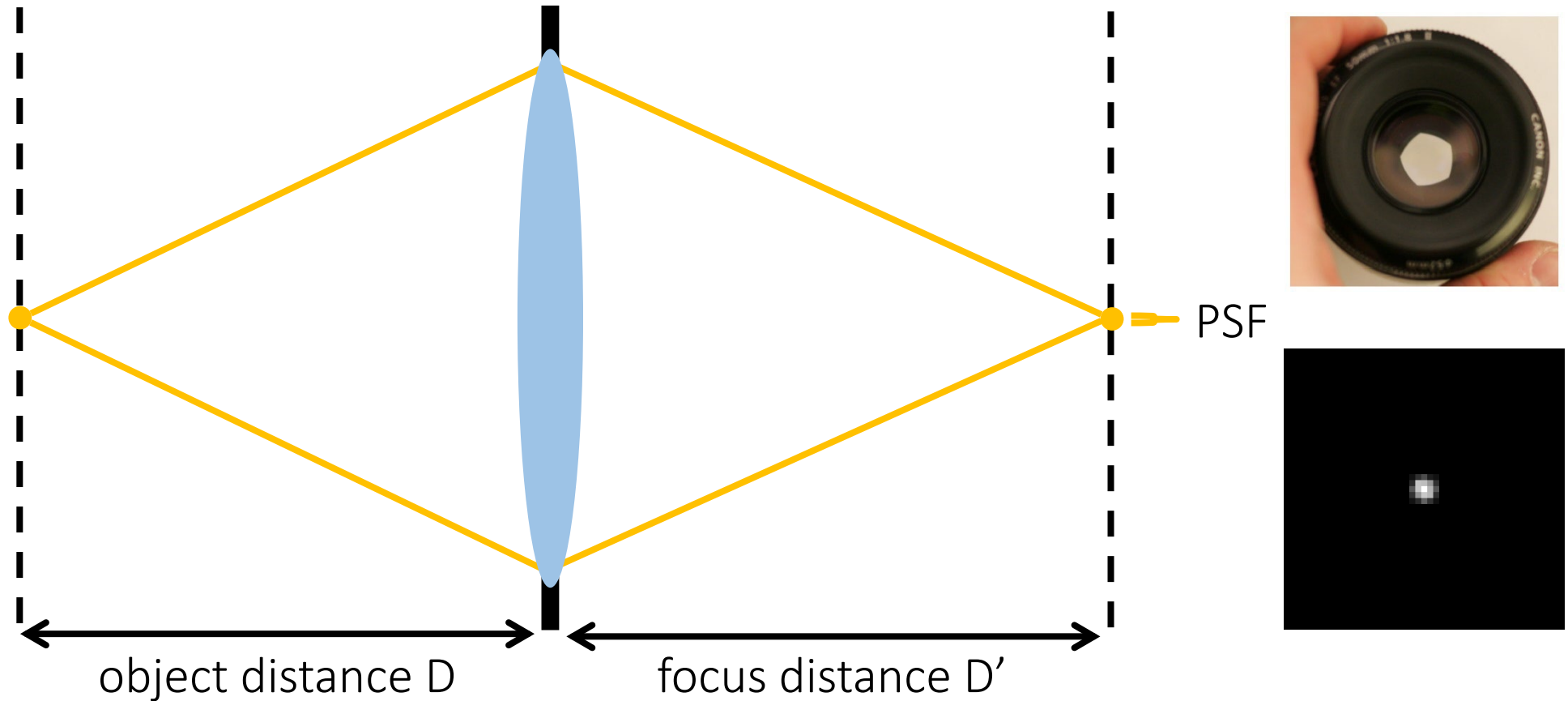
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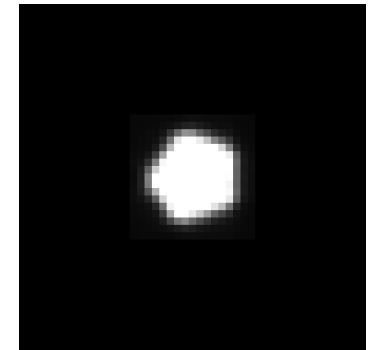
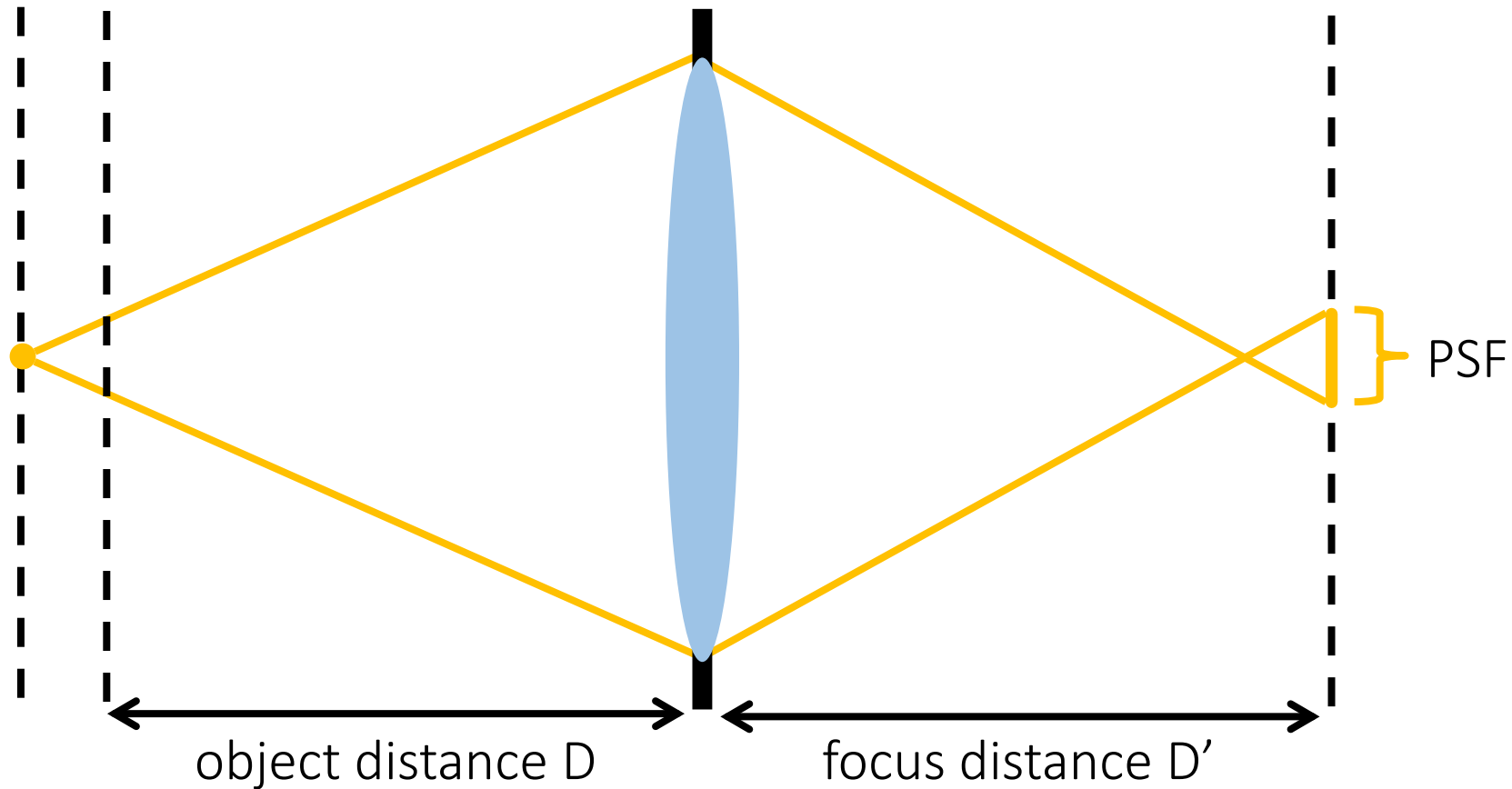
Why do they go hand-in-hand?

The depth determines the **scale** of the blur kernel.



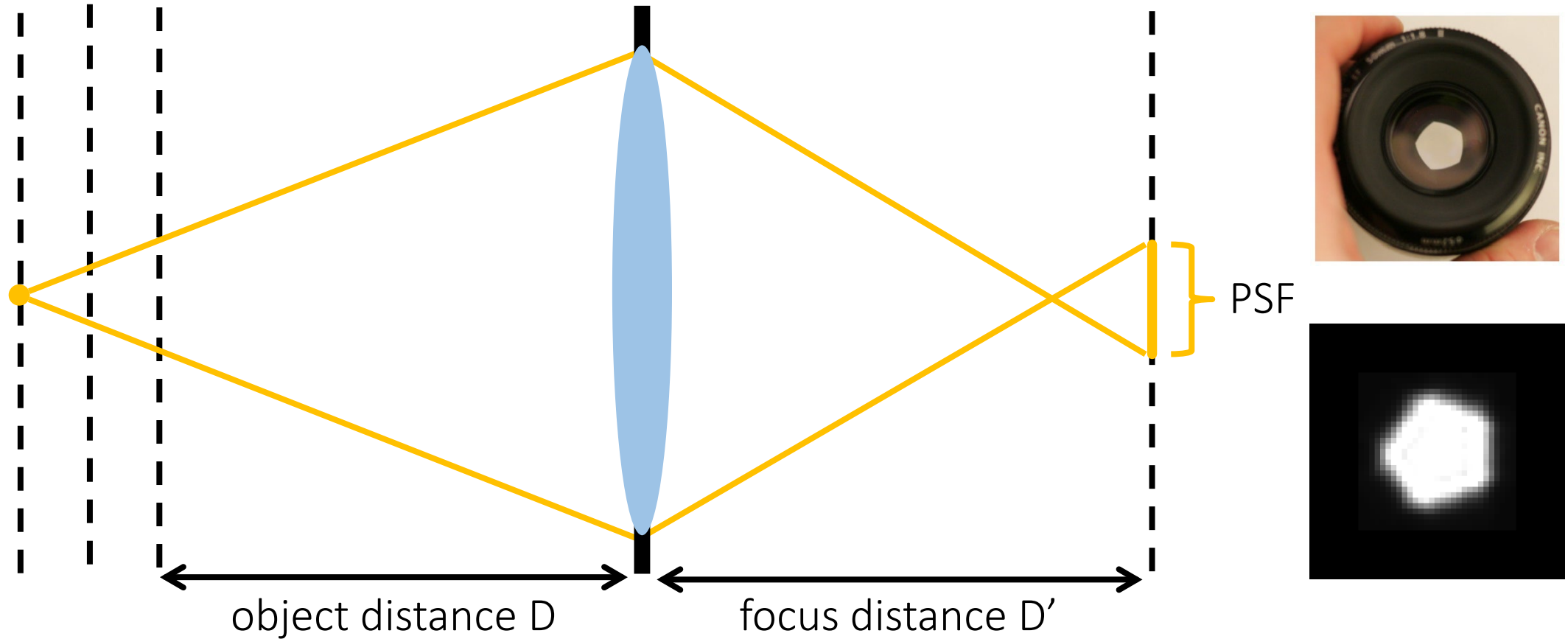
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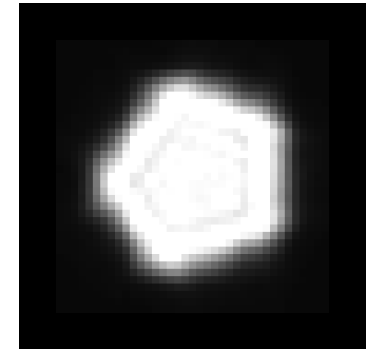
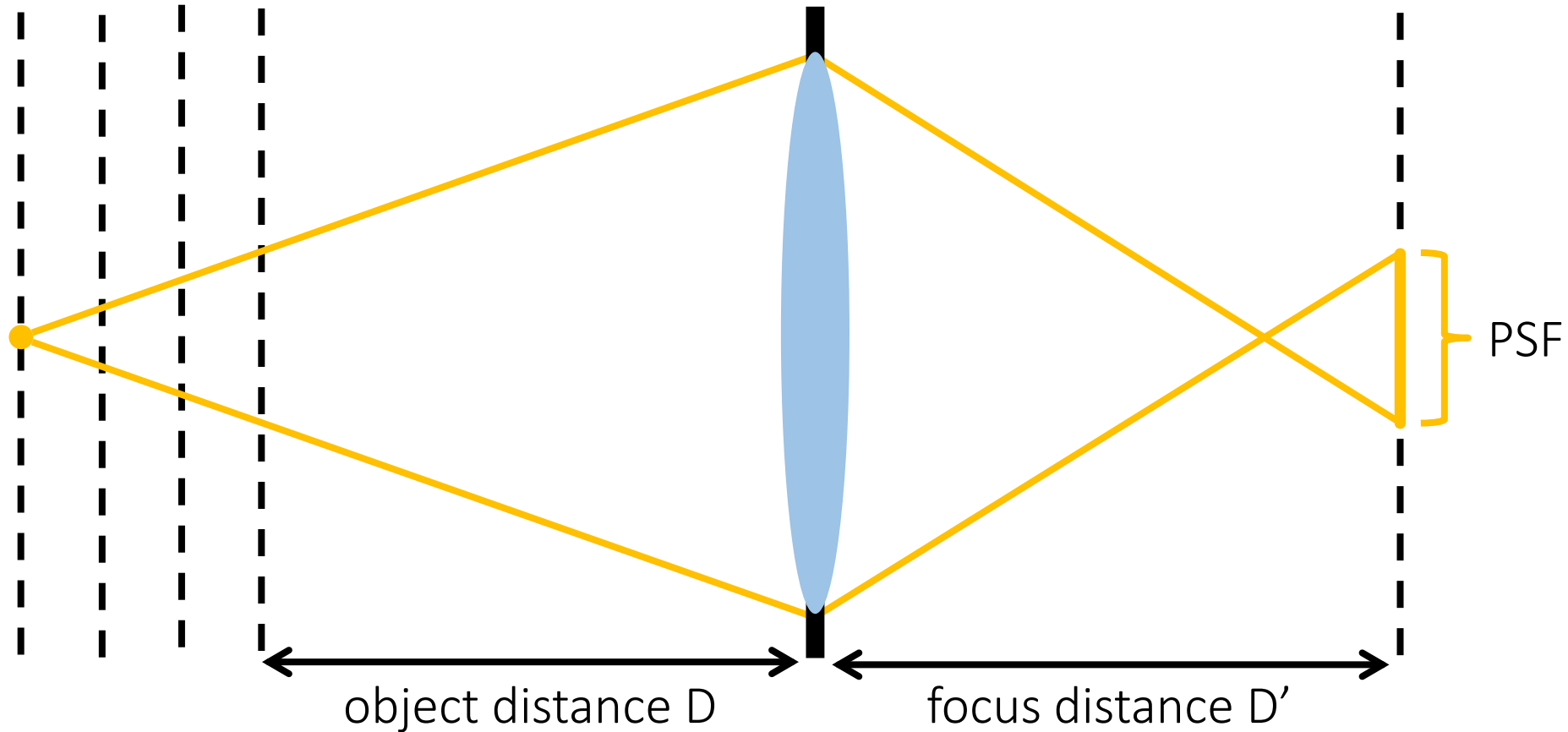
Why do they go hand-in-hand?

The depth determines the **scale** of the blur kernel.



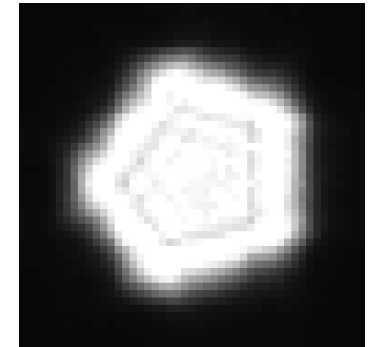
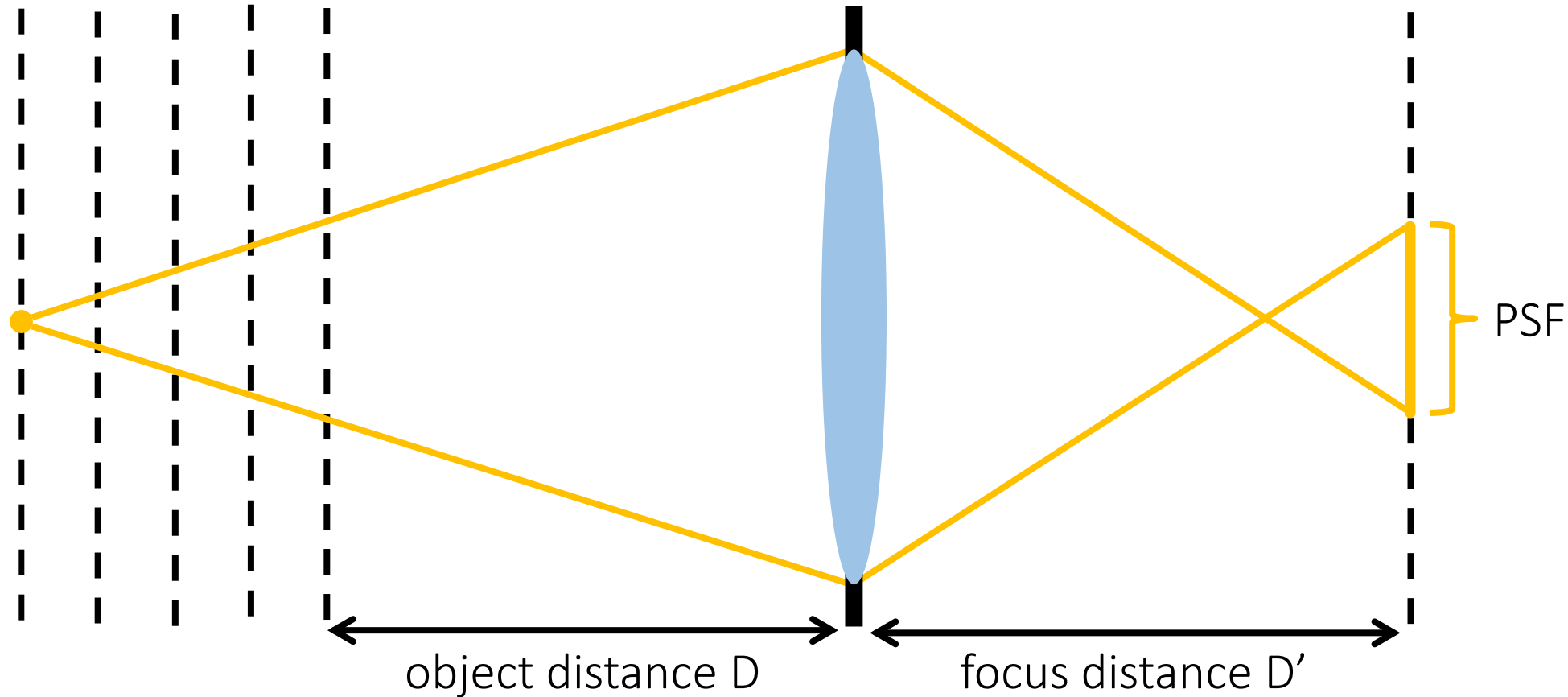
Why do they go hand-in-hand?

The depth determines the **scale** of the blur kernel.



Why do they go hand-in-hand?

The depth determines the **scale** of the blur kernel.



Why do they go hand-in-hand?

Defocus blur in a conventional camera is **depth-dependent**.

- Defocus blur = local convolution with a depth-dependent kernel.

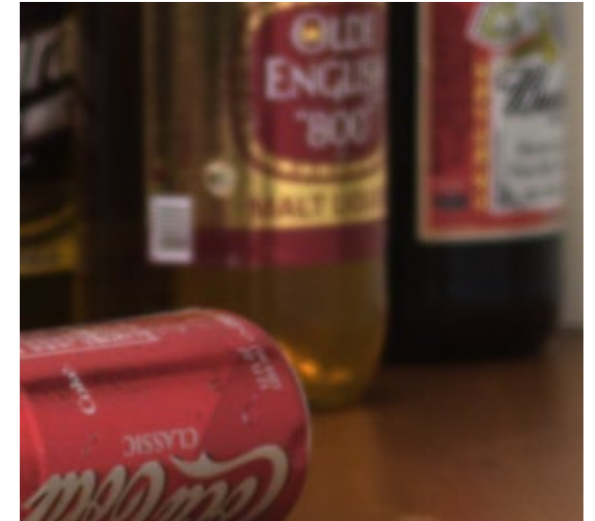
This has two consequences:

- ☹ • Defocus blur is typically spatially-varying
- 😊 • There is depth information encoded in the blur pattern

Removal of defocus blur

How do we obtain an all-in-focus image?

- Select the aperture “wisely” to prevent loss of relevant information and enable recovery of in-focus image.
- Perform deconvolution to recover the in-focus image. However:



input defocused image

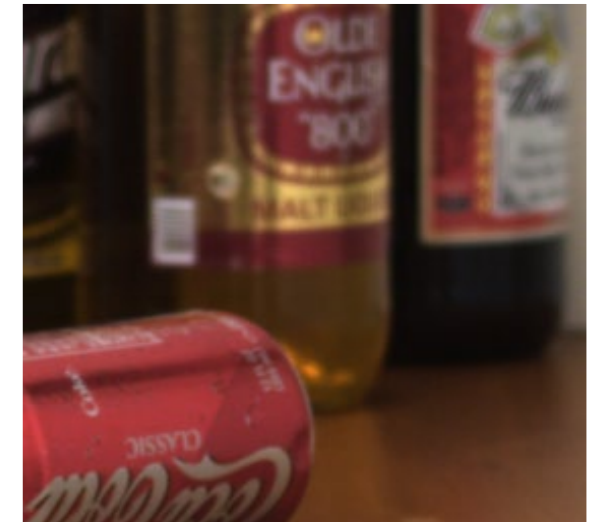
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The PSF is depth-dependent =>

- Measure PSF at different depths, yielding different kernels (that will be used for deconvolution)
- *If depth of patch is known:*
 - Deconvolve each image patch with the corresponding kernel (the PSF)
- *Else (if depth of patch is not known):*
 - Deconvolve each image patch with all kernels
 - Select the correct scale by evaluating the deconvolution results



input defocused image

Removal of defocus blur

How do we obtain an all-in-focus image?

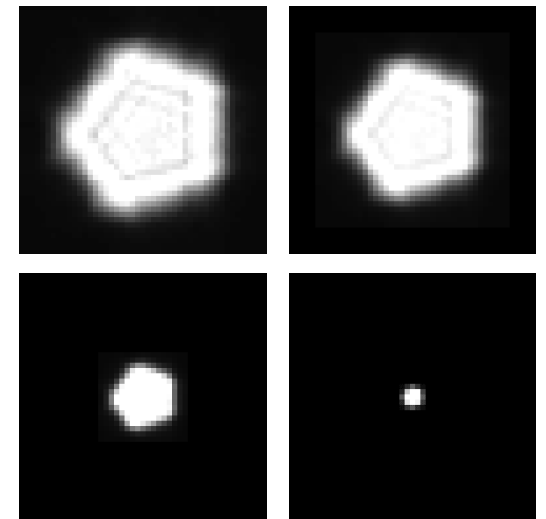
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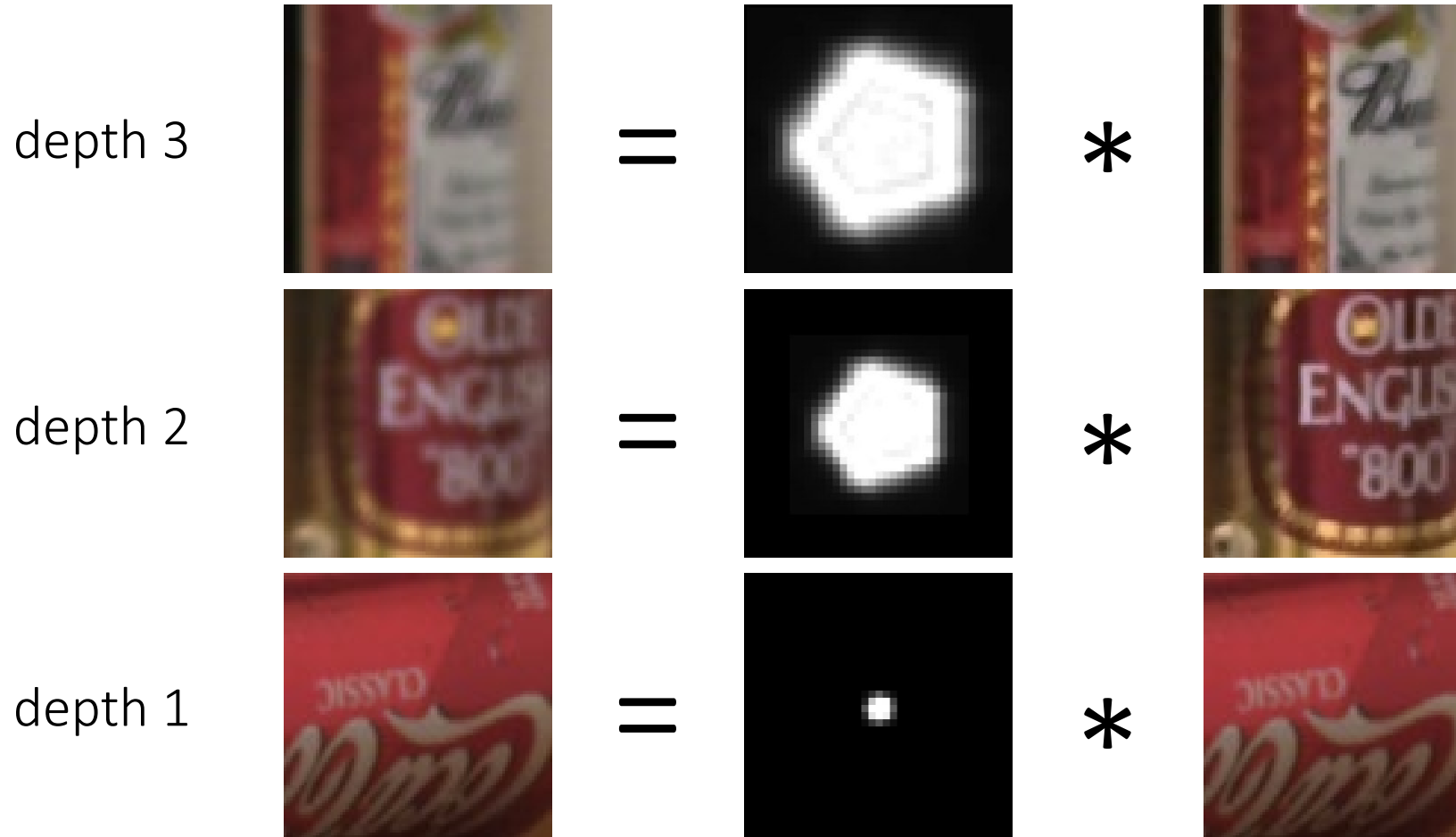
input defocused image



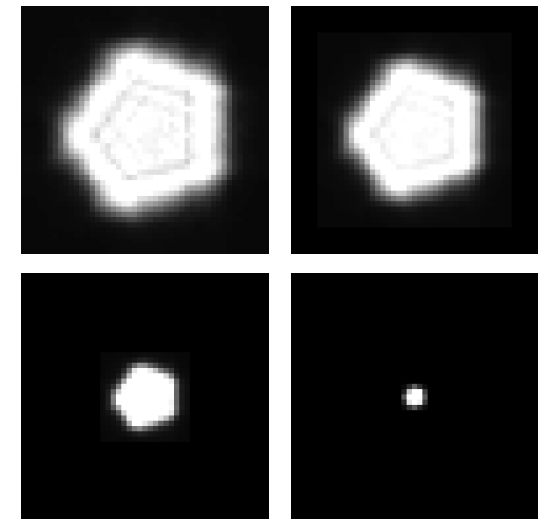
measured PSFs at
different depths

Removal of defocus blur

Objects are blurred with different PSFs depending on their depth => Defocus blur is typically spatially-varying.



input defocused image

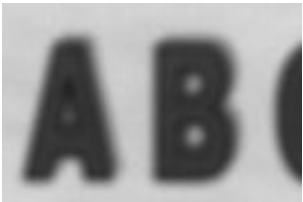
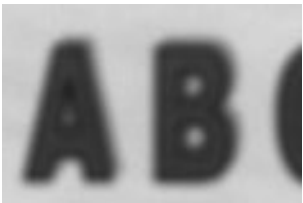
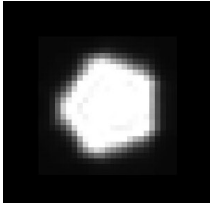
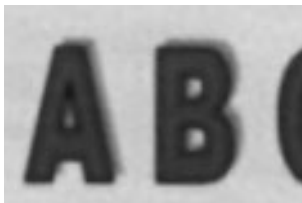
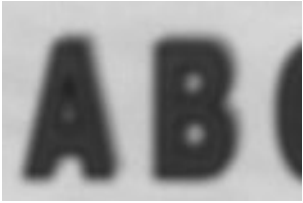
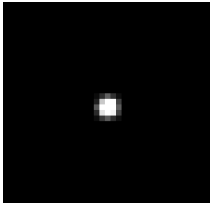


measured PSFs at different depths

Removal of defocus blur

How do we obtain an all-in-focus image?

- Deconvolve each image patch with all kernels
- Select the correct scale by evaluating the deconvolution results

	$\ast^{-1}$		$=$	
	$\ast^{-1}$		$=$	
	$\ast^{-1}$		$=$	

How do we
select the
correct scale?

Removal of defocus blur

Problem with conventional apertures:

Results at different scales can look very similar.

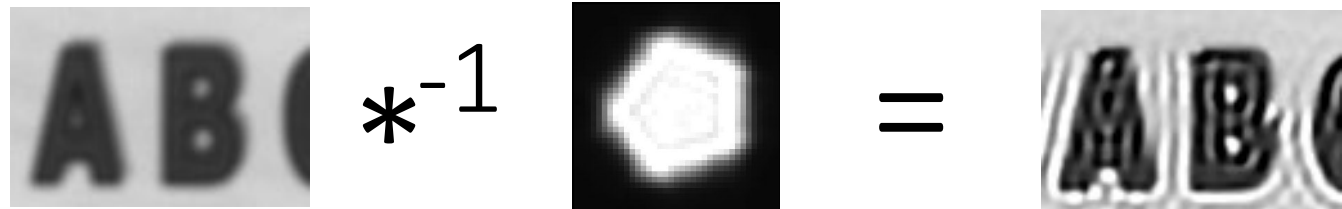


Diagram illustrating the result of defocus blur removal using a conventional aperture. The input image (blurred 'ABC') is convolved with a large, blurry aperture (represented by a large, bright, irregular shape). The resulting image shows significant ringing artifacts, indicating a wrong scale. The text "wrong scale" is written in blue, and a red 'X' is placed next to it.

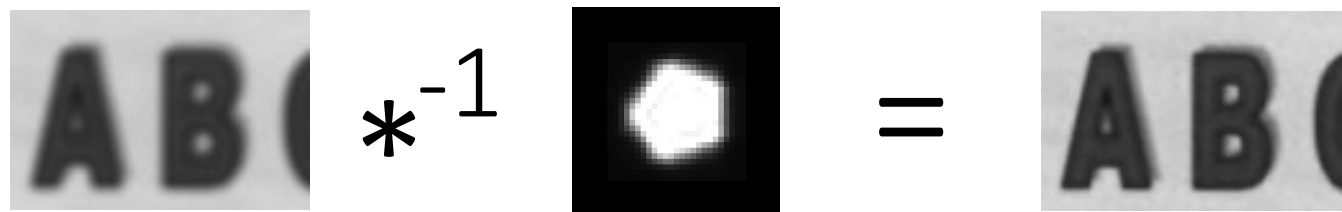


Diagram illustrating the result of defocus blur removal using a conventional aperture. The input image (blurred 'ABC') is convolved with a small, sharp aperture (represented by a small, bright, irregular shape). The resulting image shows a sharp 'ABC', indicating a correct scale. The text "correct scale" is written in blue, and a question mark is placed next to it.

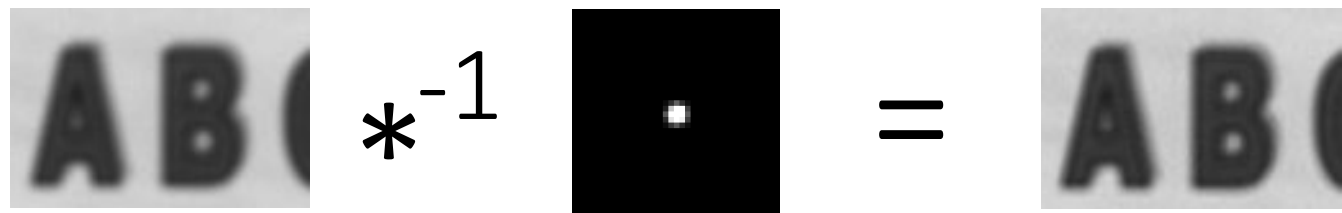
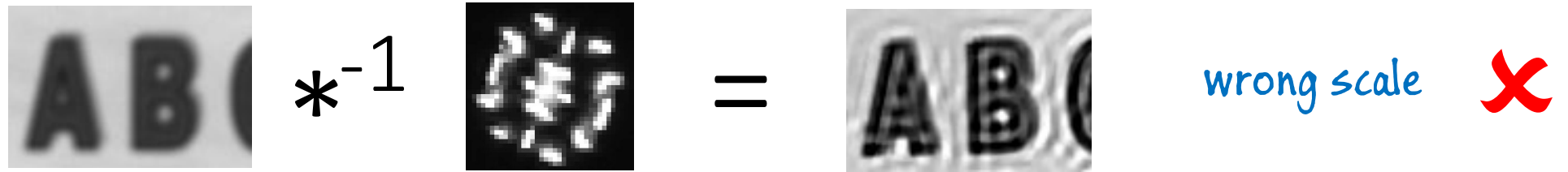


Diagram illustrating the result of defocus blur removal using a conventional aperture. The input image (blurred 'ABC') is convolved with a very small, sharp aperture (represented by a very small, bright, irregular shape). The resulting image shows a sharp 'ABC', indicating a correct scale. The text "correct scale" is written in blue, and a question mark is placed next to it.

Removal of defocus blur

Solution:

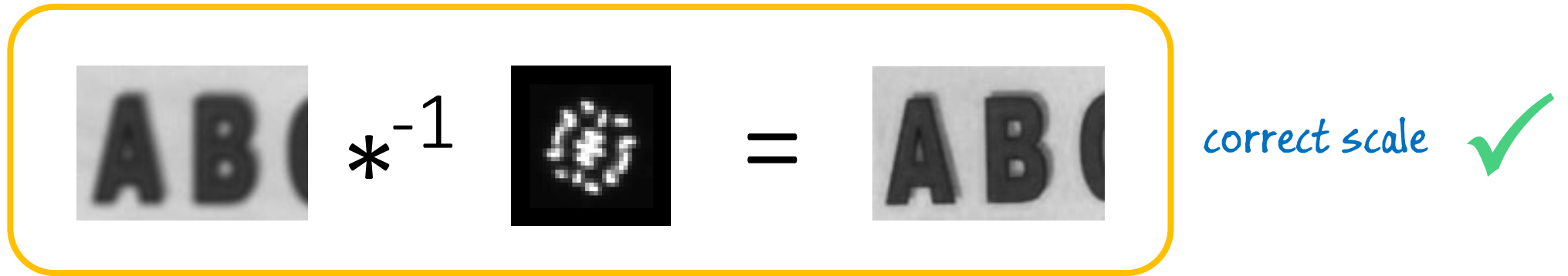
Change the aperture so that detection of the correct scale is easier.



This diagram illustrates an incorrect attempt to remove defocus blur. It shows a blurred image of the letters 'ABC' being convolved with an inverted aperture that is too large. The resulting image is still blurred, indicating that the scale of the aperture was wrong.

$$\text{Blurred ABC} *^{-1} \text{Large Aperture} = \text{Blurred ABC}$$

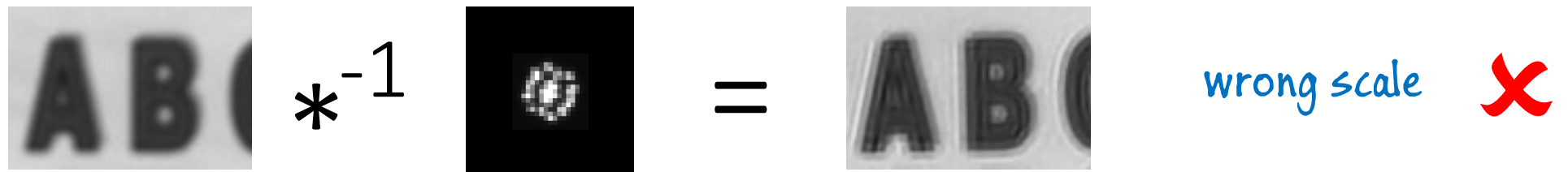
wrong scale ✗



This diagram shows the correct method for removing defocus blur. The blurred image of 'ABC' is convolved with an inverted aperture of the correct size. The resulting image is sharp, indicating that the scale was correct. This entire row is highlighted with a yellow rounded rectangle.

$$\text{Blurred ABC} *^{-1} \text{Correct Aperture} = \text{Sharp ABC}$$

correct scale ✓



This diagram illustrates another incorrect attempt to remove defocus blur. It shows a blurred image of the letters 'ABC' being convolved with an inverted aperture that is too small. The resulting image is still blurred, indicating that the scale of the aperture was wrong.

$$\text{Blurred ABC} *^{-1} \text{Small Aperture} = \text{Blurred ABC}$$

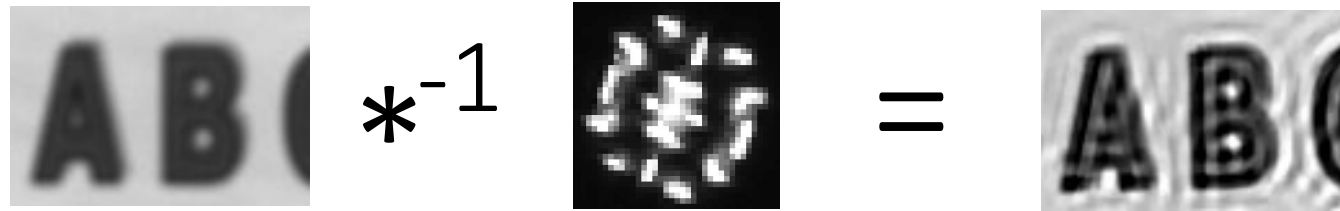
wrong scale ✗

Removal of defocus blur & depth from defocus

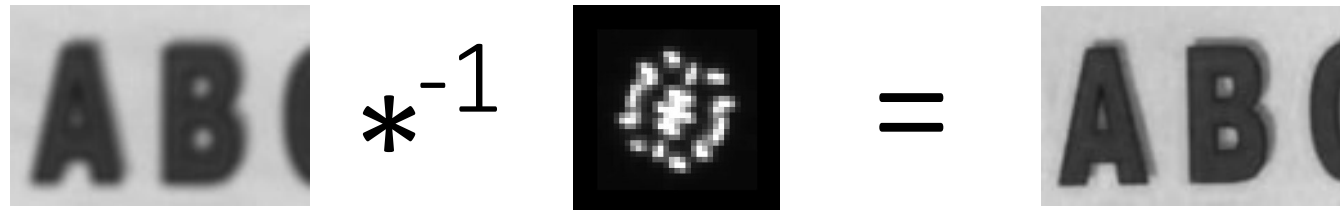
Solution:

This is another use of coded apertures. It enables obtaining depth information from a single coded image (e.g., [Levin et al. 2007]).

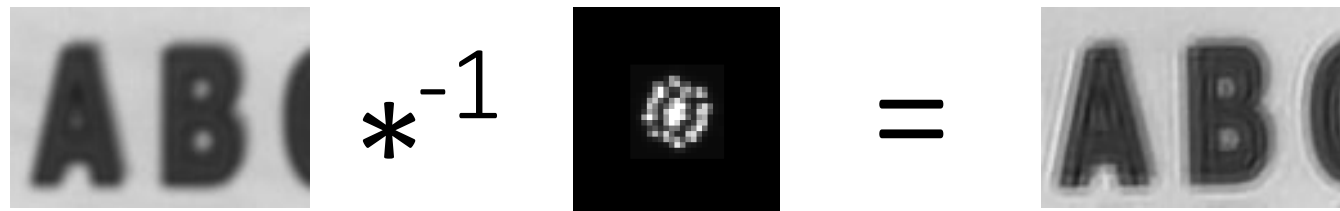
Change the aperture so that detection of the correct scale is easier.



wrong scale ✗



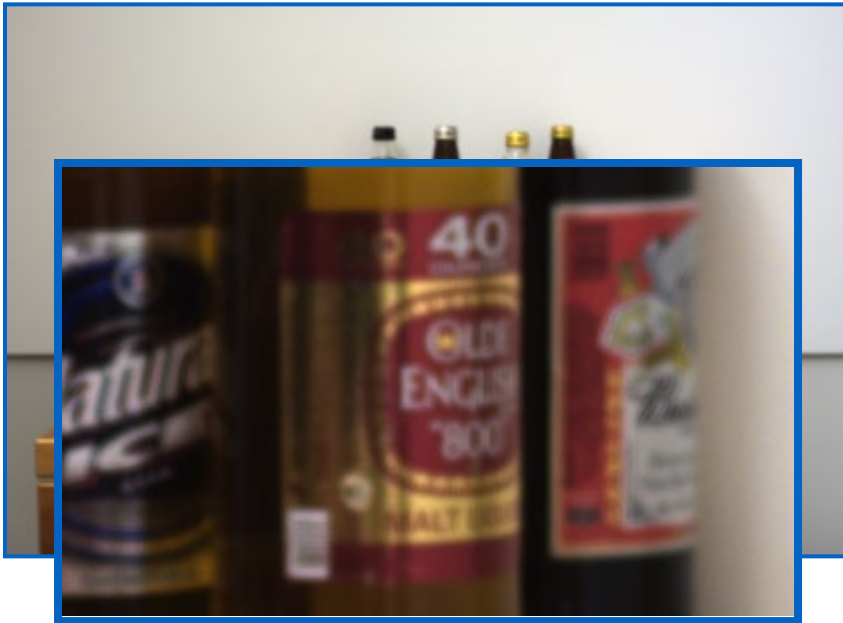
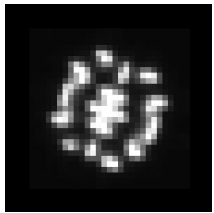
correct scale ✓



wrong scale ✗

Result

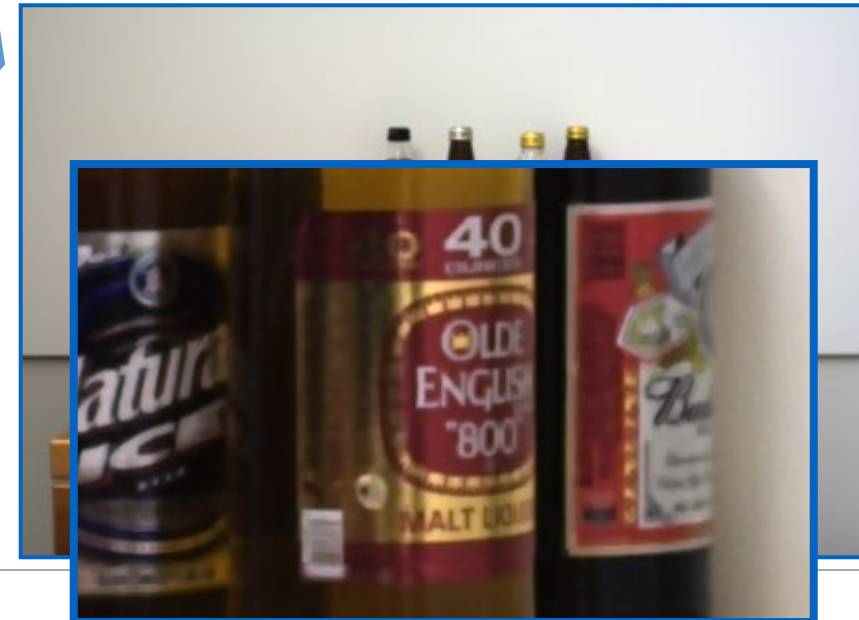
Single input image:



Output #1: Depth map



Output #2: All-focused image



Depth estimation



Refocusing



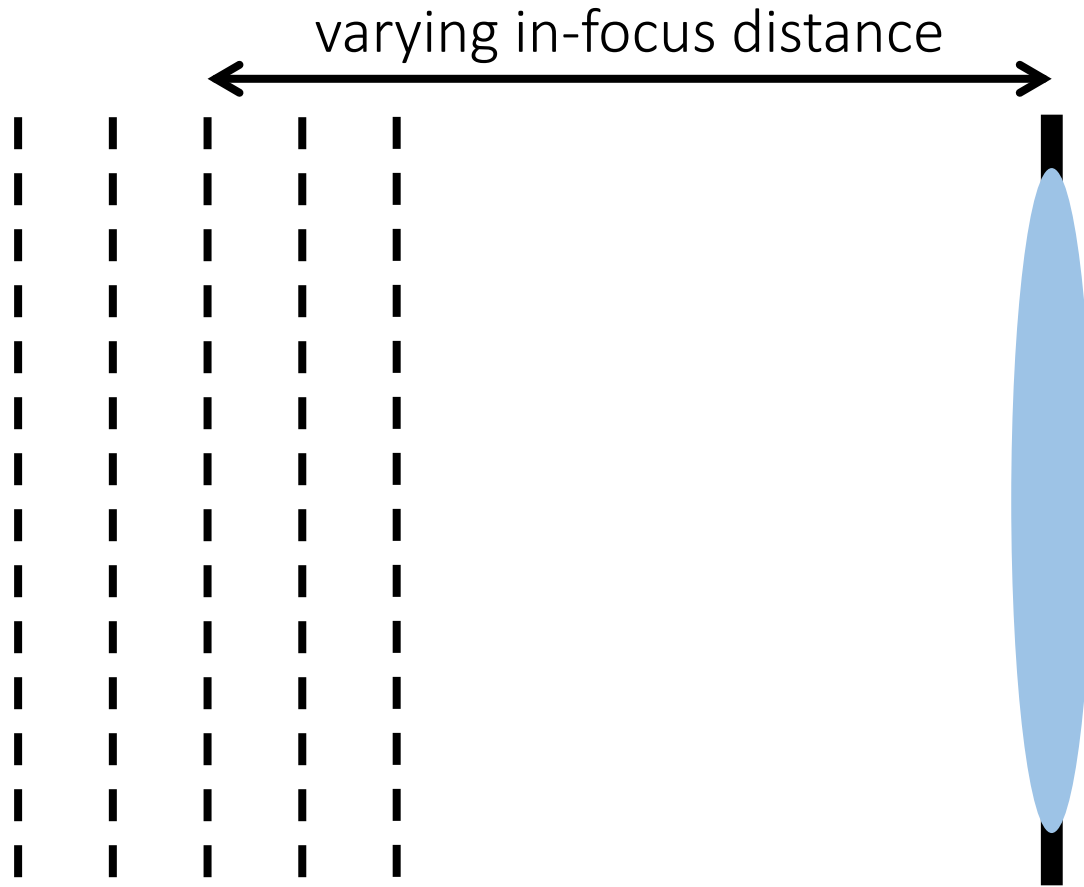
Refocusing



Refocusing



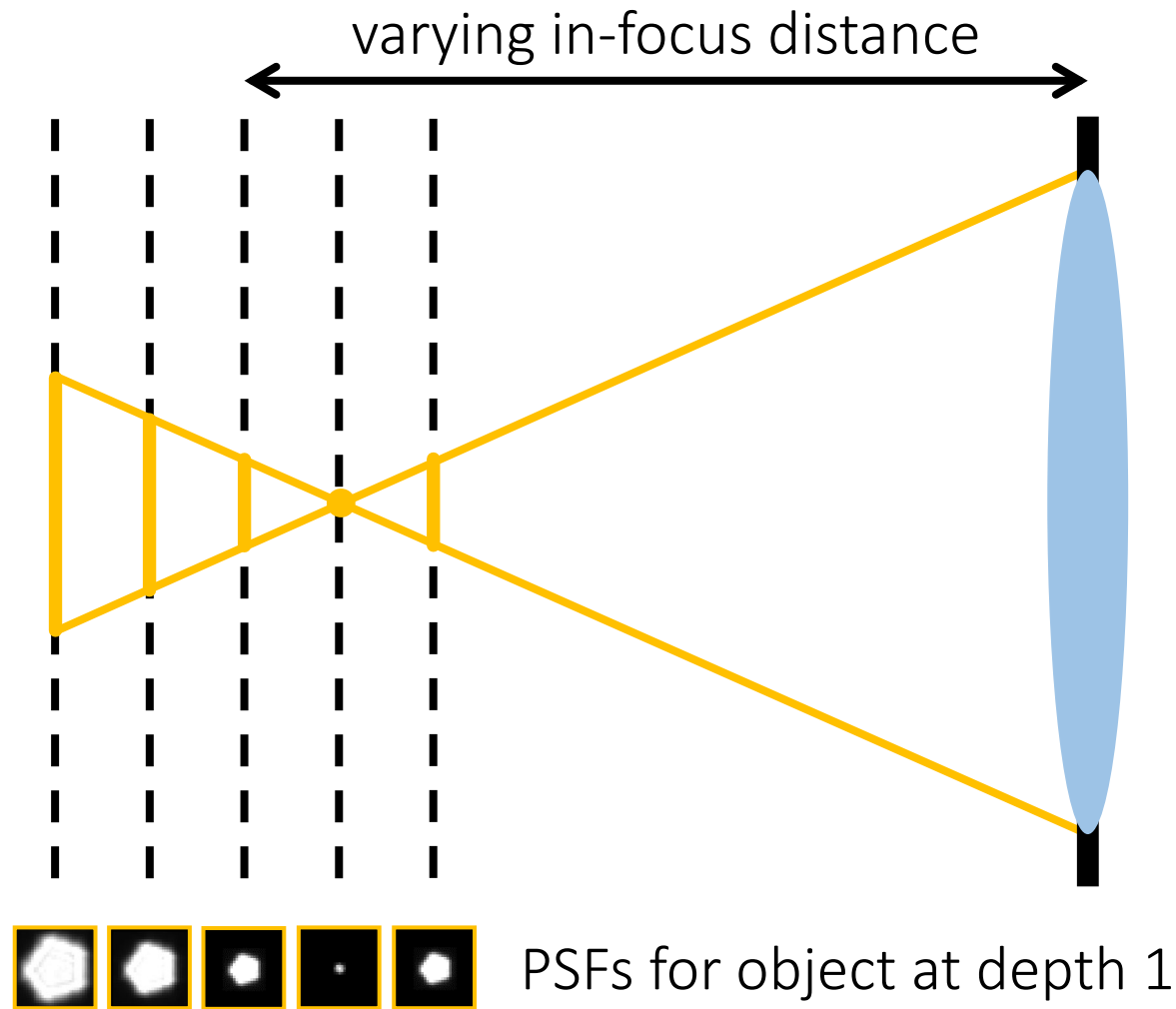
An alternative to coded aps. for defocus deblurring



It is difficult to deal with different PSFs at different depths...

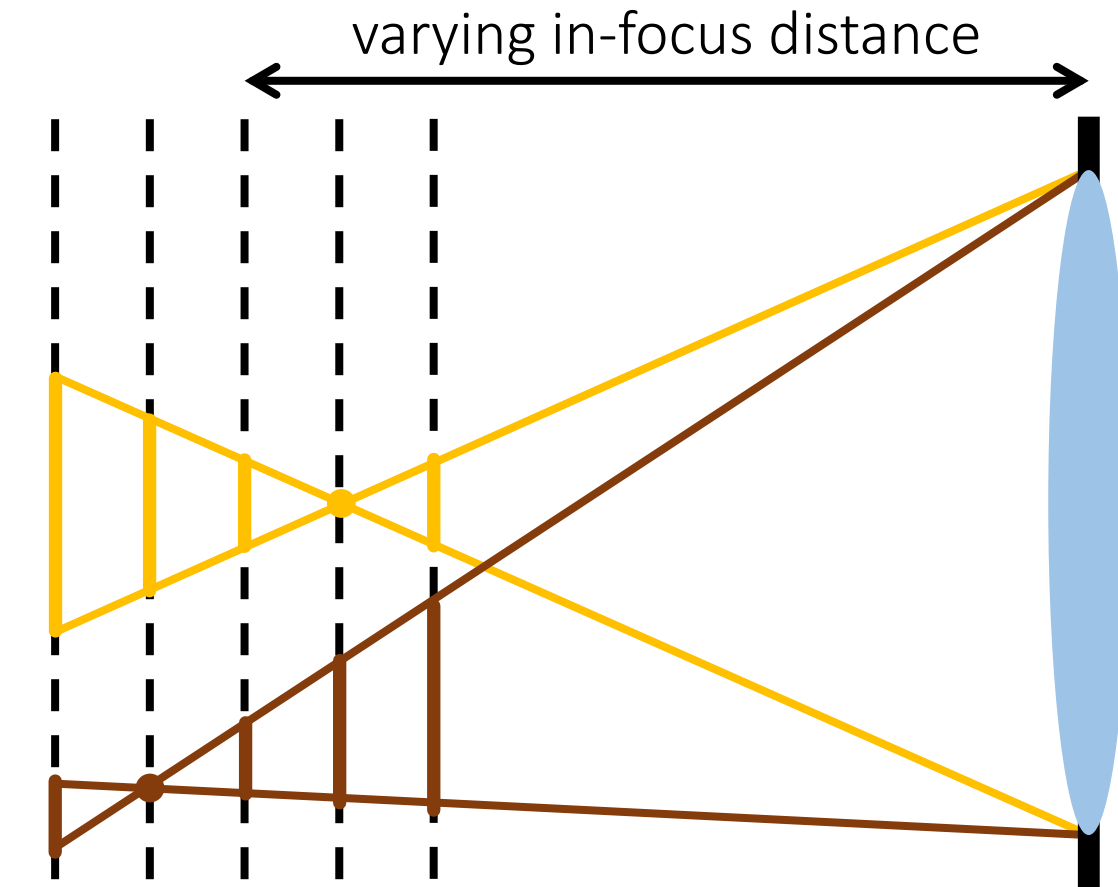
At every focus setting, objects at different depths are blurred by different PSFs

An alternative to coded aps. for defocus deblurring



At every focus setting, objects at different depths are blurred by different PSFs

An alternative to coded aps. for defocus deblurring



At every focus setting, objects at different depths are blurred by different PSFs

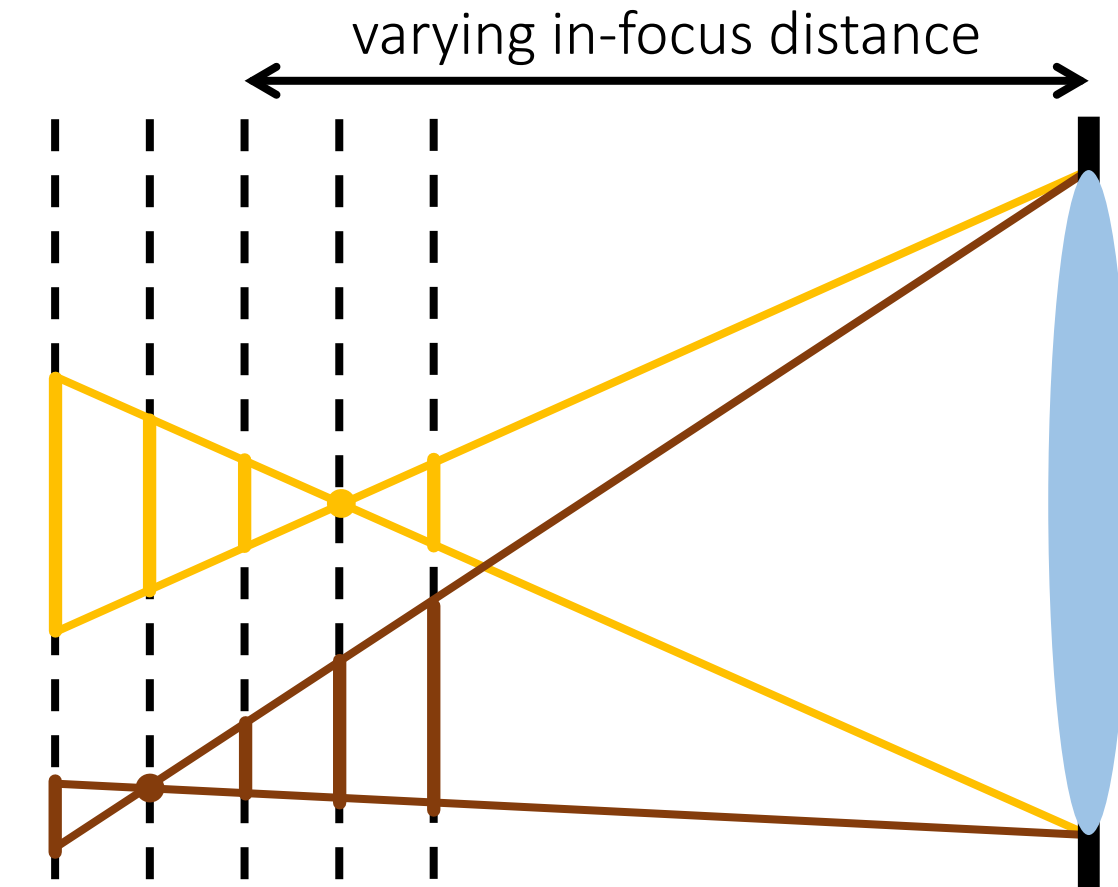


PSFs for object at depth 1



PSFs for object at depth 2

An alternative to coded aps. for defocus deblurring



At every focus setting, objects at different depths are blurred by different PSFs



PSFs for object at depth 1

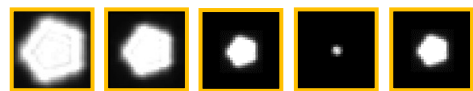
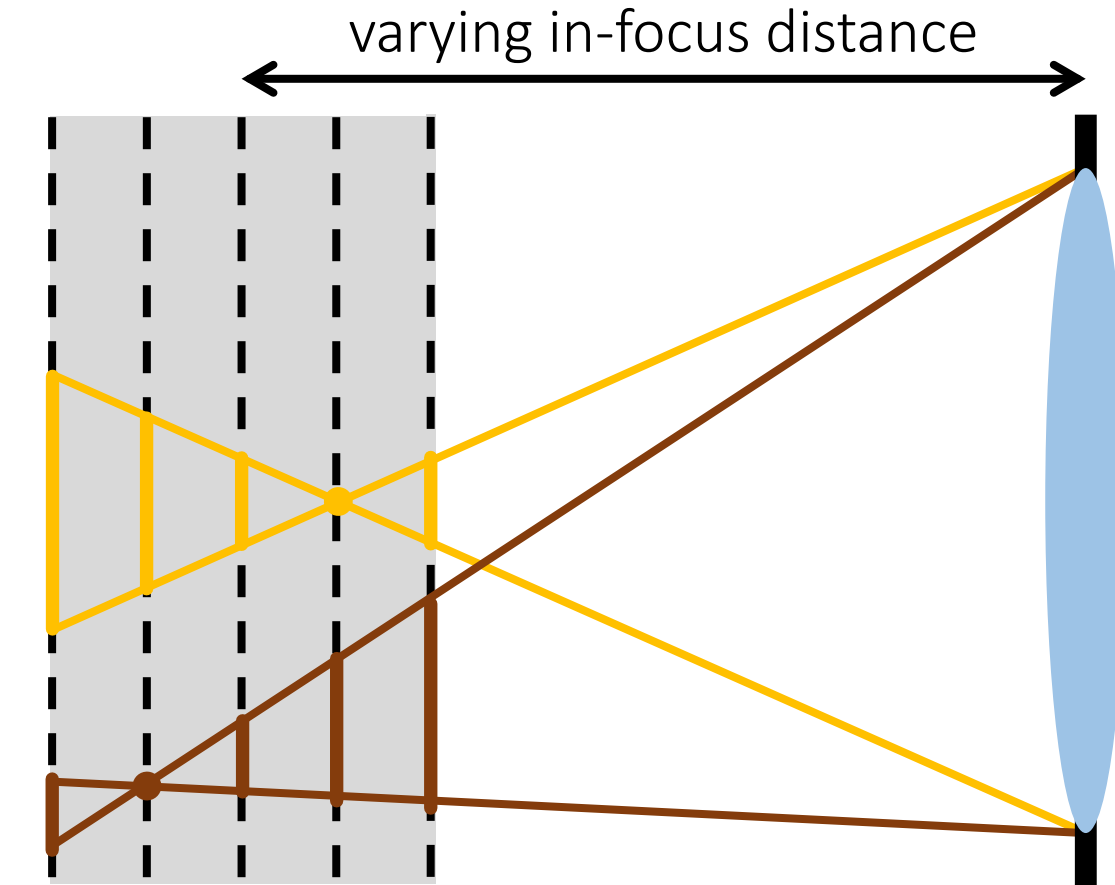


PSFs for object at depth 2

As we sweep through focus settings, each point is blurred by all possible PSFs.

Sources: [CMU 15-463](#)

An alternative for defocus deblurring: **Focal sweep**

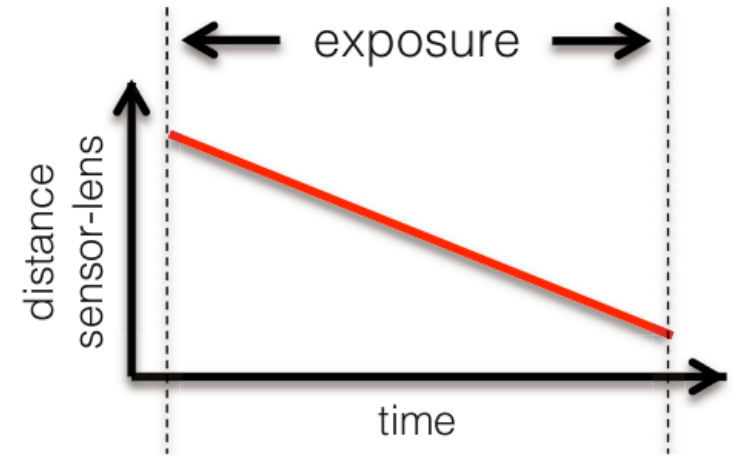


PSFs for object at depth 1



PSFs for object at depth 2

linear motion:

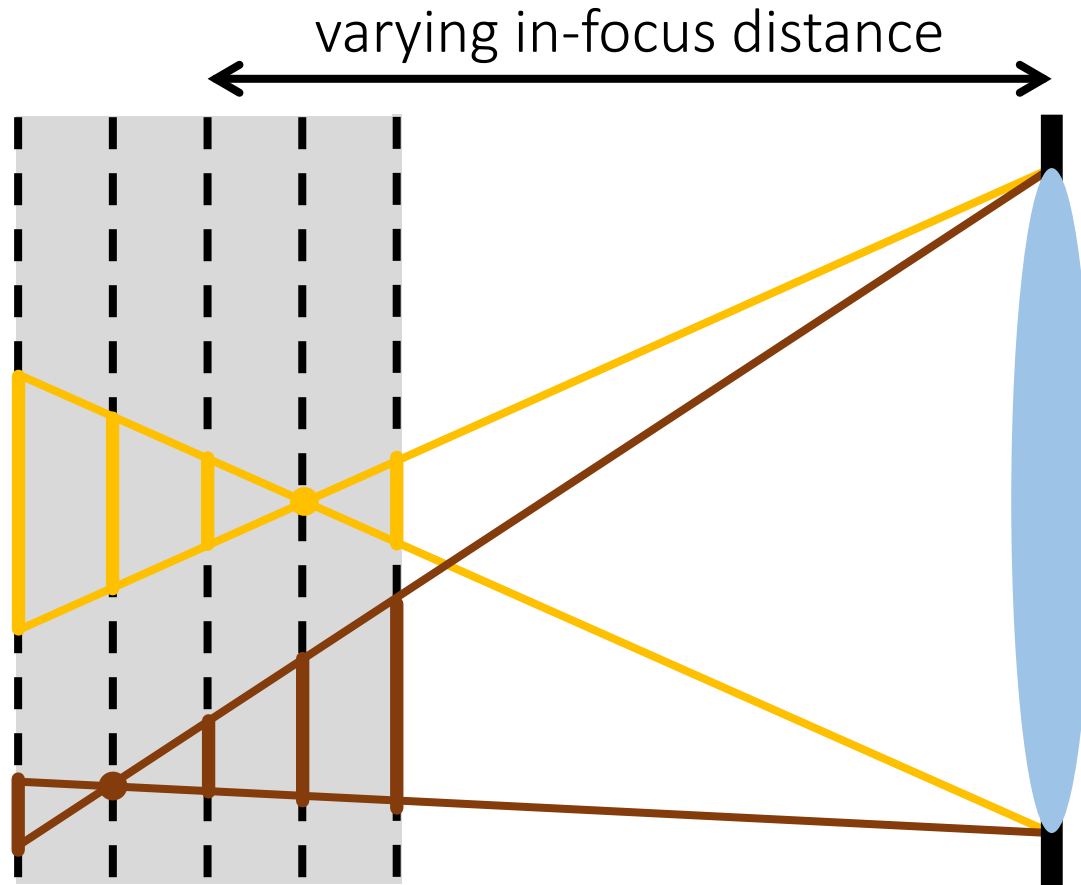


Idea: Go through all focus settings during a single exposure.

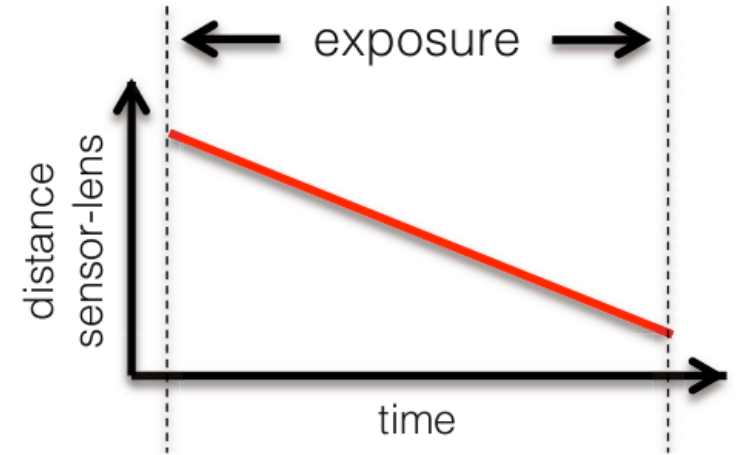
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Sources: [CMU 15-463](#)

An alternative for defocus deblurring: Focal sweep



linear motion:

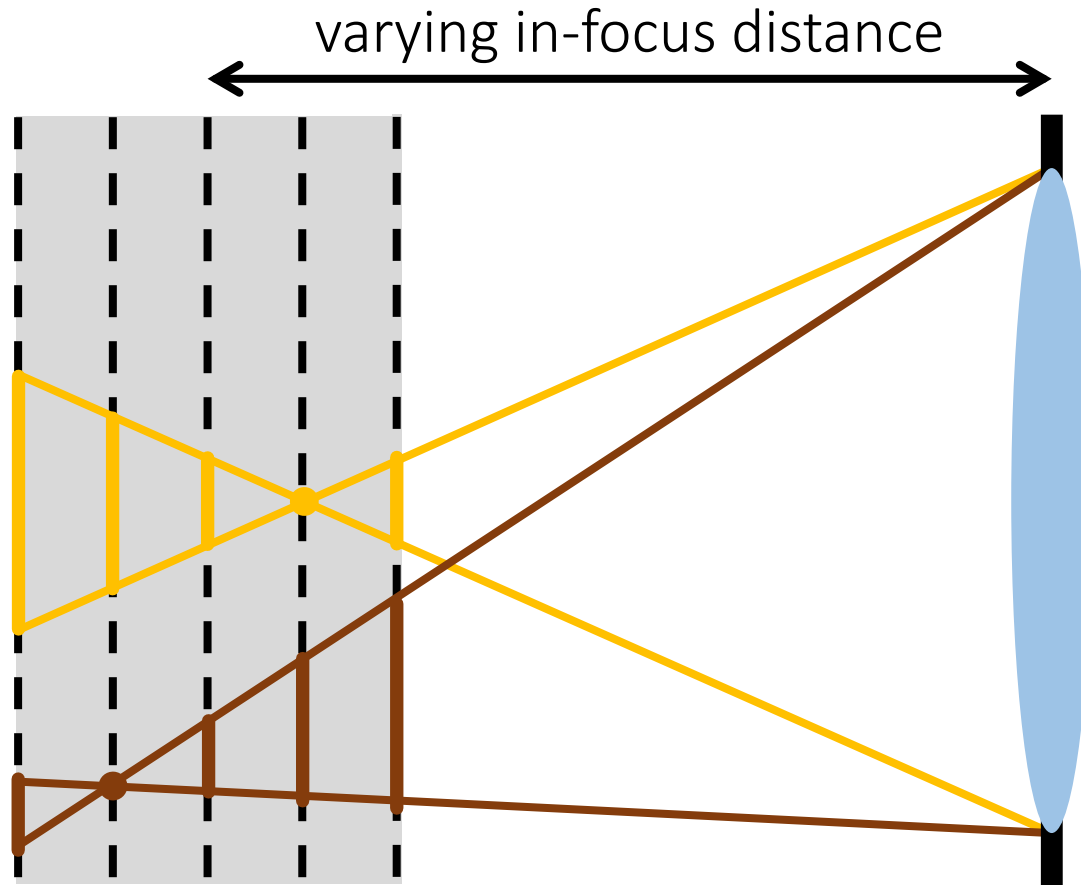


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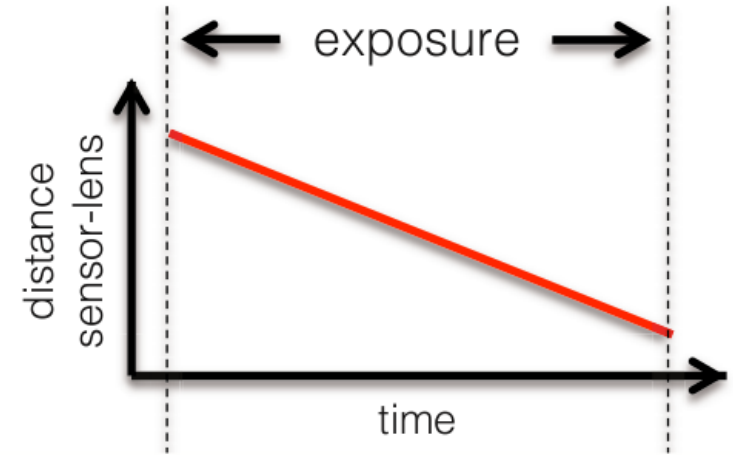
$$\int \text{[PSF sequence]} dt = \text{[Effective PSF]} \quad \text{effective PSF for object at depth 1}$$

$$\int \text{[PSF sequence]} dt = \text{[Effective PSF]} \quad \text{effective PSF for object at depth 2}$$

An alternative for defocus deblurring: Focal sweep



linear motion:



Idea: Go through all focus settings during a single exposure.

$$\int \begin{matrix} \text{[blurred PSF]} & \text{[blurred PSF]} & \text{[blurred PSF]} & \text{[sharp PSF]} & \text{[blurred PSF]} \end{matrix} dt = \begin{matrix} \text{[blurred PSF]} & \text{[blurred PSF]} & \text{[blurred PSF]} & \text{[sharp PSF]} & \text{[blurred PSF]} \end{matrix} \text{ effective PSF for object at depth 1}$$

$$\int \begin{matrix} \text{[sharp PSF]} & \text{[sharp PSF]} & \text{[blurred PSF]} & \text{[blurred PSF]} & \text{[blurred PSF]} \end{matrix} dt = \begin{matrix} \text{[blurred PSF]} & \text{[blurred PSF]} & \text{[blurred PSF]} & \text{[sharp PSF]} & \text{[blurred PSF]} \end{matrix} \text{ effective PSF for object at depth 2}$$

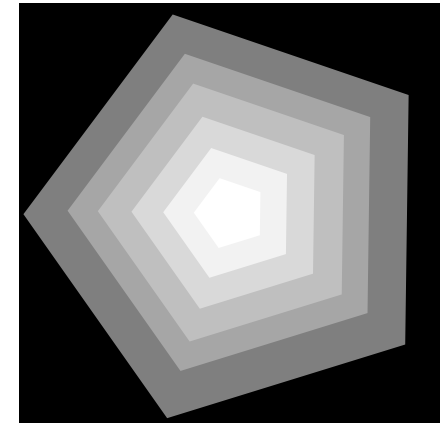
Let's have a look at these PSFs...

Focal sweep

The effective PSF is:

- Depth-invariant – All points are blurred the same way regardless of depth.
- Never sharp – All points will be blurry regardless of depth.

What are the implications of this?



Zhou, C., Miao, D., & Nayar, S. K. (2012). Focal sweep camera for space-time refocusing.

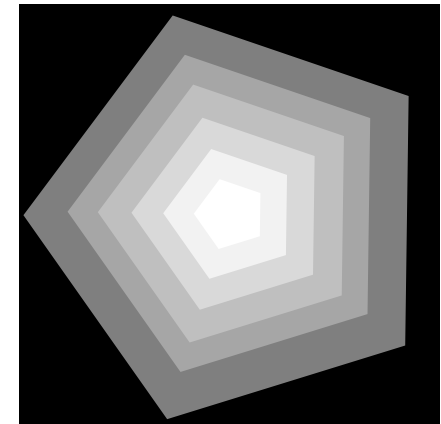
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- The captured image will not be sharp anywhere; but
- we can use simple (global) deconvolution for deblurring.
- Depth-invariance of the PSF means that we have lost all depth information.



Zhou, C., Miao, D., & Nayar, S. K. (2012). Focal sweep camera for space-time refocusing.

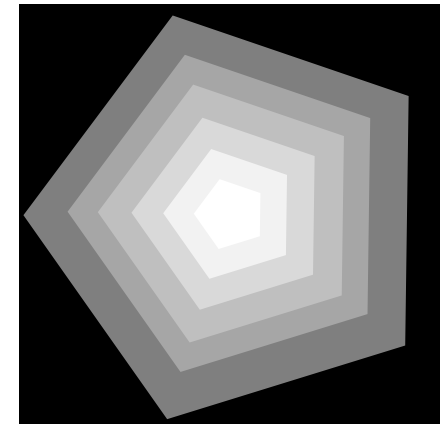
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Useful for: Extended Depth Of Field (EDOF)

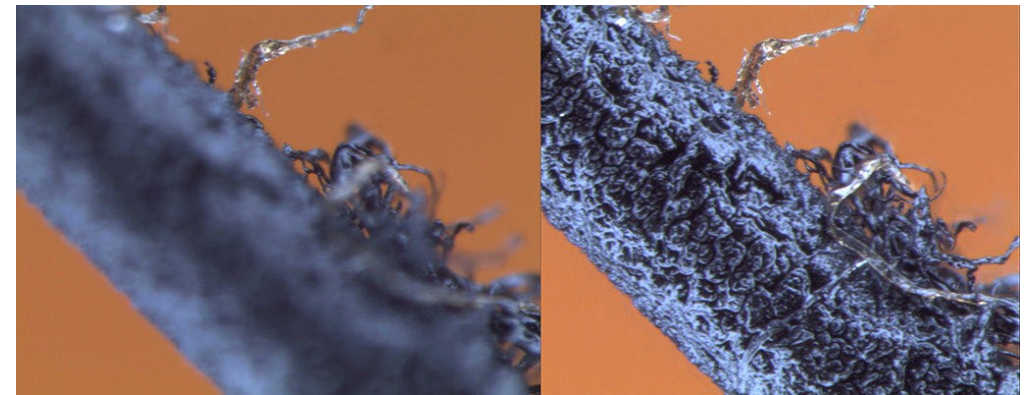
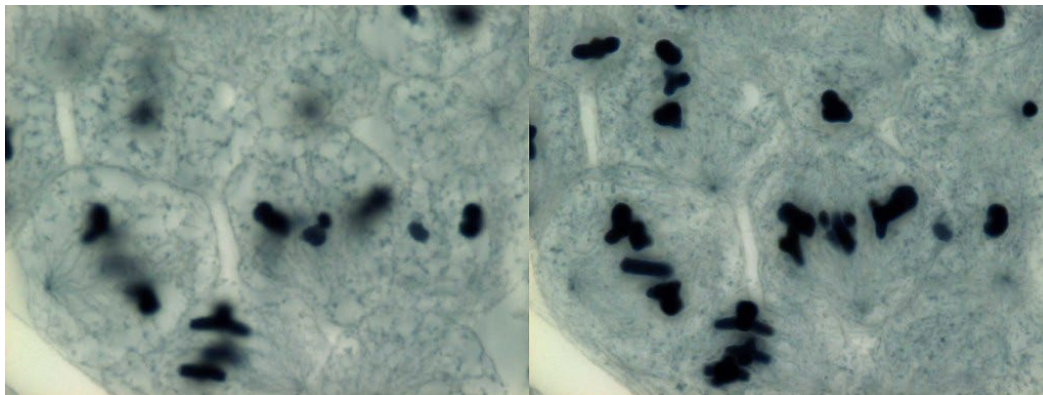
Zhou, C., Miao, D., & Nayar, S. K. (2012). Focal sweep camera for space-time refocusing.

Extended Depth of Field (EDOF)

(reminder)

- Depth of field (DOF) of an imaging system: Range of depths over which scene points appear sharp in an image.
- Often, large DOF is desirable (so that everything is in focus).
- The lens (aperture) of an imaging system limits its DOF.
- DOF can be increased by reducing aperture size.

Ok, so if you want extended DOF, reduce the aperture...



Images: Meyer Inst.

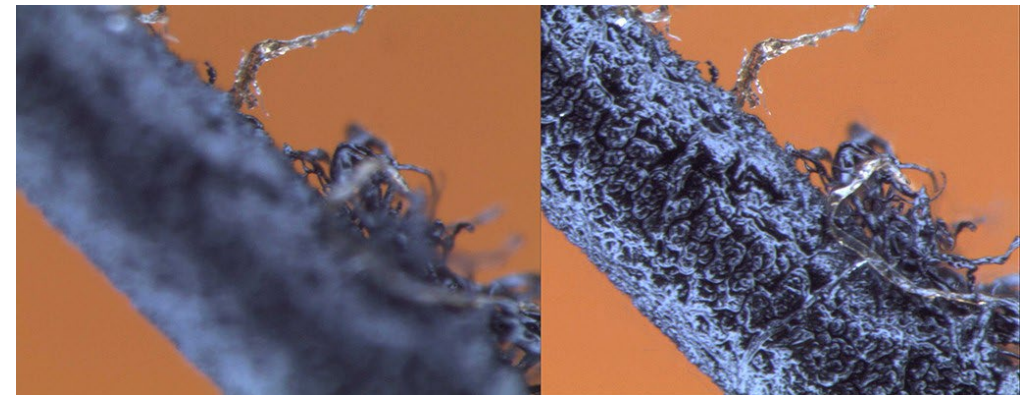
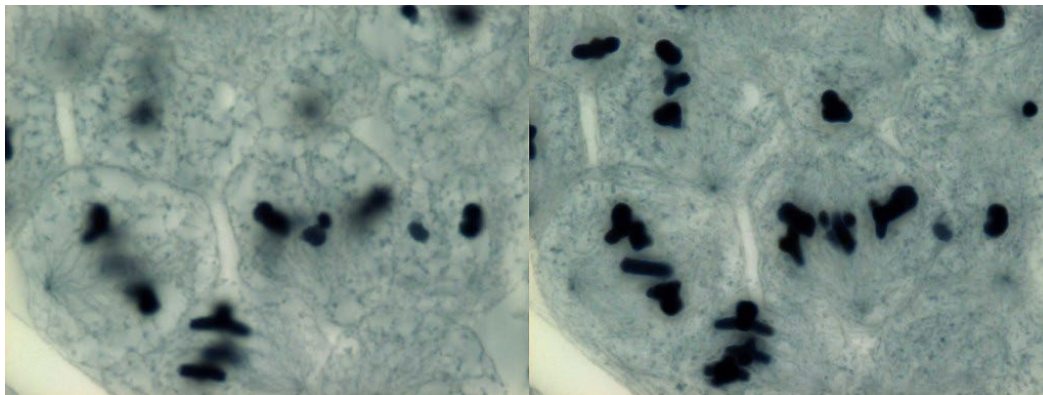
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Ok, so if you want extended DOF, reduce the aperture...

Problems! Reduced light throughput, noise.



Images: Meyer Inst.

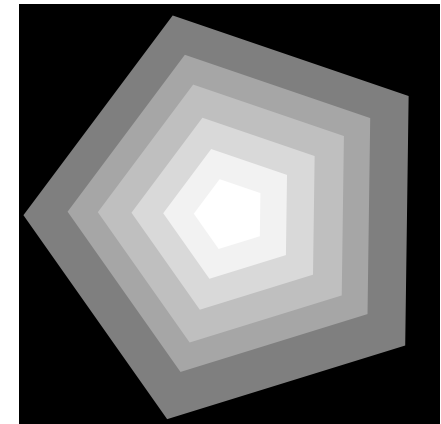
Focal sweep

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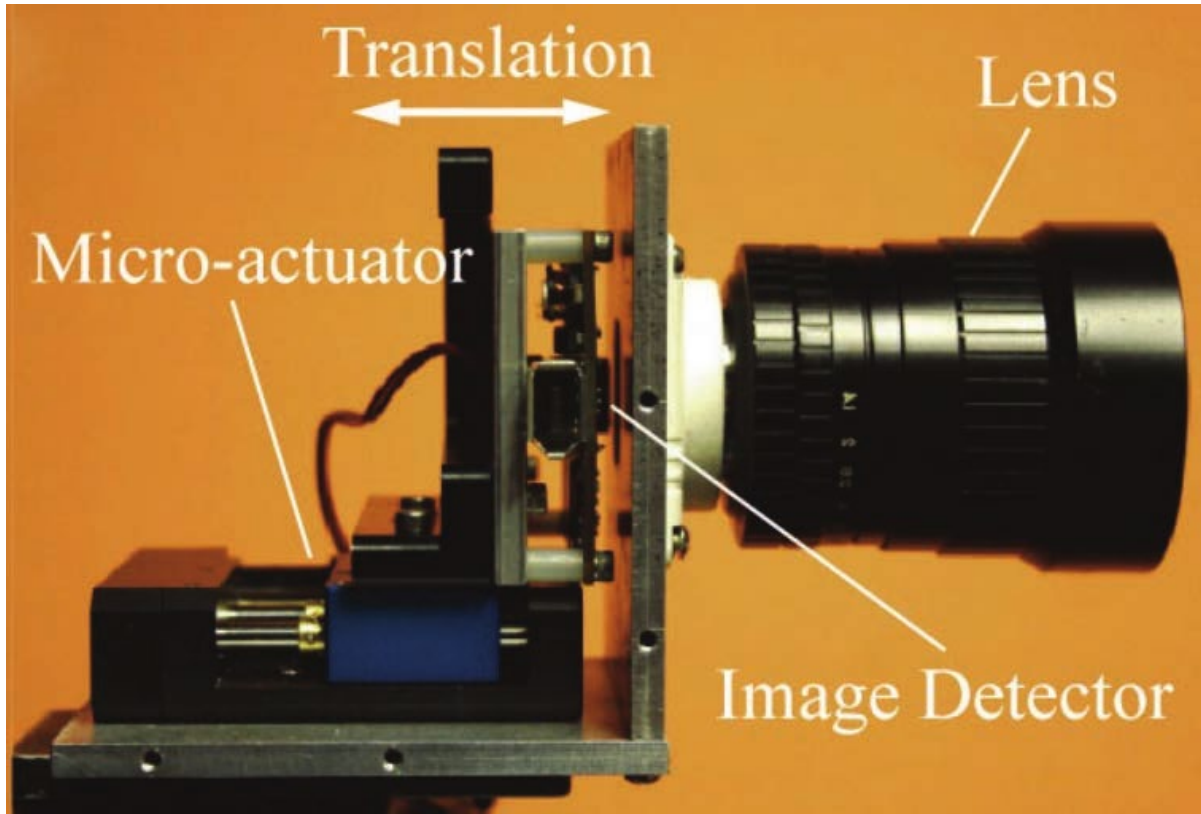


Useful for: Extended Depth Of Field (EDOF)

=> Focal sweep enables EDOF while maintaining high SNR.

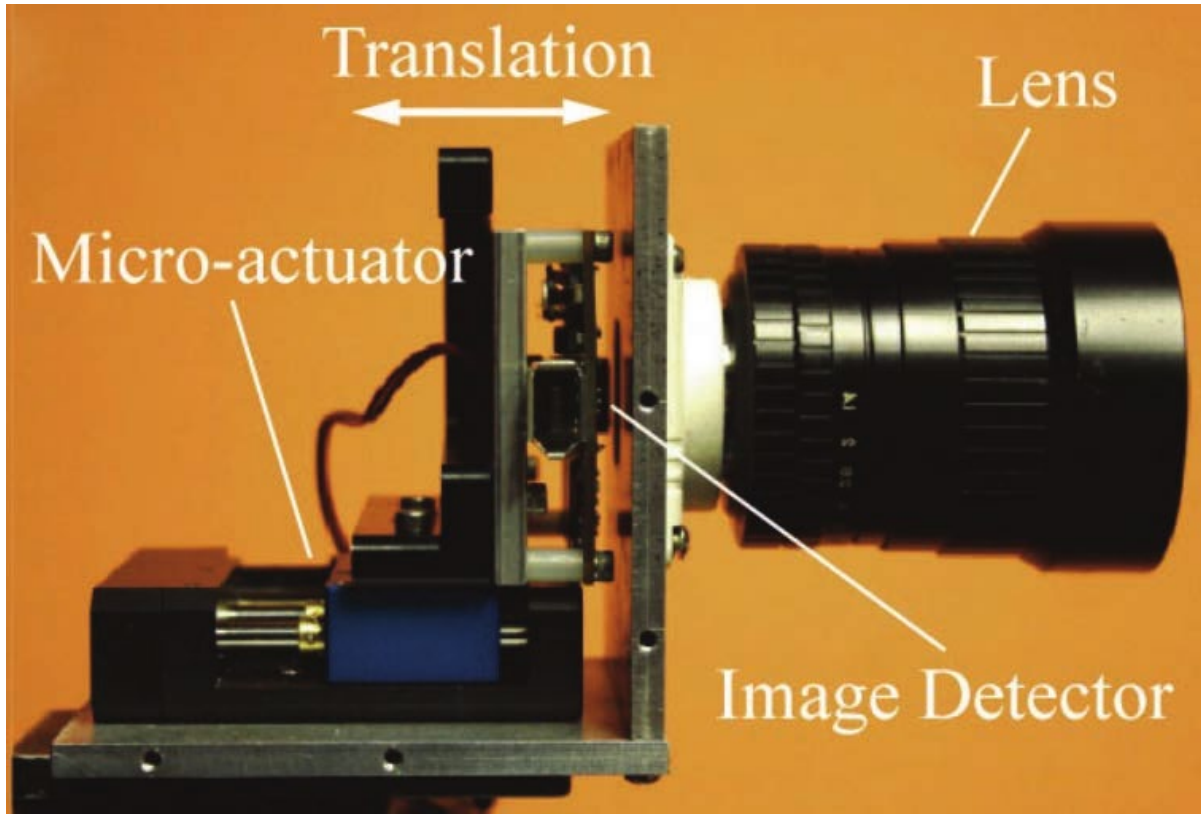
Focal sweep: Implementation

Focal sweep: Implementation



Use translation stage to move sensor relative to fixed lens during exposure

Focal sweep: Implementation



Use translation stage to move sensor relative to fixed lens during exposure



Rotate focusing ring to move lens relative to fixed sensor during exposure

Focal sweep: Results and comparison

conventional photo
(small DOF)



Sources: [CMU 15-463](#)

Focal sweep: Results and comparison

conventional photo
(small DOF)



conventional photo
(large DOF, noisy)



Focal sweep: Results and comparison

captured focal sweep
always blurry!



conventional photo
(small DOF)



EDOF image

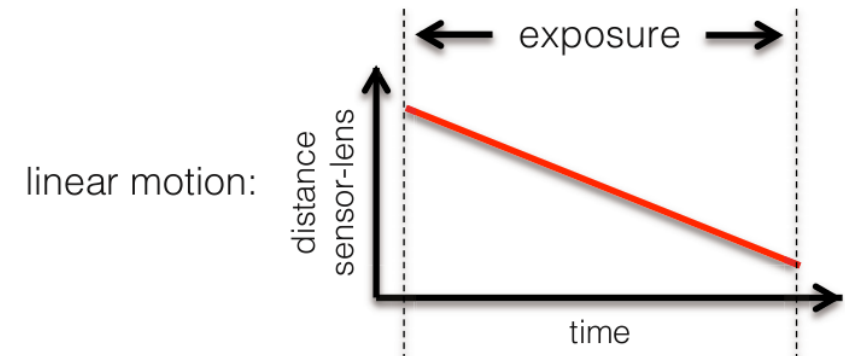
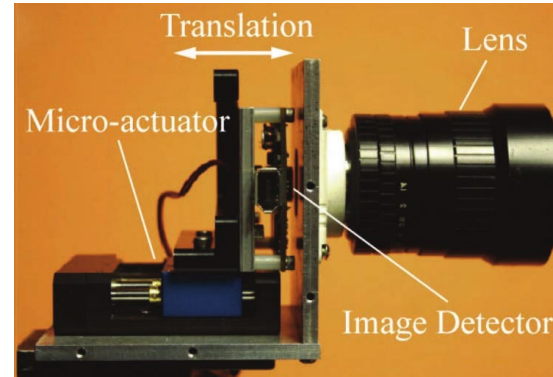


conventional photo
(large DOF, noisy)



Focal sweep: Problems

- Existence of moving parts (vibrations, motion blur).
- Perfect depth invariance requires very constant speed.
- Loss of depth information.

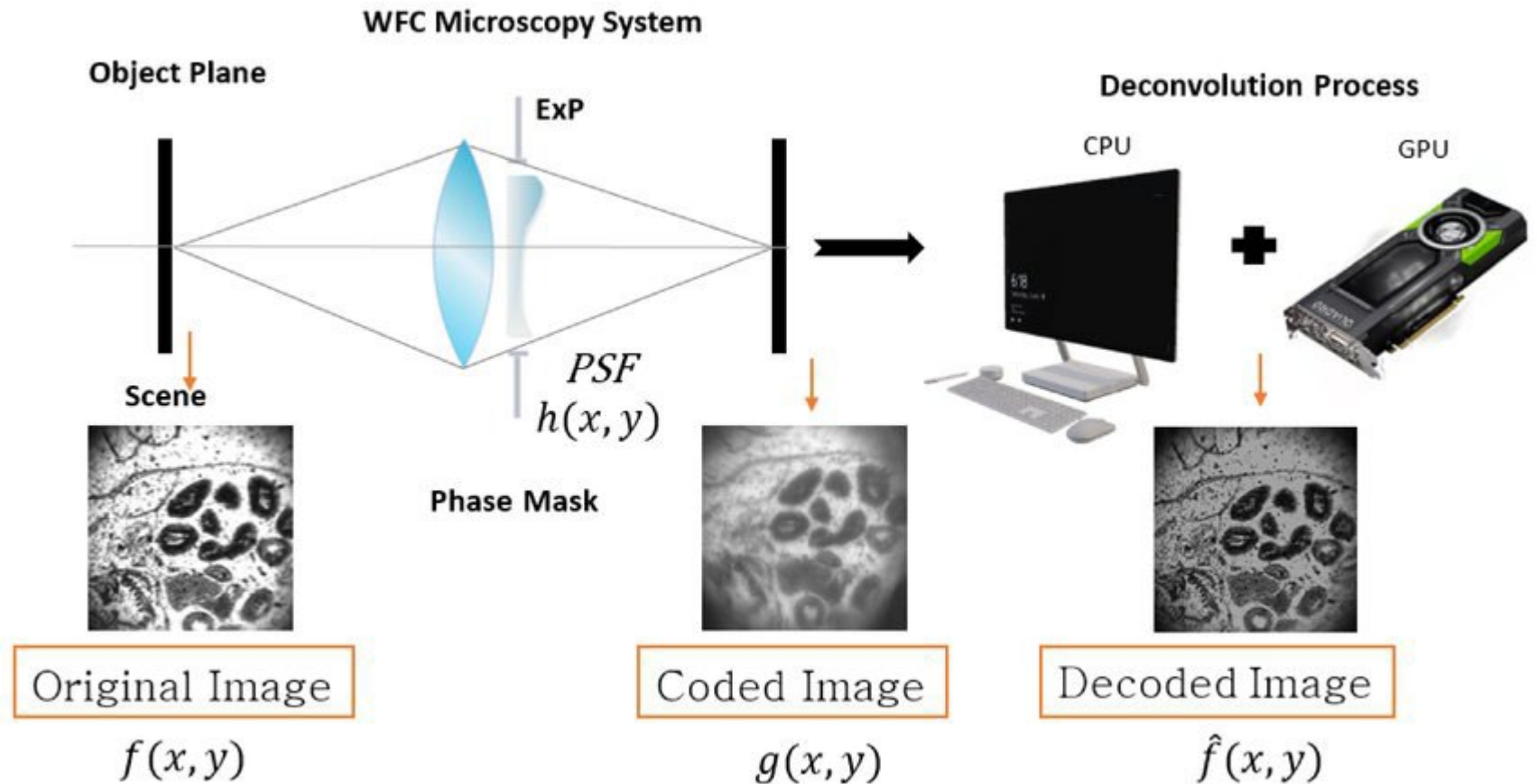


A small recap

- We can use deconvolution to obtain a deblurred image.
- We can optimize the aperture to improve deconvolution for defocus deblurring.
- The blur kernel is depth-dependent → It is hard to use deconvolution, but at the same time we have some depth information.
- We can aim at a depth-independent kernel (e.g., focal sweep; it has its own problems).

Why so worried about noise and DOF?

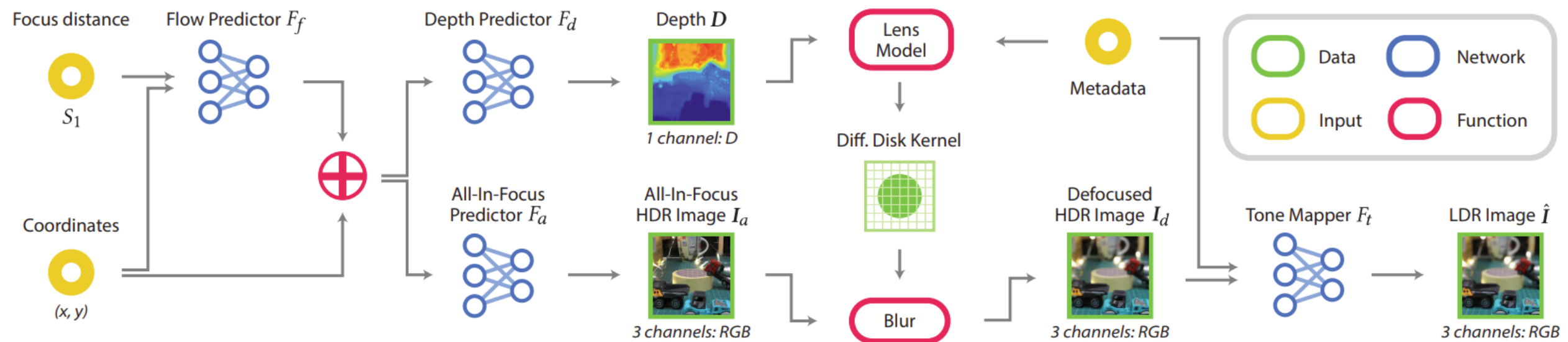
Imaging is much wider than conventional photography. Think, for instance, microscopy! (surveillance, quality monitoring, astronomy...)



Why so worried about noise and DOF?

Open research problem with no definitive solution yet!

More modern alternatives usually rely on other techniques....



<https://taf.mpi-inf.mpg.de/>

C. Wang, A. Serrano, X. Pan, K. Wolski, B. Chen, K. Myszkowski, & T. Leimkühler (2023). **An Implicit Neural Representation for the Image Stack: Depth, All in Focus, and High Dynamic Range.** *ACM Transactions on Graphics (TOG)*, 42(6), 1-11.

Today

coded photography

The introduction of codes (or other types of generalized optics) at some point of the imaging pipeline can be used for much more...

- Defocus deblurring
- Depth from defocus

These two often go hand-in-hand

- Scene motion & motion deblurring

Scene motion and motion blur

Motion blur – An initial example



Most of the scene is static

Can is moving linearly from left to right

Motion blur

It is due to relative movement between the camera and the scene (or some objects in it) during the exposure time, i.e., while the shutter is open.

- Camera moving, scene static
- Camera static, scene moving
- Camera and scene moving

camera shake



Source: Pinterest.



Source: ExpertPhotography.

Motion blur

It is due to relative movement between the camera and the scene (or some objects in it) during the exposure time, i.e., while the shutter is open.

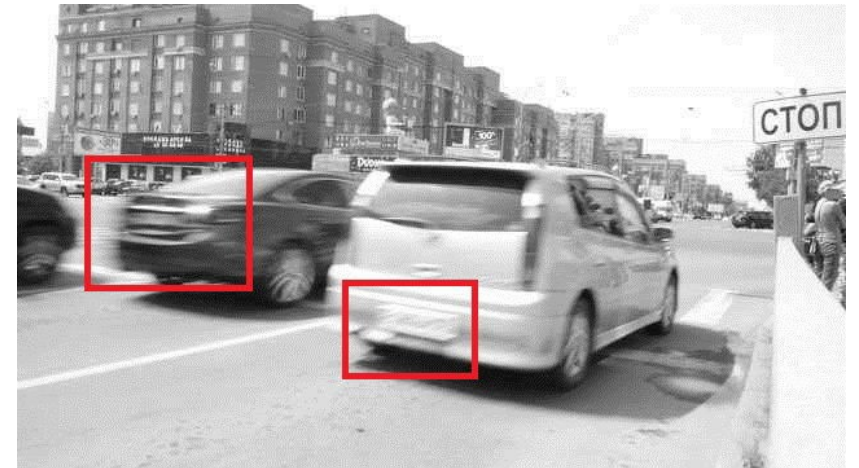
- Camera moving, scene static

camera shake



Source: Pinterest.

- Camera static, scene moving



Source: OpenCV.

- Camera and scene moving

Motion blur – Image formation model



blurry image of
moving object



motion blur kernel



sharp image of
static object

What does the motion blur kernel depend on?

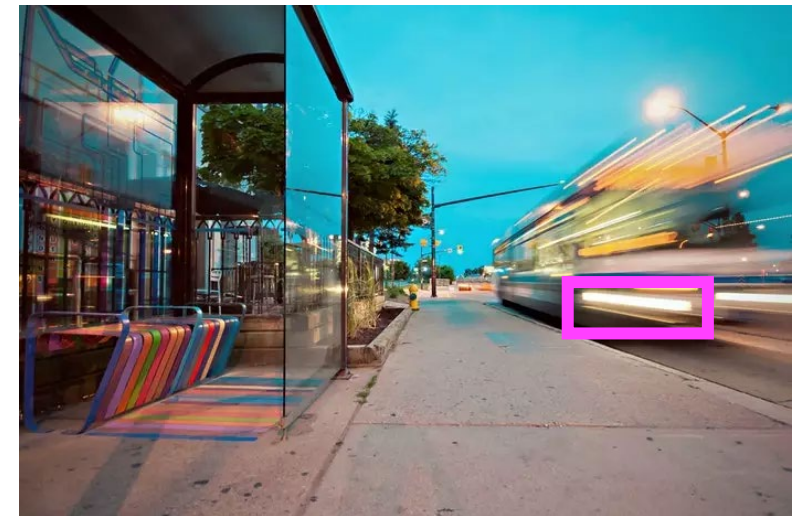
Motion blur – Image formation model

What does the motion blur kernel depend on?

- Kernel shape depends on the trajectory of the motion.
- Kernel size depends on the exposure time and relative velocity.



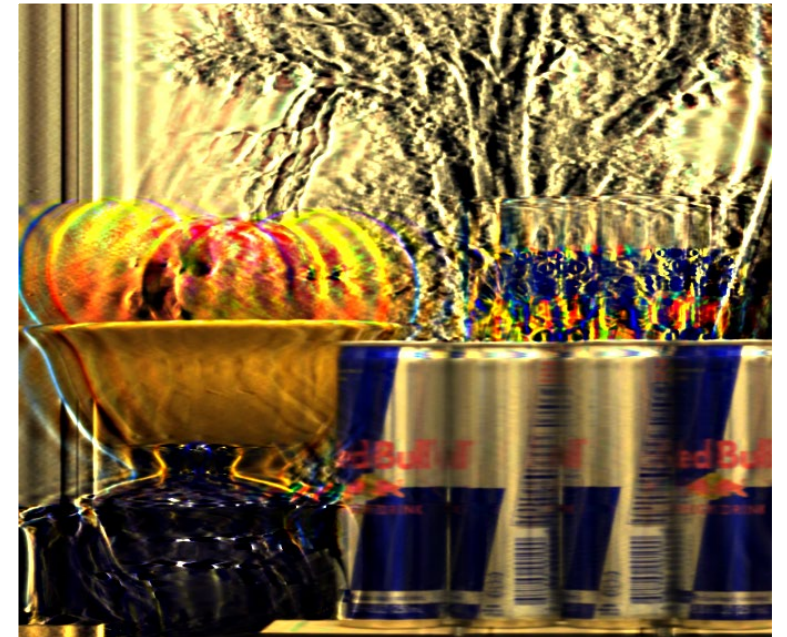
Source: Pinterest.



Source: ExpertPhotography.

Challenges of motion deblurring

- Blur kernel not invertible.
Cancellation of certain image frequencies.
- Blur kernel unknown.
Unknown motion trajectory.
- Blur kernel different for different objects.
Spatially-varying kernel (velocity-dependent).



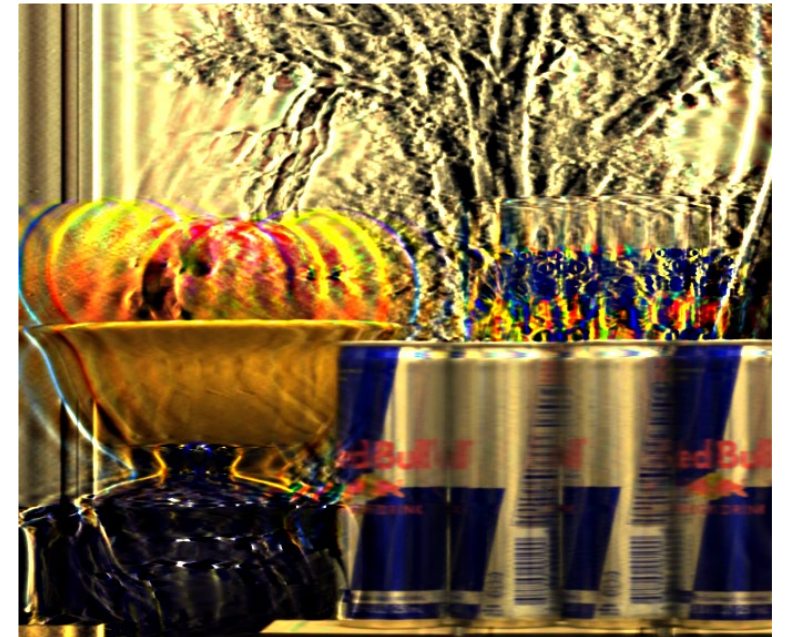
Challenges of motion deblurring

- Blur kernel not invertible.

*Similar to the coded apertures case with their zeros at certain frequencies...
Can we make the kernel invertible?*

- Blur kernel unknown.

- Blur kernel different for different objects.



Coded exposure: Flutter-shutter

Code exposure (i.e., shutter speed) to make motion blur kernel better conditioned.

traditional
camera



blurry image of
moving object

=



*



sharp image of
static object

flutter-shutter
camera



blurry image of
moving object

=



*



sharp image of
static object

Coded exposure: Flutter-shutter

(reminder)

Code exposure (i.e., shutter speed) to make motion blur kernel

traditional
camera



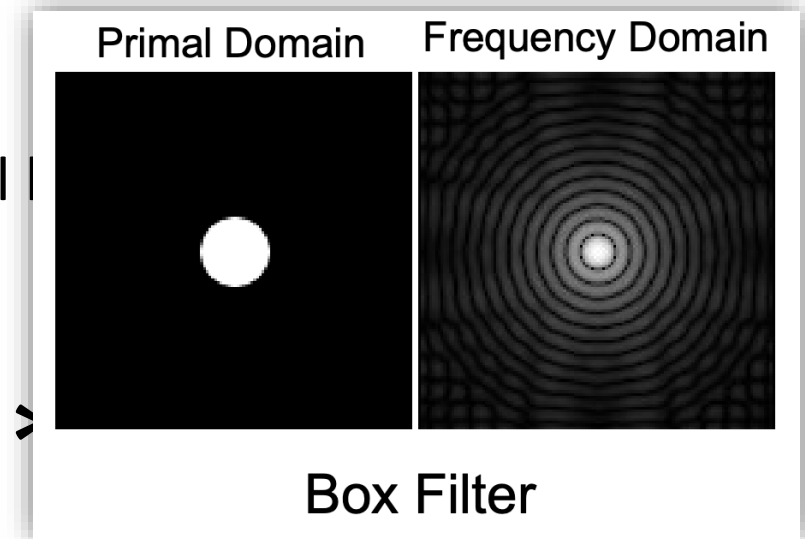
blurry image of
moving object

=



motion blur kernel

>



sharp image of
static object

flutter-shutter
camera



blurry image of
moving object

=



motion blur kernel

*



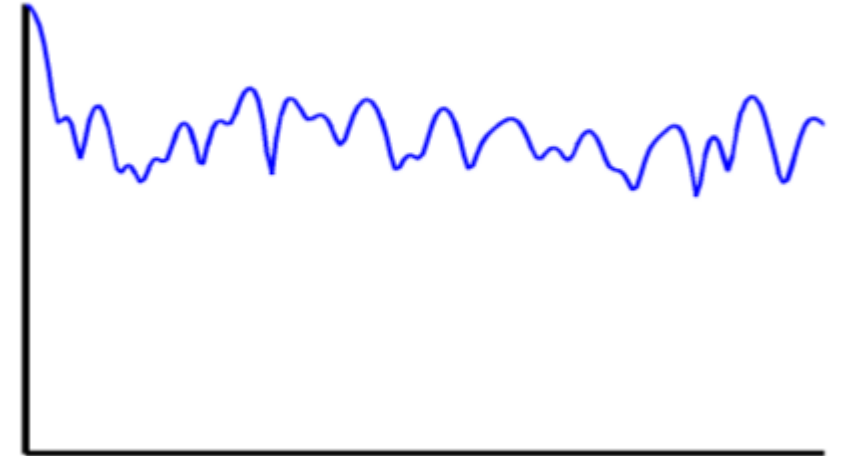
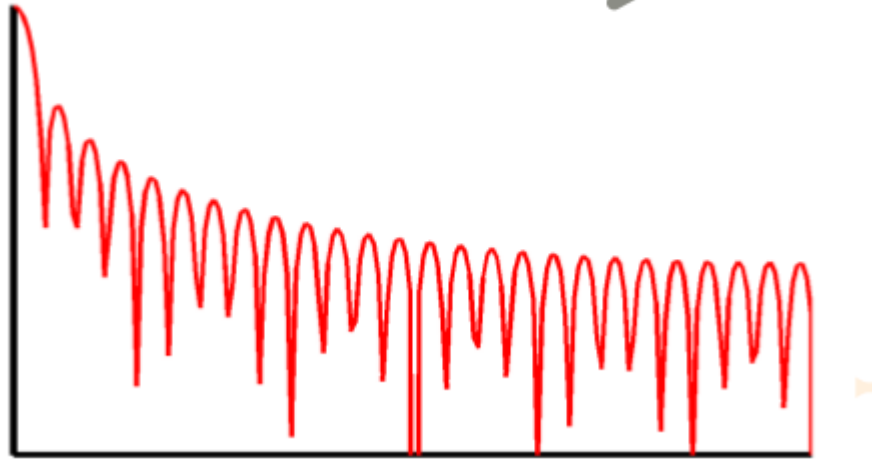
sharp image of
static object

Coded exposure: Flutter-shutter

motion blur kernel
in time domain



motion blur kernel
in Fourier domain

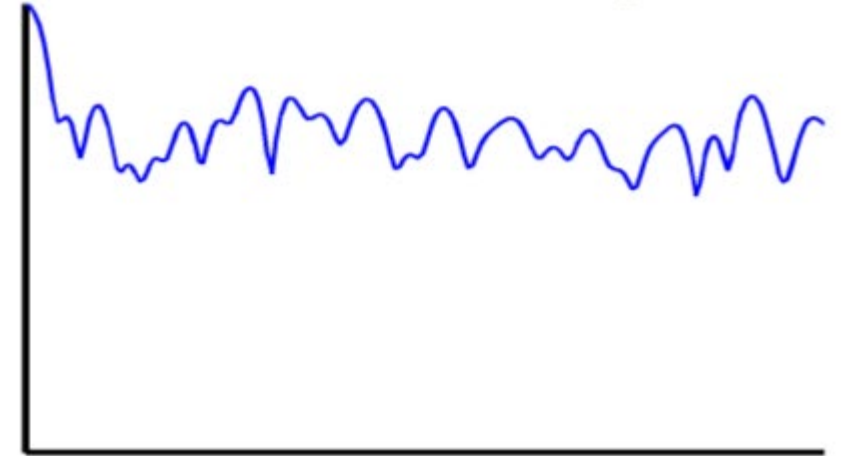
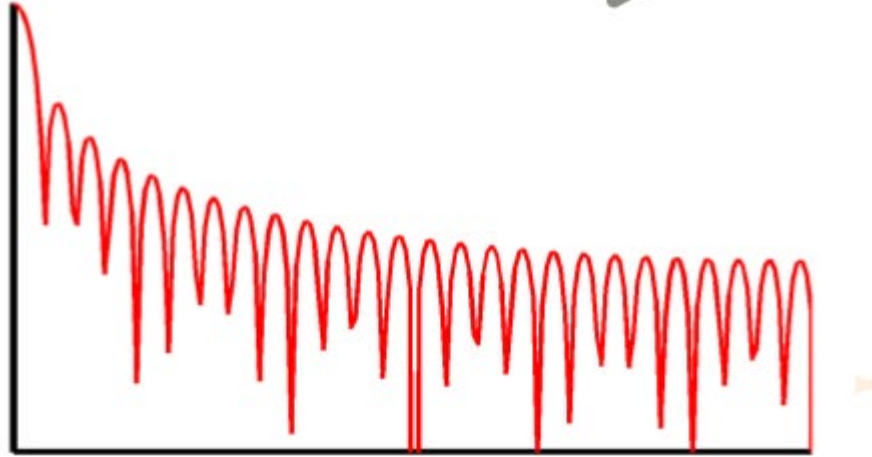


Coded exposure: Flutter-shutter

motion blur kernel
in time domain

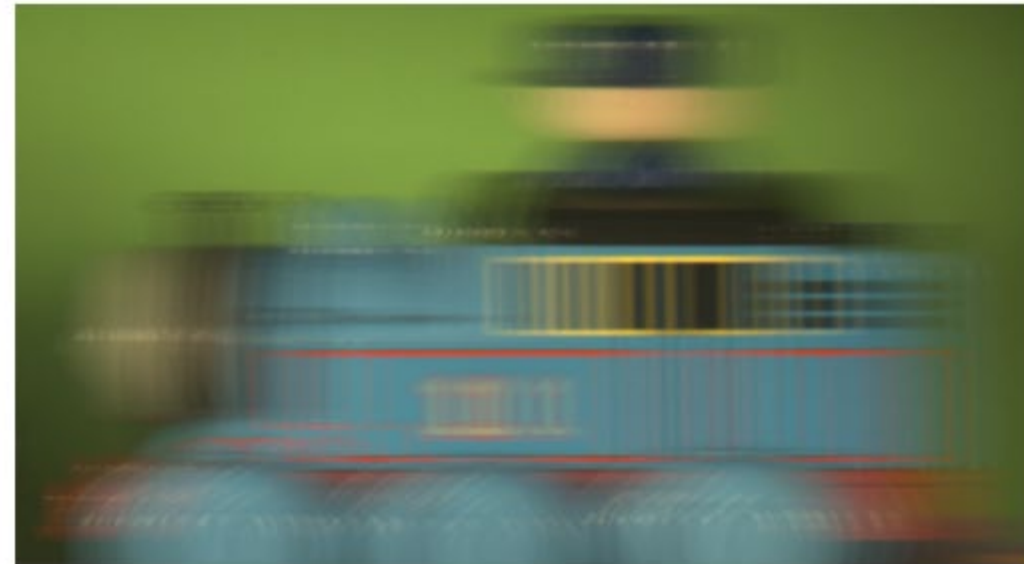
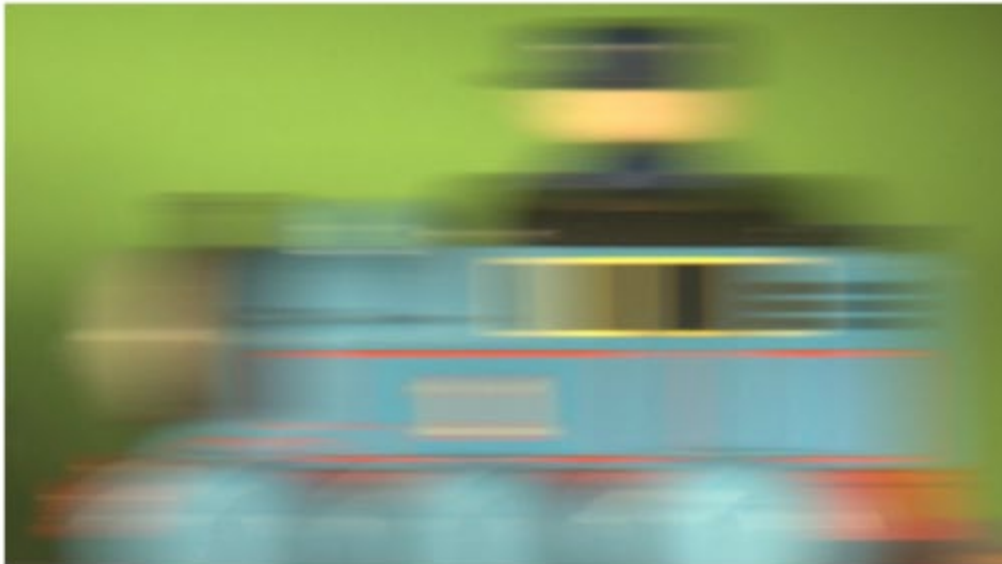


motion blur kernel
in Fourier domain



Captured
image

Why is flutter-
shutter better?

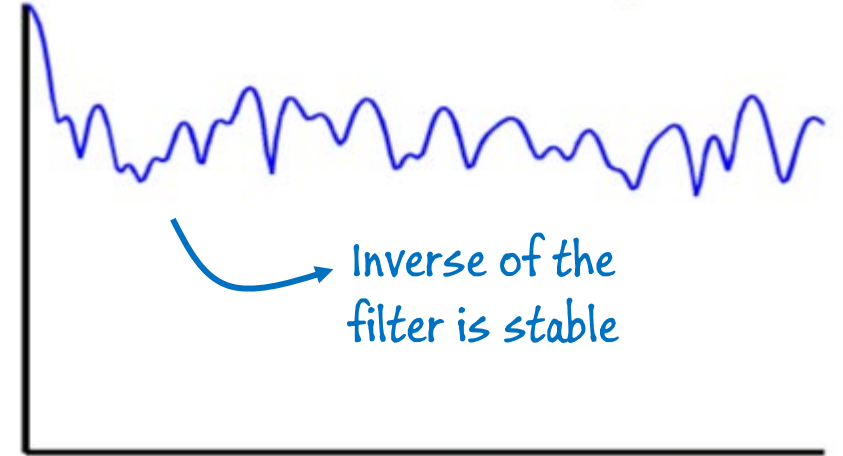
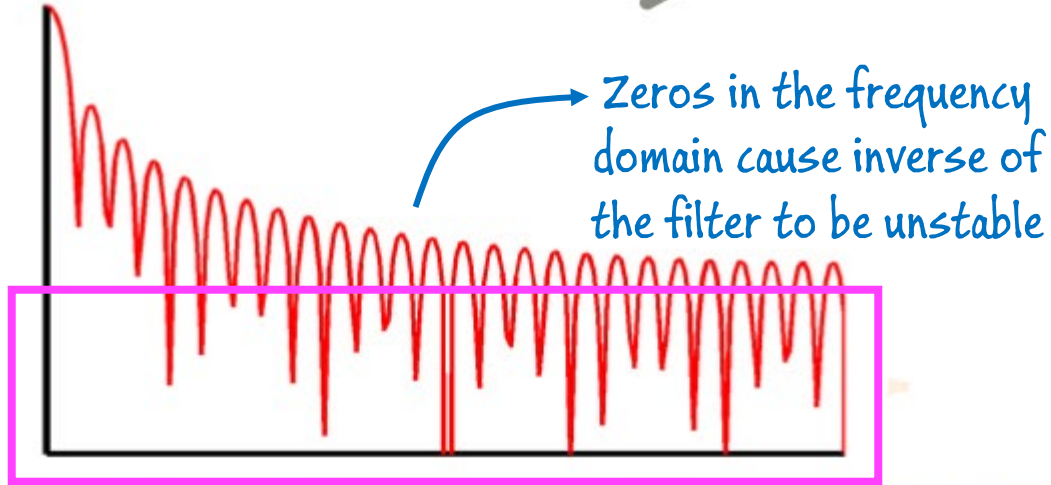


Coded exposure: Flutter-shutter

motion blur kernel
in time domain

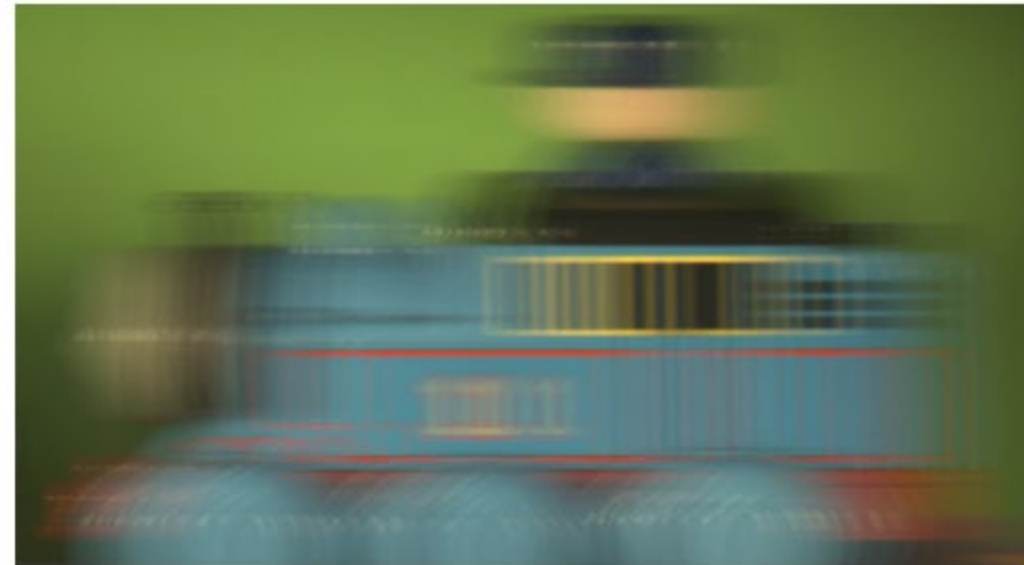
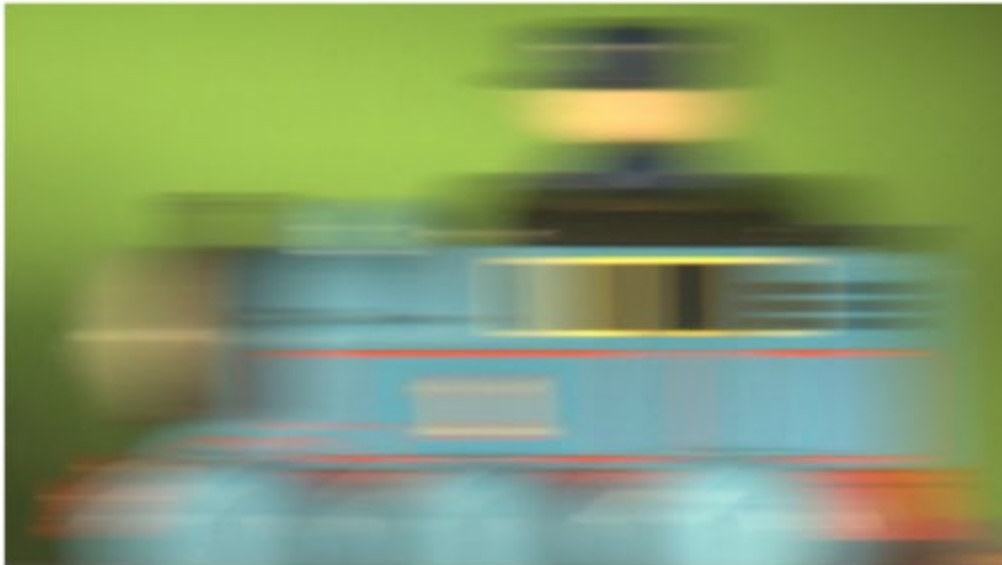


motion blur kernel
in Fourier domain



Captured
image

Why is flutter-
shutter better?

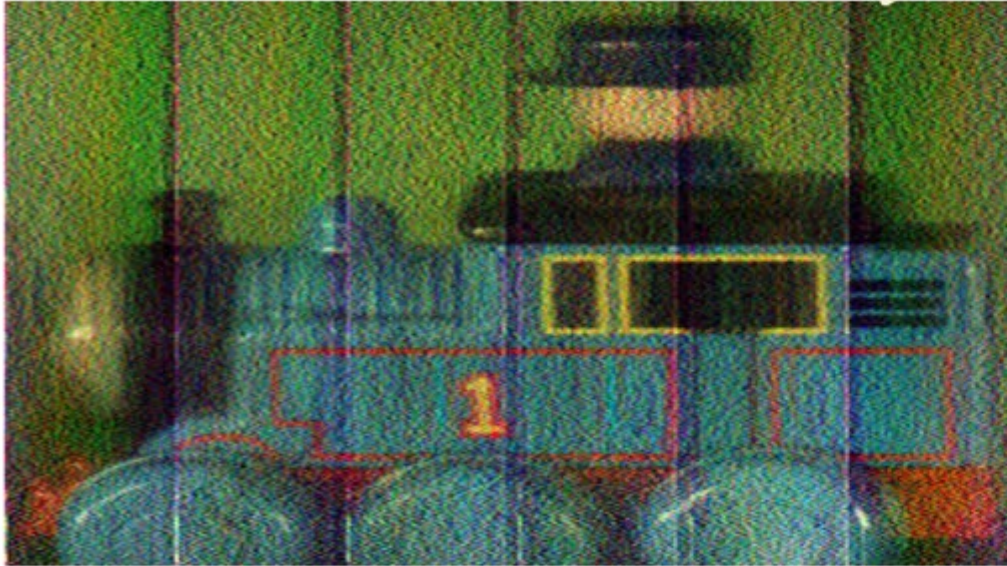


Coded exposure – Motion deblurring comparison

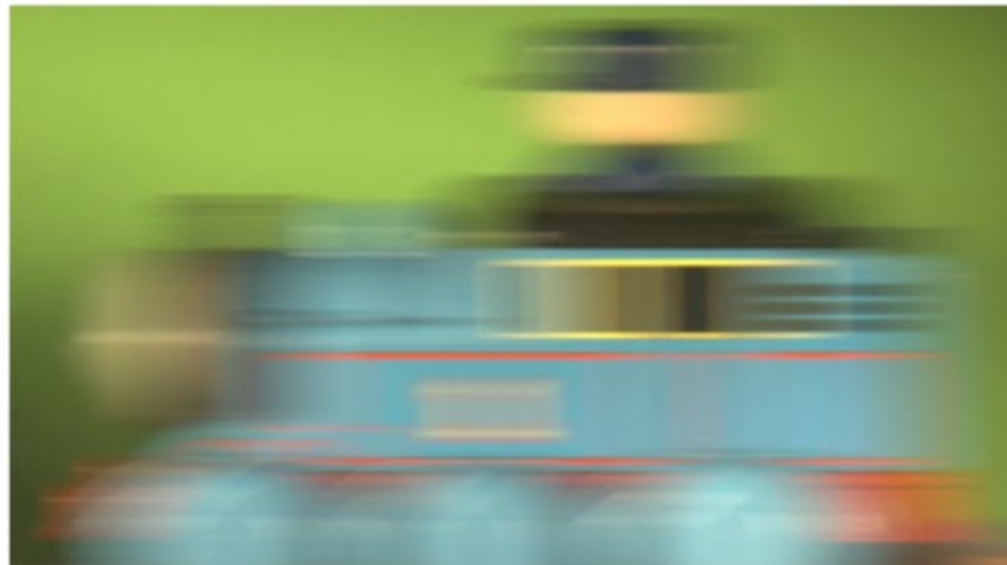
conventional photography

flutter-shutter photography

deconvolved
output

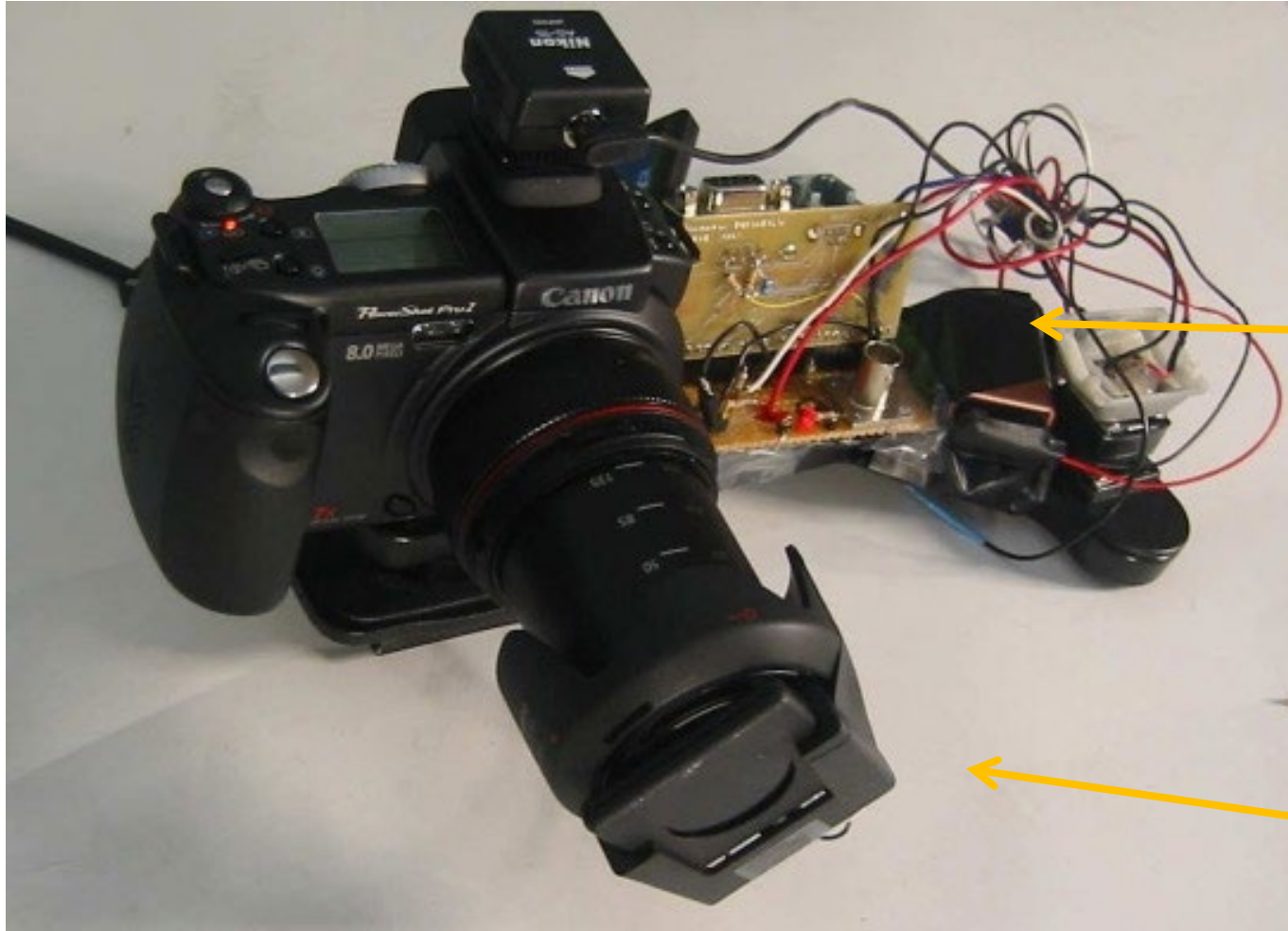


blurry
input



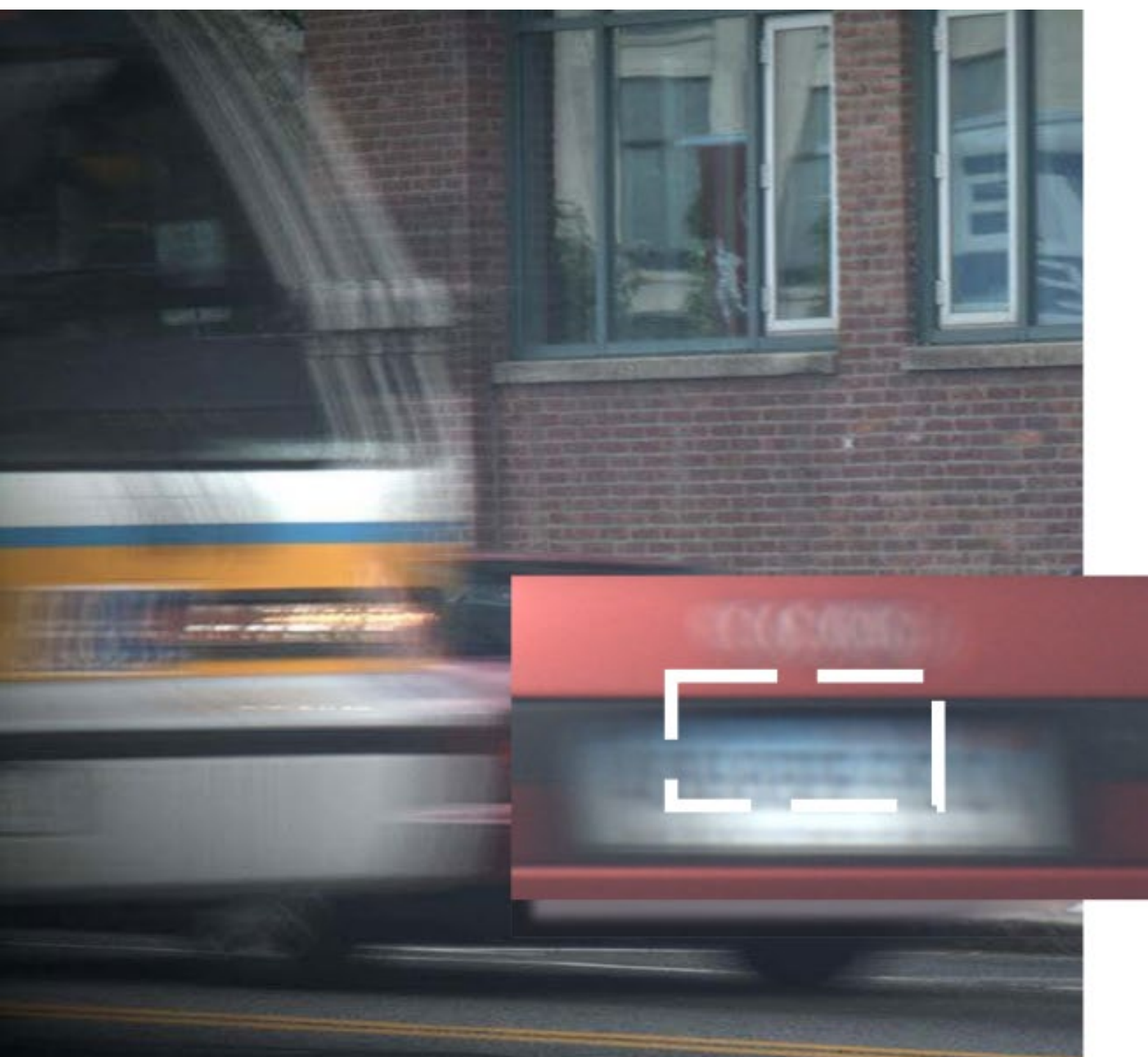
Coded exposure: Implementation

Coded exposure: Implementation

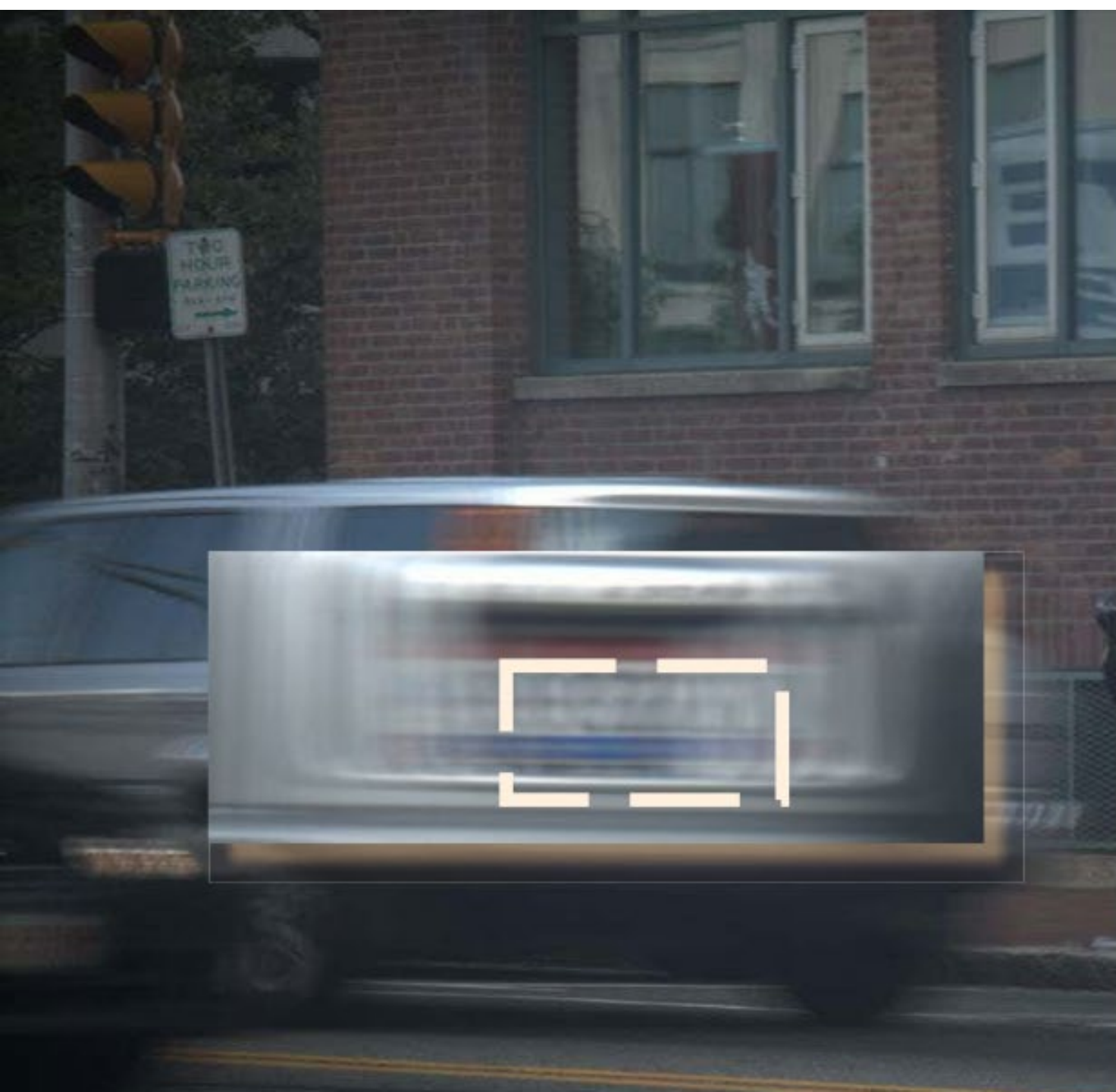


electronics for external
shutter control

very fast external
shutter



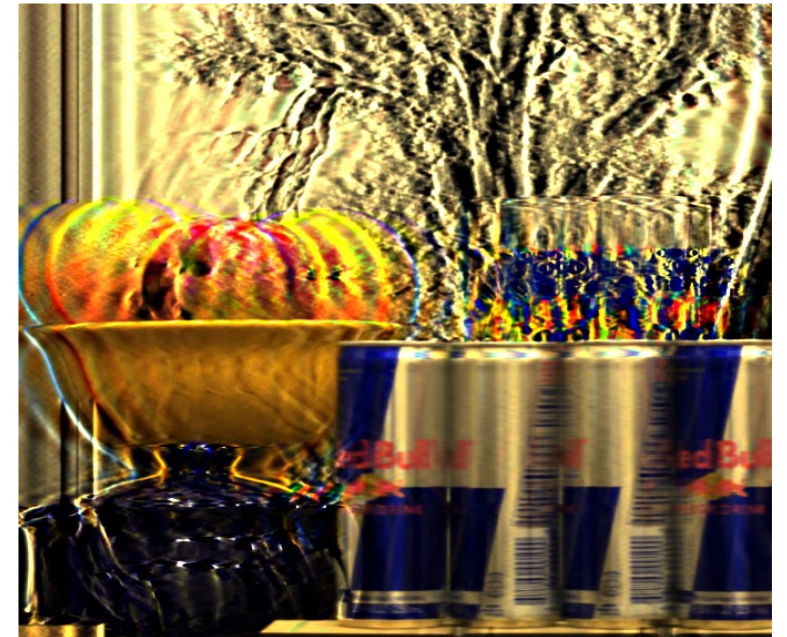
License Plate Retrieval



License Plate Retrieval

Challenges of motion deblurring

- Blur kernel not invertible → *Coded exposure*
- Blur kernel unknown
→ *Blind deconvolution, kernel estimation...*
Out of the scope of the course, but let me give you an intuition.
- Blur kernel different for different objects



Removing blur due to camera shake



Removing blur due to camera shake



Removing blur due to camera shake



image from static camera

x



PSF from camera motion

$* \quad c \quad =$



image from shaky camera

b

Removing blur due to camera shake

Blind deconvolution



image from static camera

x



PSF from camera motion

$* \quad c \quad =$



image from shaky camera

b

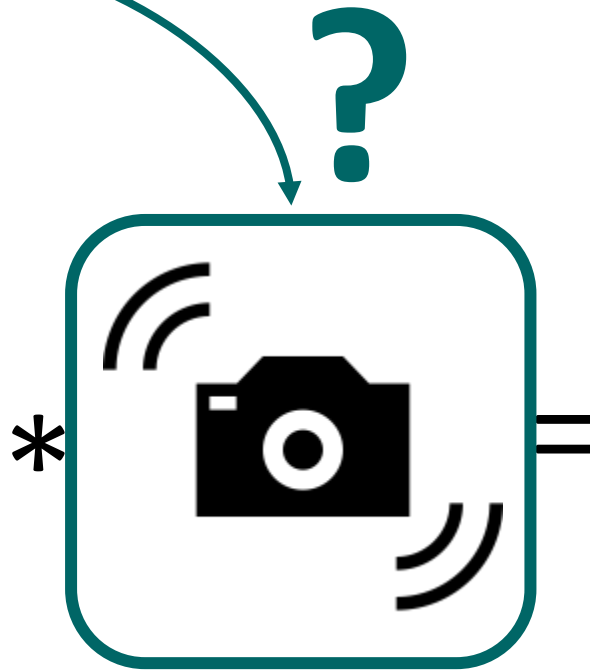
Removing blur due to camera shake

Blind deconvolution



image from static camera

x



PSF from camera motion

$* \quad c \quad =$



image from shaky camera

b

Removing blur due to camera shake

Blind deconvolution If we know **b**, can we recover **x** and **c**?

Fergus et al., "Removing camera shake from a single image", SIGGRAPH 2006



image from static camera

x

$*$



PSF from camera motion

$*$

c

$=$

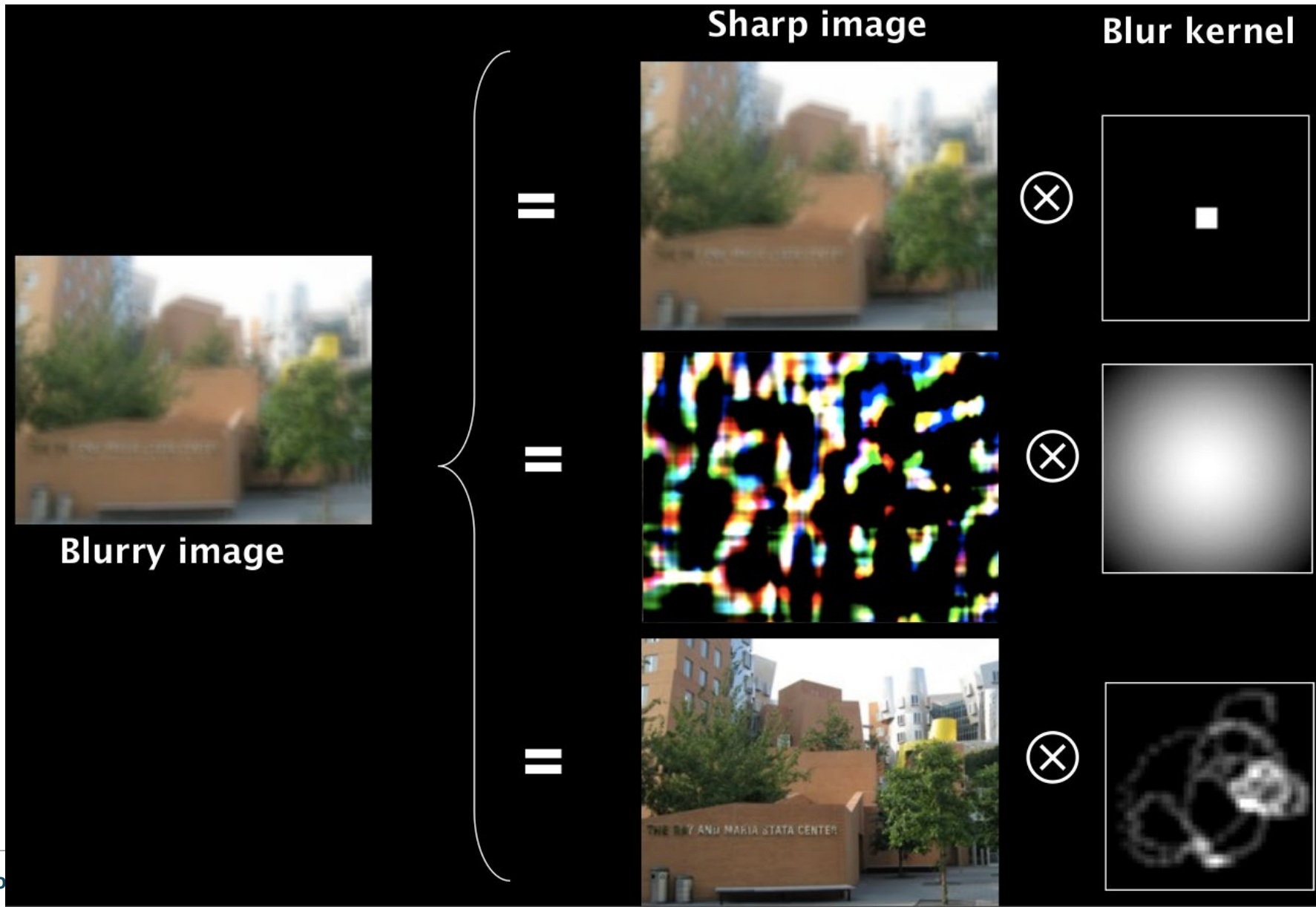
$=$



image from shaky camera

b

Multiple possible solutions



How do we detect this one?

Use prior information

Among all the possible pairs of images and blur kernels, select the ones where:

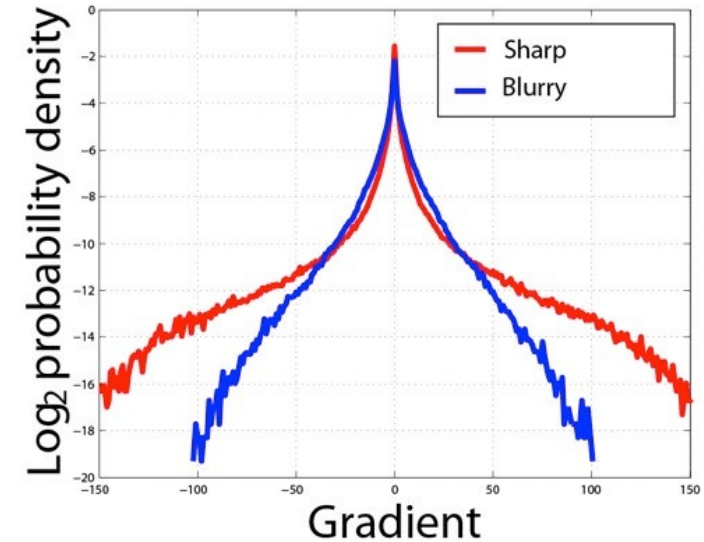
- The image “looks like” a natural image.
- The kernel “looks like” a motion PSF.

Use prior information

Among all the possible pairs of images and blur kernels, select the ones where:

- The image “looks like” a natural image.

Gradients in natural images follow a characteristic “heavy-tail” distribution.



- The kernel “looks like” a motion PSF.

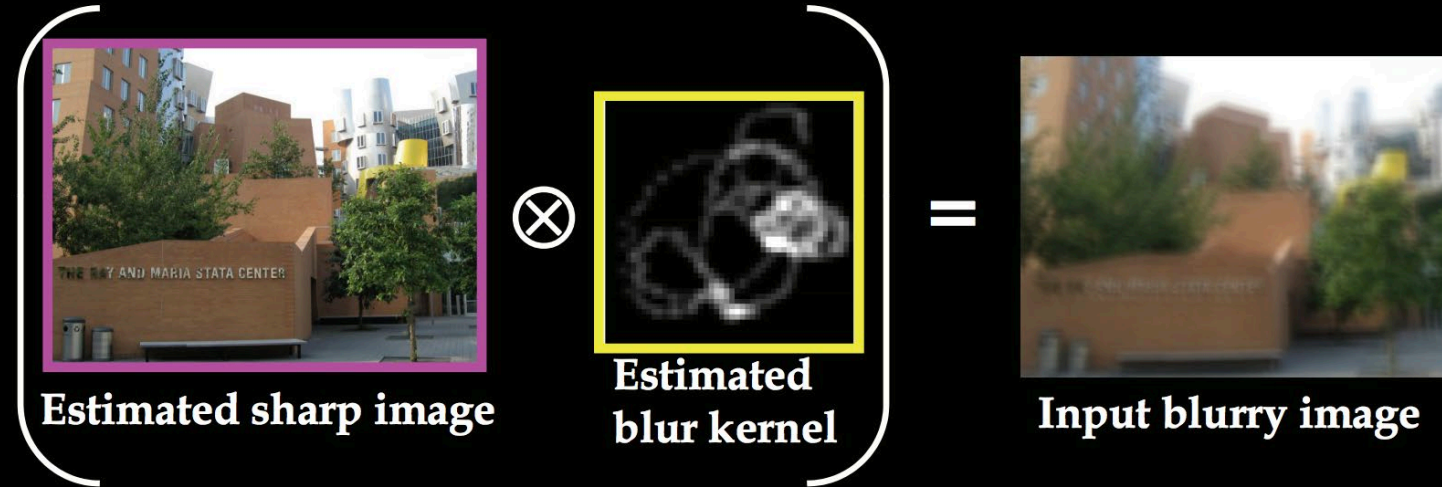
Shake kernels are very sparse, have continuous contours, and are always positive



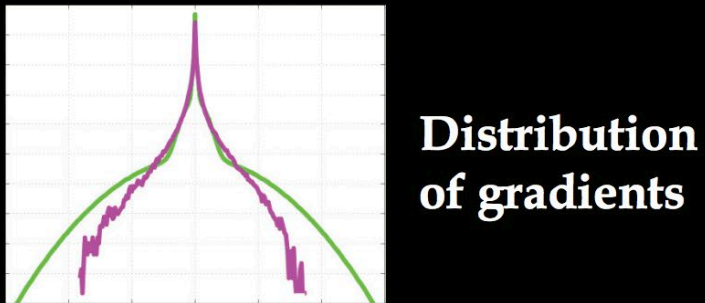
How do we use this information?

So we now have *three* sources of information:

1. Reconstruction constraint:



2. Image prior:



3. Blur prior:



Regularized blind deconvolution

Solve regularized least-squares optimization

$$\min_{x,c} ||b - c * x||^2 + ||\nabla x||^{0.8} + ||c||_1$$

What does each term in this summation correspond to?

Regularized blind deconvolution

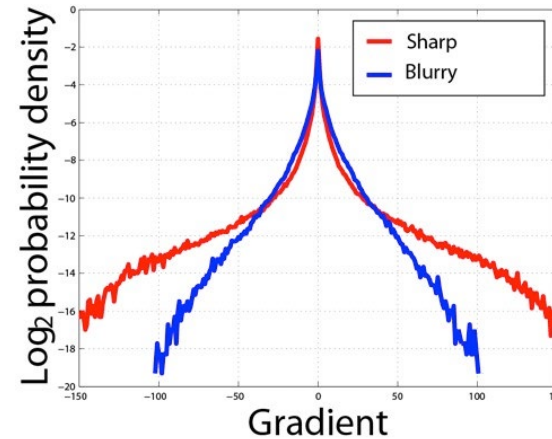
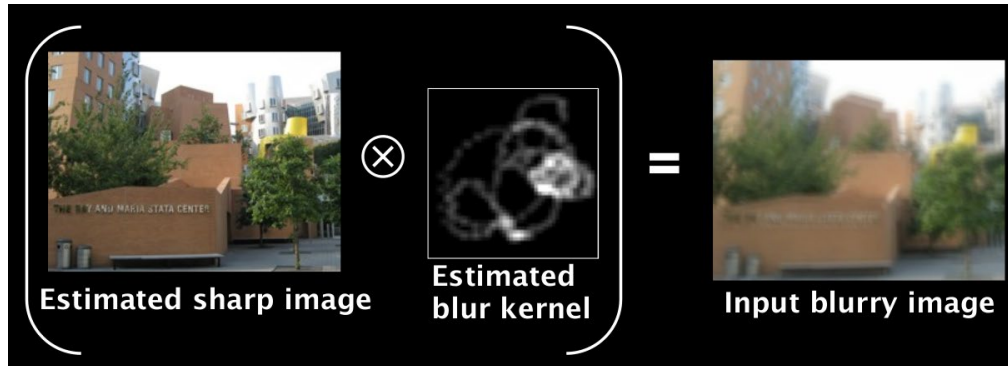
Solve regularized least-squares optimization

$$\min_{x,c} ||b - c * x||^2 + ||\nabla x||^{0.8} + ||c||_1$$

data term

natural image prior

shake kernel prior



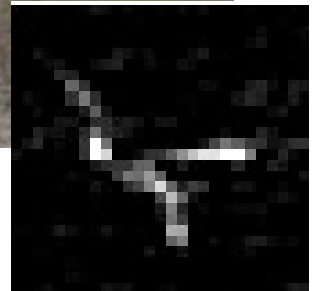
Note: Solving such optimization problems is complicated.

A demonstration

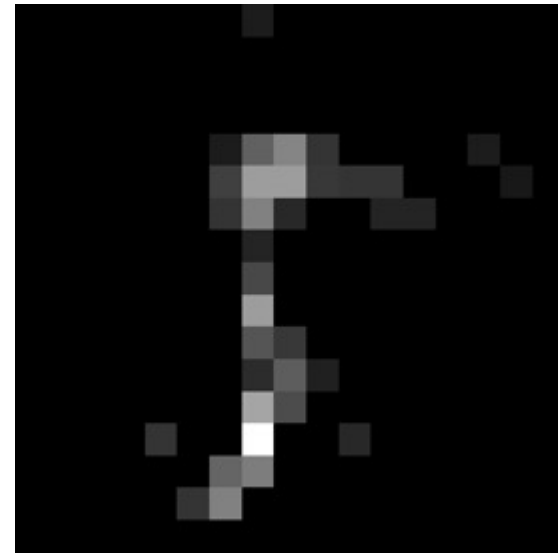
input



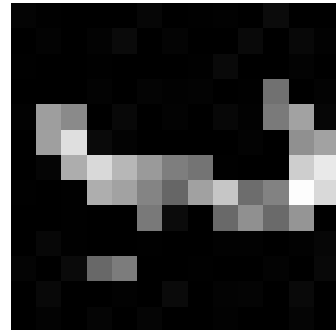
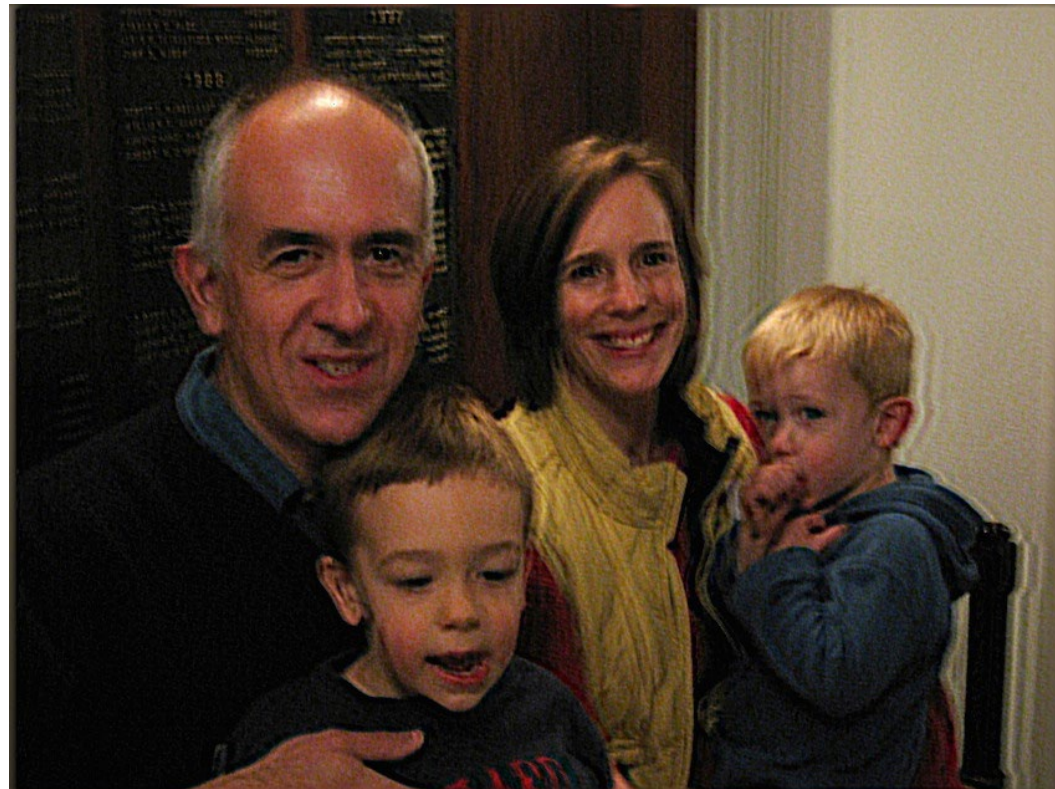
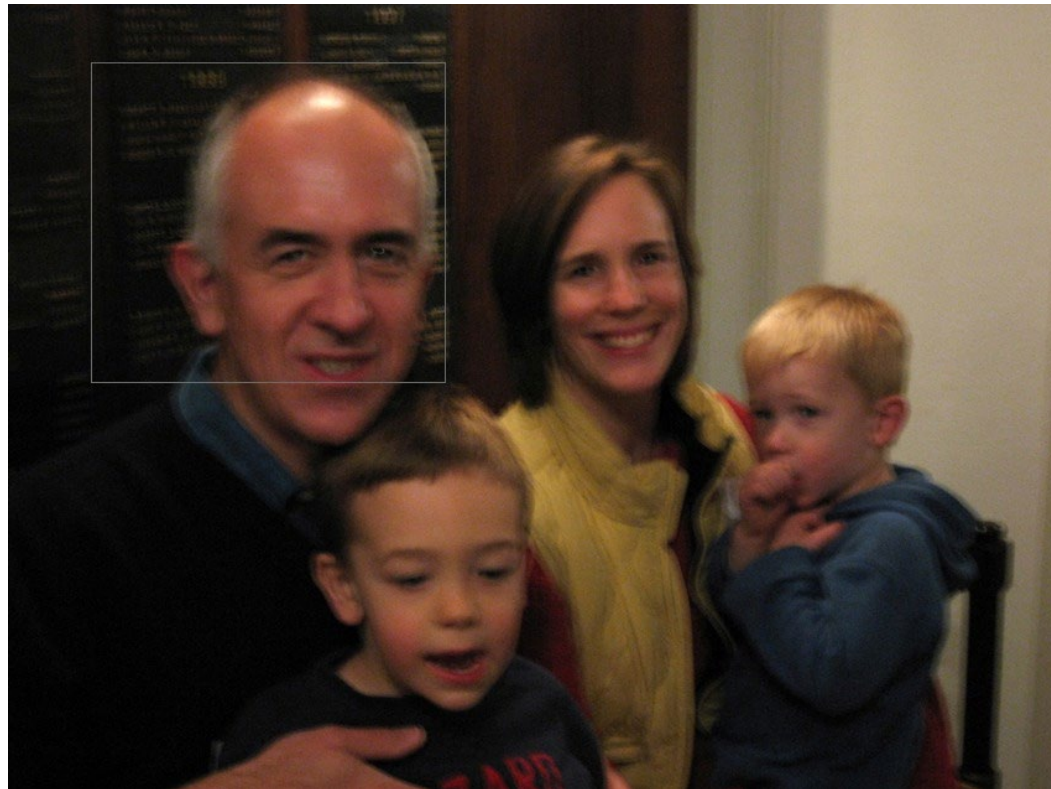
Solution



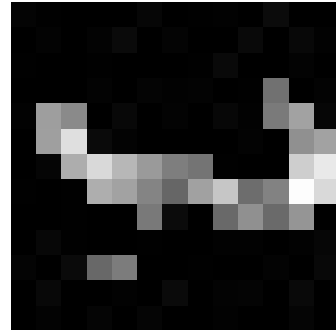
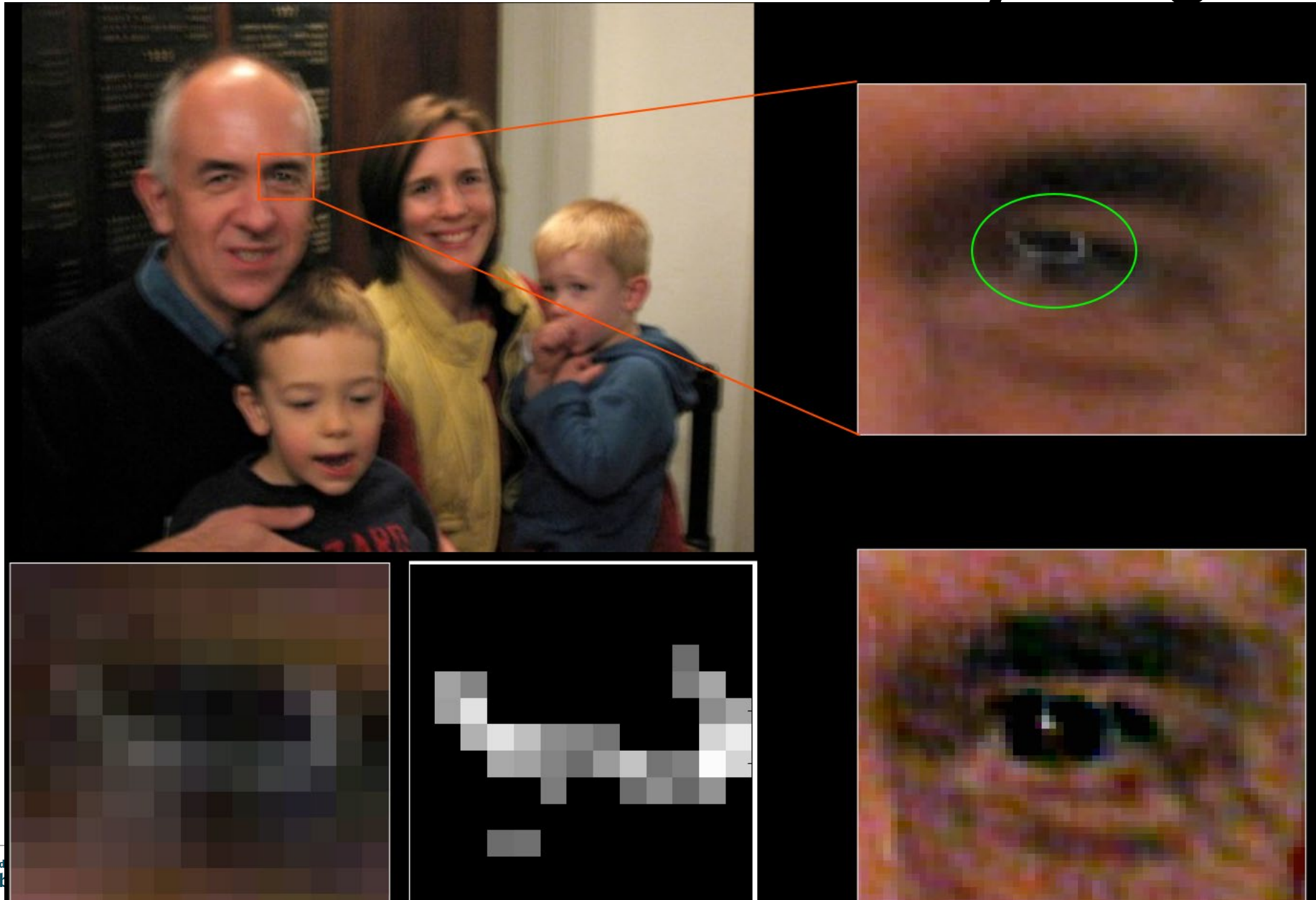
Results on real shaky images



Results on real shaky images

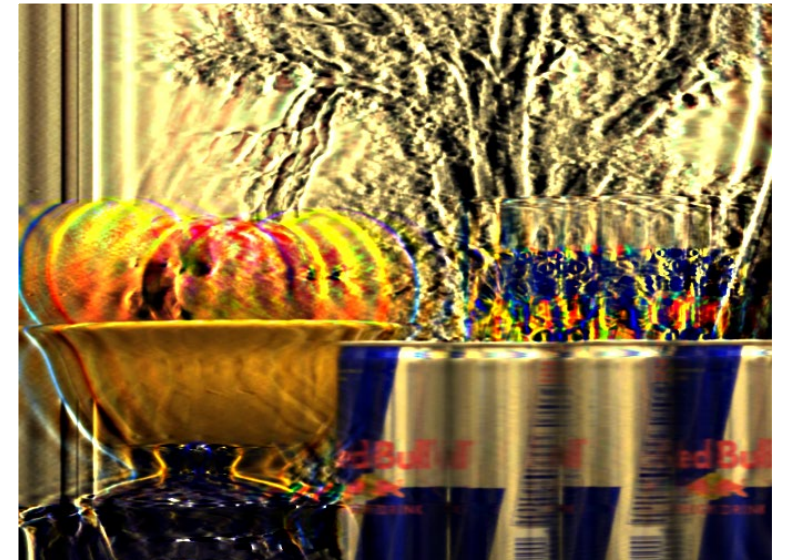


Results on real shaky images



Challenges of motion deblurring

- Blur kernel not invertible → *Coded exposure*
- Blur kernel unknown → *Blind deconvolution, kernel est.*
- Blur kernel different for different objects
There are approaches similar in spirit to the problem of spatially-varying kernel due to different depths → They attempt to have a motion-invariant kernel at the cost of capturing everything blurry (introducing additional motion)
→ *Out of the scope of the course.*



Levin, A., Sand, P., Cho, T. S., Durand, F., & Freeman, W. T. (2008). Motion-invariant photography. *ACM Transactions on Graphics (TOG)*, 27(3), 1-9.

Today

coded photography

The introduction of codes (or other types of generalized optics) at some point of the imaging pipeline can be used for much more...

- ✓ • Defocus deblurring
- ✓ • Depth from defocus

} These two often go hand-in-hand

- ✓ • Scene motion & motion deblurring

Today

Let's now widen our perspective, into a more general vision...

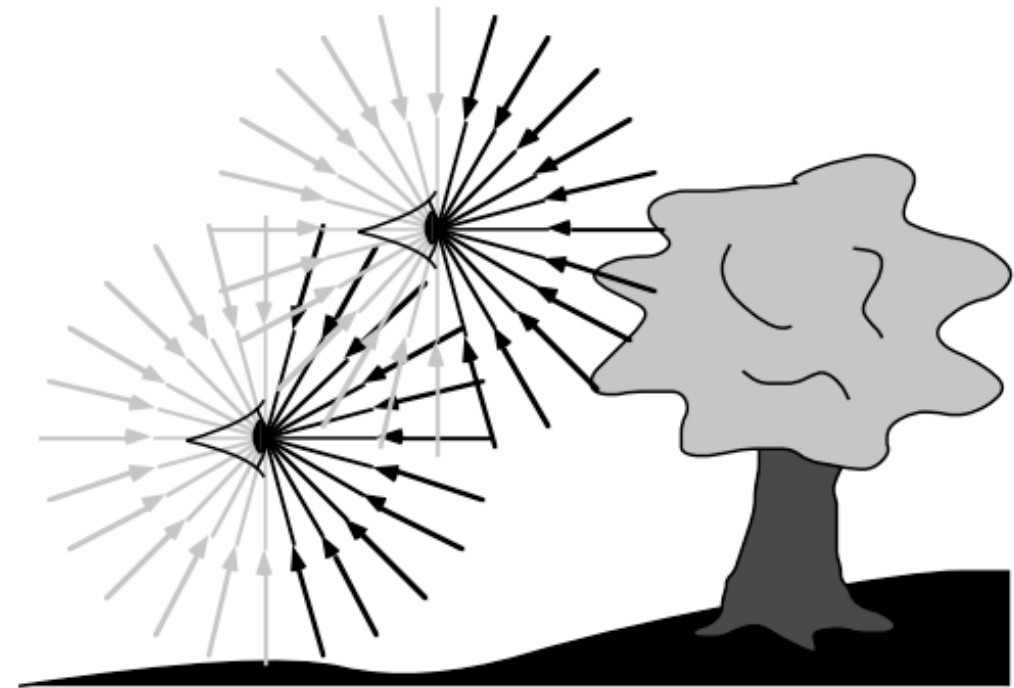
- ✓ • Coded photography
 - Plenoptic function and plenoptic imaging
 - Compressive sensing

Plenoptic function

A representation of the light present in a scene, that encompasses most properties relevant to image acquisition.

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

“Such a representation would contain, implicitly, a description of every possible photograph that could be taken of a particular space-time chunk of the world (neglecting polarization and other wave-based effects).”



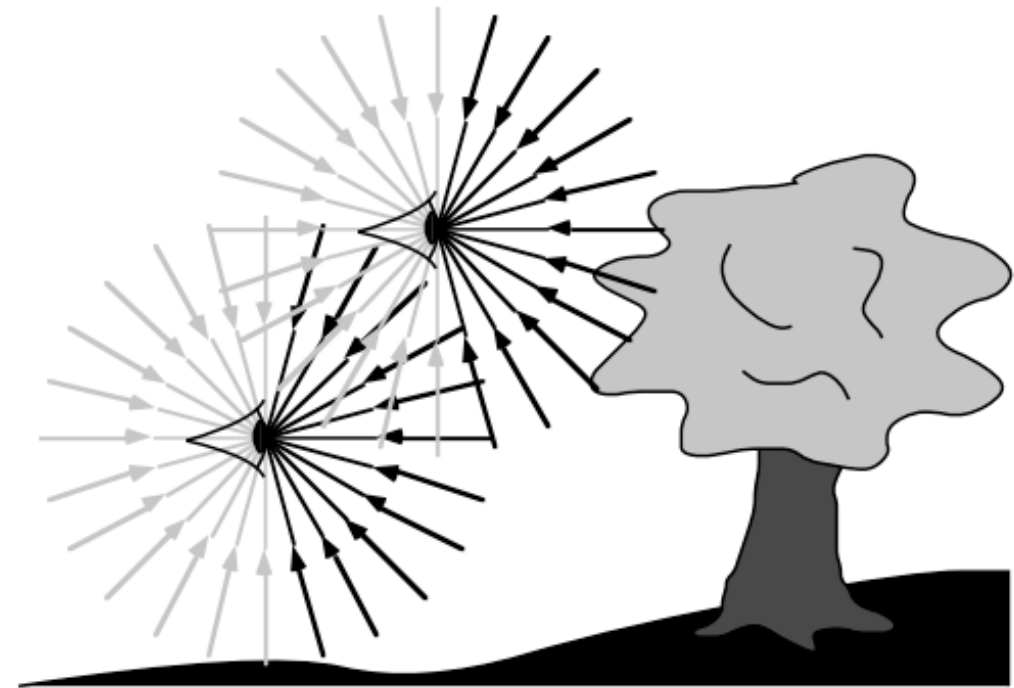
The Plenoptic Function and the Elements of Early Vision [Adelson and Bergen 1991]

Plenoptic function

A representation of the light present in a scene, that encompasses most properties relevant to image acquisition.

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

- It is an idealized concept, and one does not expect to completely specify it for a natural scene.
- An observer takes *samples* from this function.



The Plenoptic Function and the Elements of Early Vision [Adelson and Bergen 1991]

Plenoptic imaging

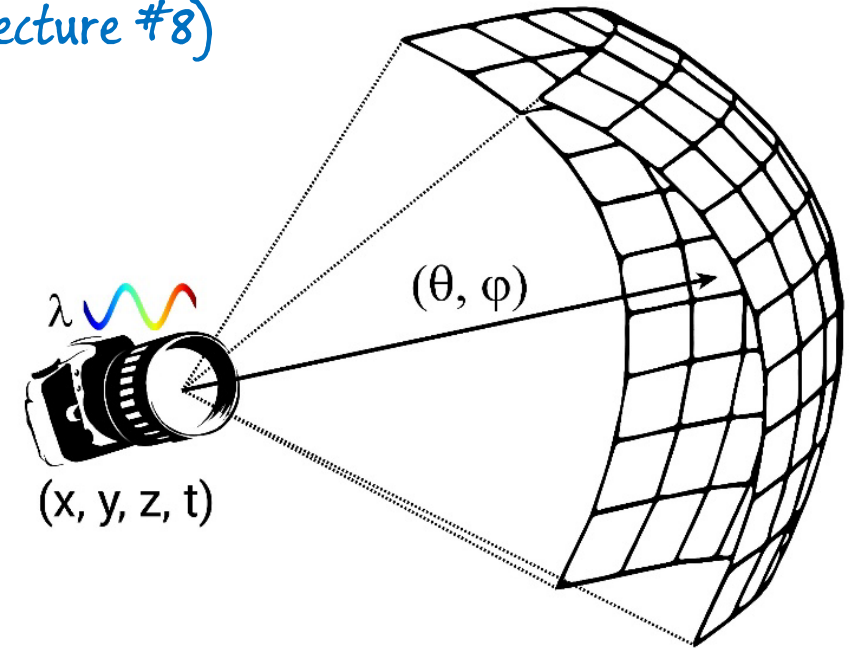
$$P = P(\theta, \phi, \lambda, t, \underline{x}, \underline{y}, \underline{z})$$

Transient imaging (Lecture #8)

Hyperspectral imaging (Lecture #6)

Light field imaging (Lecture #4)

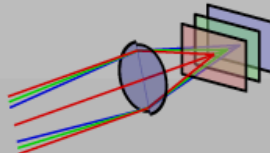
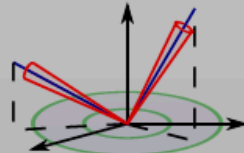
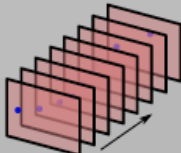
High dynamic range imaging (Lecture #5)



Plus common tools and techniques:
Coded photography (Lecture #3)
Compressive sensing (Lecture #3)
Computational illumination (Lecture #7)

Plenoptic imaging

There is a wealth of techniques for each dimension; we will explore a number of them throughout the semester.

Plenoptic Dimension	 Dynamic Range	 Color Spectrum	 Space Focal Surfaces	 Directions Light Fields	 Time
Acquisition Approach					
Single Shot Acquisition	Assorted Pixels Gradient Camera Adaptive DR Imaging	Color Filter Arrays Assorted Pixels Dispersive Optics	Coded Apertures Focal Sweep Field Correction	Plenoptic Cameras w/ Lenses, Masks, or Mirrors Compound Eye Cameras	Assorted Pixels Flutter Shutter Reinterpretable Imager Sensor Motion
Sequential Image Capture	Exposure Brackets Generalized Mosaics HDR Video	Narrow Band Filters Generalized Mosaicing Agile Spectrum Imaging	Focal Stack Jitter Camera Super-Resolution	Programmable Aperture Camera & Gantry	High-Speed Imaging Temporal Dithering
Multi-Device Setup	Split Aperture Imaging Optical Splitting Trees	Multi-Camera Arrays Optical Splitting Trees	Multi-Camera Arrays	Multi-Camera Arrays	Multi-Camera Arrays Hybrid Cameras

May involve one or more cameras, and one or more captured images.

Source: Wetzstein et al. Plenoptic imaging.

Take-home messages

Conventional photography mimics our eyes, and samples certain dimensions of the plenoptic function.

In it, optics capture something close to the final image, computation enhances this captured info.

Computational imaging aims at sampling more dimensions of the plenoptic function.

In it, generalized optics encode the visual information into an intermediate representation, computation decodes this representation into the final image(s).

Defocus deblurring, extended depth of field, depth recovery, and motion deblurring are enabled through computational imaging.

*Different techniques for each – often there is **feedback** between dimensions.*

References & further reading

Levin et al., “Image and depth from a conventional camera with a coded aperture,” SIGGRAPH 2007.

Veeraraghavan et al., “Dappled photography: Mask enhanced cameras for heterodyned light fields and coded aperture refocusing,” SIGGRAPH 2007.

The two papers introducing coded aperture for depth and refocusing, the first covers deblurring in more detail, whereas the second deals with optimal mask selection and includes very interesting lightfield analysis.

Nagahara et al., “Flexible depth of field photography,” ECCV 2008 and PAMI 2010.

The focal sweep paper.

Dowski and Cathey, “Extended depth of field through wave-front coding,” Applied Optics 1995.

The wavefront coding paper.

Levin et al., “4D Frequency Analysis of Computational Cameras for Depth of Field Extension,” SIGGRAPH 2009.

The lattice focal lens paper, which also includes a discussion of wavefront coding.

Cossairt et al., “Diffusion Coded Photography for Extended Depth of Field,” SIGGRAPH 2010.

The diffusion coded photography paper.

Raskar et al., “Coded Exposure Photography: Motion Deblurring using Fluttered Shutter,” SIGGRAPH 2006.

The flutter shutter paper.

Levin et al., “Motion-Invariant Photography,” SIGGRAPH 2008.

The motion-invariant photography paper.

Gehm et al., “Single-shot compressive spectral imaging with a dual-disperser architecture,” Optics Express 2007.

This paper introduces the use of coded apertures for hyperspectral imaging, instead of depth and refocusing.

Masia et al., “Perceptually optimized coded apertures for defocus deblurring,” Computer Graphics Forum 2012.

Presents an optimization function which takes into account perceptual aspects.

Zhou and Nayar., “What are good apertures for defocus deblurring?,” ICCP 2009.

Algorithms for obtaining near-optimal aperture patterns.

Heide et al., “High-quality computational imaging through simple lenses”, SIGGRAPH 2013

Exploits the idea of using computation to get rid of optical artifacts caused by cheap lenses

Fergus et al., “Removing camera shake from a single image”, SIGGRAPH 2006

Example of motion deblurring covered in this talk

Zhou et al., “Coded aperture pairs for depth from defocus”, ICCV 2009.

Both papers show how to build on the basic approach to extract depth (also: see next topic in the course)